Introduction to Sustainable Economics

Additional attainments for the Master's in Sustainability Transformations

Table of contents

Preface 4

I Sustainable Economics 7

1 How Can We Think About Economics? 8
- 1.1 What Is Economics? 8
  - 1.1.1 Understandings of economics in history 10
  - 1.1.2 Understandings of economics in the present 11
- 1.2 History of Economic Thought 13
- 1.3 How Do These Different Understandings of Economics Come About? 17
- 1.4 Styles of Thought and Thought Collectives 19
  - 1.4.1 Thought collectives in economics 20
- 1.5 Pluralist Economics 21
  - 1.5.1 Scavenger hunt through pluralist economics 21
- 1.6 Sustainable Economics: The Approach of This Script 23

Self-test Quiz 24

2 Scope of Economics 28
- 2.1 Prosperity in a Changing World 29
  - 2.1.1 Gross domestic product (GDP) as an indicator of (material) prosperity 30
  - 2.1.2 Alternative indicators of prosperity 32
  - 2.1.3 Economic growth 34
- 2.2 Forms of Provision in the Economy 36
- 2.3 (Re)production Factors 38
  - 2.3.1 Land – can we replace nature with technology at will? 38
  - 2.3.2 Capital – why do some people own capital and others do not? 40
  - 2.3.3 Labour – why do I earn as much as I do – and why is some work, such as care work, often not paid at all? 41
- 2.4 Types of Goods 47
  - 2.4.1 Properties of goods 47
  - 2.4.2 External effects – why does a flight often cost less than a train ride? 49
- 2.5 Abstract Markets or Capitalist Production System 50
  - 2.5.1 How the market works in neoclassical economics 52
  - 2.5.2 The market as central coordinating mechanism 53
  - 2.5.3 Markets as arenas of power and distribution – capitalist production system 59
  - 2.5.4 What actually makes an economic system capitalist? 60
  - 2.5.5 Are financial crises accidents – or built into the system? 63

Self-test Quiz . . . 64

3 Problem Analysis: What Is Going Wrong? 68
3.1 Ecological Limits – How Much Earth Do We Actually Have – and How Much Do We Use? . . . 68
3.1.1 Planetary boundaries and climate crisis . . . 69
3.1.2 Different understandings of the nature-economy relationship . . . 73
3.1.3 Climate crisis – market failure or systemic problem . . . 78
3.2 Social Limits: Is the Pie Growing for Everyone – or Only for a Few? . . . 87
3.2.1 Social limits in the 21st century . . . 88
3.2.2 Justice and inequality . . . 90
3.2.3 Types and causes of inequality . . . 93
3.2.4 Division of labour and comparative advantage . . . 94
3.2.5 Power relations, labour share, and wealth inequality . . . 95
3.2.6 National and global inequality . . . 97
Self-test Quiz . . . 98

4 Strategies for a Sustainable Economy: Efficiency, Consistency, Sufficiency 102
4.1 Efficiency Strategy . . . 103
4.2 Consistency Strategy . . . 104
4.3 Sufficiency Strategy . . . 106
Self-test Quiz . . . 106

5 Role of the State and Policy Paradigm for Sustainability 109
5.1 Why "Market vs. State" Is Historically Not a Given . . . 110
5.2 Market-Liberal Paradigm: The State as Night-Watchman . . . 111
5.2.1 Critique of the market-liberal paradigm: markets are not neutral – they constitute . . . 111
5.3 Welfare-Capitalist Paradigm: The State as Stabilizer and Co-Shaper . . . 112
5.3.1 Critique of the welfare-capitalist paradigm: from "correcting" to "shaping" 112
5.4 Post-Growth Paradigm: The State as Guardian of Limits – or Part of the Problem? 113
5.4.1 Critique of the post-growth paradigm: the state as ideal collective capitalist 114
5.5 Beyond Market and State . . . 115
Self-test Quiz . . . 115

6 Sustainable Economics: A Synthesis 119
6.1 Six Core Elements . . . 120

7 Approaches to Sustainable Economics 121
7.1 Approaches of Sustainable Economics . . . 121
7.1.1 Welfare-capitalist paradigm . . . 122
7.1.2 Post-growth paradigm . . . 122

References 124

Preface

Version 1.0, October 2026

This e-learning course is intended for students to work through it at their own pace. There are no sessions held in presence. However, students can reach out to the lecturer for questions and support at any time via email. The course is structured along seven chapters that build on each other. Many sections are interactive and contain links and videos to consult. For more information on the course please see the information document of the course.

The course is divided into three blocks. In the introductory block, students are introduced to basic concepts and the theoretical assumptions of economics (Chapters 1-2). This is followed by a block on the problem analysis addressing the ecological and social challenges faced by the current economic and societal system (Chapter 3). In the final block, participants explore strategies, the role of the state, and economic policy paradigms, and bring these together in a synthesis that provides starting points for shaping a sustainable economy, and get to know concrete approaches from practice (Chapters 4-7).

Below you will find the contents of the course. Following the script will guide you through all the chapters and learning material in a sensible order. Since this course requires a high degree of autonomy in acquiring the material, we also encourage you to contact us via email if you want to discuss questions (see information document).

Learning Outcomes

Students can:

- understand that there are different theories in economics and are able to name them.
- can identify the foundations of the emerging field of sustainable economics.
- can distinguish between the problem analyses and solution approaches of sustainable economics and those of today's dominant school of thought (neoclassical and environmental economics).
- are able to critically question both traditional and new economic approaches (concepts) and assess the conflicts inherent in them.
- can reflect on the strengths and weaknesses of different theoretical approaches, both in general and in relation to concrete problems.
- can analyse how alternative economic models (e.g., circular economy, post-growth economics) function and evaluate their potential and limitations for economic policy.
- acquire basic knowledge about measures and transformation scenarios for implementing alternative economic models beyond the focus on maximizing profits and goods production.

Assessment

The Assessment is based on two parts.

The first part is a written exam. The exam covers all topics discussed in the script up to (and including) Chapter 3 (problem analysis). The exam can be conducted anytime you feel ready to take it. If you feel ready to take the exam, contact us (nicola.bezzola@unibe.ch) to arrange a time to take the exam. It is held in person. Once you contact us, we organise a room for you to take the exam. The exam is in a written format on ilias and will take 30 minutes. You need 60% of the points to pass the exam. If you fail the first attempt, you can take a second attempt.

The second part of this assessment requires you to write a reflective essay on the material you have studied and your understanding of economics and the economy. We invite you to consider how the course material has (or hasn't) changed your understanding of economics and the economy in the context of sustainable transformation. Use the questions below to guide your reflection paper. However, you can also refer to other sources if you find them relevant for your reflection. You may use any resources. Students must sign a declaration of authorship. The written paper should comprise 6,000–7,500 characters (excluding spaces, the reference list, the title page, footnotes, headers and footers, lists, tables, figures and appendices). This corresponds to approximately three pages.

Guiding questions

- How has your understanding of economics changed as a result of taking this course? What implications does this have for your understanding of sustainable transformation?
- How has your understanding of the economy changed as a result of this course? What are the implications for your understanding of sustainable transformation?
- What role does an understanding of economics play in the context of a sustainable transformation?

Please submit your reflection paper via ilias.

Both parts of the assessment can be done and submitted at any time. The course counts as completed as soon as students have worked through all chapters and passed both parts of the assessment. The course is graded with pass or fail.

Please note that the exam and the reflection paper must be completed **by 30 May 2027 at the latest**.

Reading guide

Definitions and further readings

Examples and reflections

Exercises and links to ILIAS

Part I

Sustainable Economics

1 How Can We Think About Economics?

Guiding question

What actually is economics – and why are there such different ideas about what counts as "economic" and how economic activity should be thought about?

1.1 What Is Economics?

The concept of economics is omnipresent in our everyday lives. Economic topics take a prominent place, for example, in newspapers, in our everyday discussions, in political debates, and on social media. And on the evening news, topics are repeatedly put into context by economists from an *economic perspective*. But what exactly is this economic perspective? It quickly seems as if everyone knows exactly what is meant by it. Intuitively, we often associate it with terms such as markets, prices, and finance. In fact, however, neither in the history of the term nor in the present has it been conclusively settled what is meant by economics and thus by the subject matter of economics as a discipline (in German, *Volkswirtschaftslehre*). Since economics is a social science, we are not dealing with laws of nature that we discover and describe. **Rather, there are different perspectives and theories on how we can look at particular questions**. In this context, we speak of schools of economic thought, which differ in their understanding of economics. The distinction is never clear-cut, since there is always overlap between schools of thought and always differences within them. What is understood by economics thus depends strongly on the perspective we take. The debate between schools of economic thought is consequently a constant societal and scientific process of negotiation. Accordingly, there have been and still are different definitions of economics and its subject matter, that is, of what exactly *economics* deals with.

Before we turn in the next chapter to the subject matter of economics and to central terms and concepts, we briefly survey different understandings of economics in history and in the present. To find our own bearings in this negotiation process about the subject matter of economics, it seems central to us to have a rough overview of the different understandings of economics and their subject matter.

Excursus: Economics as a discipline of the economic sciences

The economic sciences can be divided into two areas: business administration (*Betriebswirtschaftslehre*, BWL) and economics (*Volkswirtschaftslehre*, VWL).

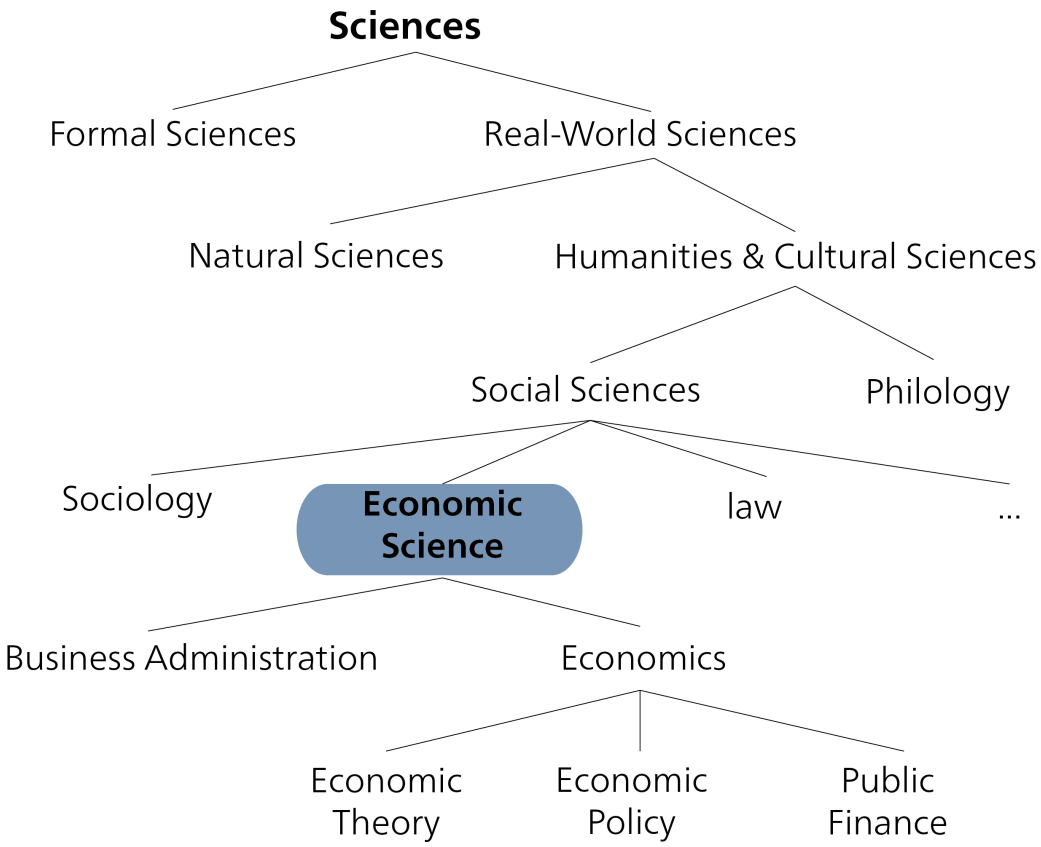


Figure 1.1: Disciplinary location of the economic sciences. Source: Own illustration after Engelkamp and Sell (2013, 3).

Business administration "is the study of the economic, organisational, technical, and financial processes in companies and the various economic institutions" (Friedli et al. 2019, 16). Researchers in this field develop a microeconomic perspective on the respective object of study. **Economics**, by contrast, does not examine what happens within individual economic actors but the interplay between them – that is, a macroeconomic perspective. This course deals with this perspective.

Economics can be divided into **three larger subfields**:

Economic theory: describes economic theories and their implications. It is divided into **microeconomics** (the behaviour of individual households and firms and their coordination through markets and prices). Microeconomics makes no statements about aggregate economic relationships but examines the behaviour of individual firms and households and their coordination process via the market by means of price adjustments (Bontrup and Marquardt 2021, 1). In contrast, **macroeconomics** deals with economy-wide phenomena and the rela-

tionships between economic sectors and different markets. It examines, for example, overarching economic indicators such as the unemployment rate or inflation.
Economic policy: deals with state interventions in the economy to increase societal welfare (Bontrup and Marquardt 2021, 2), for instance through interventions such as cartel bans or the introduction of minimum wages.
Public finance examines the various components of the state budget, such as taxes or subsidies, and their mutual relationship. In doing so, it implicitly examines questions about the role of the state in resource allocation and in the distribution of income and wealth.

1.1.1 Understandings of economics in history[1]

The term economics first appears in the history of Western thought in ancient Greece, as the study of household management. The word is composed of two terms: *oikos* (house) and *nemein* (to manage or arrange). In this understanding, economics dealt, among other things, with questions of managing one's own property, planning and assessing agricultural activities, selecting slaves, and the relationship between husband and wife. Economics was strongly embedded in politics and ethics and was meant to contribute to strengthening the *polis*, the city-state. The ultimate goal was the happy life, *eudaimonia*. In sharp contrast to economics stood so-called chrematistics, which aimed at the accumulation of money instead of the happy life and was rejected in ancient Greece.

In medieval Europe, the understanding of economics initially remained closely tied to moral and religious ideas. Economic activity was not understood as an independent area of society but was part of a divine and moral order. The goal of economic activity was not the greatest possible accumulation of wealth but securing a livelihood and the common good. This understanding was strongly shaped by Scholasticism, in particular by Thomas Aquinas (c. 1225-1274). At the centre were questions about a just price (*iustum pretium*), a just wage, and the moral legitimacy of economic activities. Trade was accepted in principle as long as it served the exchange of necessary goods. By contrast, making profit purely for its own sake was considered problematic. The charging of interest was judged particularly critically, since money itself was regarded as unproductive and was not supposed to serve the multiplication of money. The economy was thus strongly embedded in ethical and societal norms.

This began to change only with the emergence of modern capitalism, the expansion of trade, and industrialization, and an understanding of the economy as an independent area of society gradually became established. With the rise of mercantile capitalism and the Industrial Revolution, classical political economy emerged in the 18th century. Representatives such as Adam Smith (1723-1790), David Ricardo (1772-1823), or, later, Karl Marx (1818-1883) were less interested in the management of individual households than in the creation, distribution, and development of the societal wealth of entire national economies. The subject matter of economics thus shifted from household management to the analysis of the production, distribution, and accumulation of wealth. Questions such as these were at the centre: How does economic prosperity arise? How is societal income distributed between labour, capital, and land? The analyses focused on social classes – for example workers, owners of capital, and landowners – and their relationships to one another. The economy

[1] This short overview is based in part on Bäuerle et al. (2026).

was understood as a societal process whose development is shaped by institutions, property relations, and the distribution of the societal surplus.

Towards the end of the 19th century, the so-called marginalist revolution fundamentally changed the understanding of certain parts of economics. The works of representatives of this new current, such as William Stanley Jevons (1835-1882), Carl Menger (1840-1921), and Léon Walras (1834-1910), formed the basis for the emergence of neoclassical economics. At the centre of this new perspective was no longer the production of societal wealth but the individual endowed with the capacity for decision-making. The starting point of the analysis was now households and firms that make rational decisions on markets under conditions of scarce resources and maximize their utility or profit. The economy was understood as the result of the decisions of many individual economic subjects. This change meant a fundamental shift in perspective. While classical political economy looked at the national economy, that is, entire economies, and put social classes, production, and distribution at the centre, neoclassical economics takes individual markets above all as its frame of reference and focuses on individual decisions and price mechanisms. With the works of John Maynard Keynes (1883-1946), economy-wide relationships came into view again in the mid-20th century, not only in Keynesian schools of thought but also in neoclassical economics. Macroeconomics emerged on the basis of Keynes's work.

History was by no means a linear development from one understanding of economics to the next. Often, different understandings existed at the same time, influencing, contradicting, and developing one another. The perspectives outlined above are meant to make clear that the picture of what the economy is has changed considerably over time and continues to change. Moreover, present-day understandings build on these developments, so it is also important to understand where certain understandings come from and how they developed.

1.1.2 Understandings of economics in the present

With the development of economic theories, different ideas have developed about what economics actually studies. Two fundamentally different ways of understanding economics can be distinguished. Karl Polanyi already made this distinction in 1957 (Polanyi et al. 1957):

- On the one hand, economics can be understood through its **subject matter**: one determines *which section of reality* one studies – regardless of the analytical perspective. Medicine, for example, defines itself through its subject matter: the human body and its health. Polanyi calls this understanding the substantivist one. It places the economy as an empirical subject area at the centre of the investigation.
- On the other hand, economics can be understood through a **specific analytical approach**. In this approach, the analytical approach is the starting point – regardless of which area of reality it is used to study, that is, regardless of the subject matter. Polanyi calls this a formal understanding of economics. This analytical approach results from a logical derivation rather than from an empirical investigation.

In principle, almost all historical understandings defined themselves through the study of their subject matter. In ancient Greece, for example, this was the management of a household (how does one best manage the land, supplies, and slaves of a household?). In the 17th and 18th centuries, it

was the national economy (how does a country as a whole become prosperous, for example through trade and gold?). In both cases, it was clearly outlined *what* was being talked about – a specific section of reality.

With the marginalist revolution, the current economic mainstream increasingly departs from this principle, defines itself through its analytical approach, and develops a formal understanding of economics. Based on a definition by Lionel Robbins from the 1930s, economics is described as a science that studies how people choose between alternatives under conditions of scarcity in order to achieve their goals. Robbins writes:

> "Economics is the science which studies human behaviour as a relationship between ends and scarce means which have alternative uses" (Robbins 1932, 15).

Put simply: economics studies how people use scarce means (time, money, resources) to achieve their goals as well as possible – no matter *what* the goals are, although it almost always involves maximizing individual utility. A simple example: a student has only three hours in the evening (a scarce resource) and must decide whether to study for the exam, go jogging, or meet friends (alternative uses). Precisely such a decision under scarcity is at the centre of the mainstream approach and forms the analytical starting point of this understanding of economics. Economics students are not infrequently introduced to "the foundations of economic thinking" in a course. It is precisely about practising this analytical approach as the basis of economics.

From this analytical starting point, the actions of actors in the economy are then analysed, with a strong focus on markets, the price mechanism, and utility-optimizing individuals. Markets form the framework within which individuals try to optimize their utility through decisions under scarcity. This means that questions outside this framework often receive no attention. In this video, the economist Ha-Joon Chang talks about the neoclassical understanding, which, through its underlying analytical starting point, already gives a certain orientation and systematically screens out certain aspects (see also Chang and Chang 2014). And because this approach is not tied to a particular subject matter, in practice it is also applied to topics that one would not intuitively call "economic" – for example, the question of why people marry or have children (analysed as an "economic decision" under scarcity of, for example, time, money, and affection). In 1992, the economist Gary Becker received the Nobel Memorial Prize in Economic Sciences for precisely this extension of the field of application.

Many other schools of thought, by contrast, continue to define economics through its subject matter: the economy itself. They thus follow a substantivist understanding. The subject matter, and above all its boundaries (what belongs to the economy), can be defined in different ways. For Polanyi, the substantivist understanding is about examining how humans, as a society and in relation to their environment, provide the means of subsistence and distribute material prosperity. Concretely, this means questions such as "How is bread produced and distributed?" or "Who gets how much of societal wealth?" – regardless of the analytical approach used to examine these questions. As mentioned above, however, there are also different views on how exactly this subject matter should be delimited. *Feminist economics*, for example, emphasizes that activities outside the market, such as unpaid care work, are a central area of the economy and must be included in the analysis. *Ecological economics* similarly emphasizes that the economy is embedded in nature, that nature forms the basis of all economic activity, and that it must therefore be included in economic analysis. In line with the different understandings of the subject matter, different schools of thought also

focus on different aspects, such as power structures, institutions, or economy-wide and societal relationships and structures. In the section on pluralist economics (see Chapter 1.5, *Pluralist Economics*), we will look more closely at what characterizes neoclassical economics (the mainstream) and get to know some of the other schools of thought.

In this script, we follow a substantivist understanding and define economics through the subject matter of the economy. In Chapter 2 (*Scope of Economics*), we will then examine this subject matter in more detail.

Interactive exercise – Substantivist or formal?

Substantivist or formal?: available in the online version of this script.

Further reading

Ha-Joon Chang's book *Economics: The User's Guide* offers an easy introduction to topics in economics and gives a good overview of the history and current debates. Depending on your interest, individual chapters can also easily be consulted on their own.
Chang, Ha-Joon. 2014. *Economics: The User's Guide*. First U.S. edition. Bloomsbury Press.

1.2 History of Economic Thought

We have now looked briefly at different understandings of economics. In this section, we want to engage a little more with the history of economic theory and ask how economics itself has developed. The emphasis is on the emergence of the dominant school of thought, neoclassical economics.

Mandatory reading

Ötsch (2018): download and read at least chapters 4 to 9.
In this working paper, Walter Ötsch describes how economics developed from a moral science under Adam Smith (1723-1790) into a science with a biologically determined image of the human under Thomas Malthus (1766-1834) and David Ricardo (1772-1823). In this process, metaphors from the natural sciences (clock system, balance equilibrium, computer information) become increasingly important, and economics comes to be seen as "social physics". This process is very illustrative of the development of modern neoclassical economics.
Walter Ötsch is a professor at the University of Society Design (Hochschule für Gesellschaftsgestaltung) in Koblenz. As an economist and cultural historian, he researches and publishes on socio-political and economic topics such as populism and the societal role of markets.
Download and read Ötsch (2018), at least chapters 4 to 9

Excursus: History of homo oeconomicus

The history of homo oeconomicus goes back at least to the mention of the "economicus" in Xenophon's work of the same name in the fourth century BCE (Wilson and Dixon 2014, 11). In the long period until the founding of modern economics by Adam Smith in the 18th century, various conceptions of the economic man[2] were used – often strongly shaped by the respective spirit of the age.

The concept of homo oeconomicus, as a rational utility maximizer, is commonly located as early as Smith, since he recognizes self-love as a central human characteristic in behaviour. But as so often when it comes to Smith's concepts, this common portrayal is idealized. Besides self-love, Adam Smith identifies further human characteristics that are central to behaviour. Smith paints a much more complex picture of human behaviour than the reduced version of the modern homo oeconomicus does (Hill 2012). Smith's dense description of the economic man makes it impossible to model him or to work with him mathematically (Morgan 2006, 2012). This was to change over time through a deliberate reduction of the description of the economic man.

Thomas Robert Malthus and David Ricardo already used somewhat more reduced versions of the *economic man* at the beginning of the 19th century, which already show a certain model character (Morgan 1996, 2006, 2012). John Stuart Mill (1806-1873) took this further and reduced the economic man to the characteristics considered central to the economic domain: the desire to accumulate wealth, the aversion to work, and the desire for luxury goods (Robson and Mill 1996). Although the model of the economic man was reduced in Malthus, Ricardo, and Mill, it could not yet be used, and was not used, in the way it is today.

A decisive foundation for this development was laid by William Stanley Jevons (1835-1882) and Francis Edgeworth (1845-1926) towards the end of the 19th century. Alongside Alfred Marshall (1842-1924), Léon Walras (1834-1910), and Vilfredo Pareto (1848-1923), they are regarded as important founders of neoclassical economics and were part of the marginalist revolution. Like some other economists with a background in mathematics or physics, the two economists were willing to make economics a natural science. The neoclassical economists were convinced that the analytical and deductive method should also be the royal road for economics. Jevons justified the necessity of mathematics for economics simply by the fact that economics deals with quantities (Jevons 1879, 4). This group of economists took their orientation from physics and physical concepts. Irving Fisher (1867-1947), another of the first neoclassical economists, explicitly used various mechanical or hydraulic concepts for his theories and applied them to the economy.

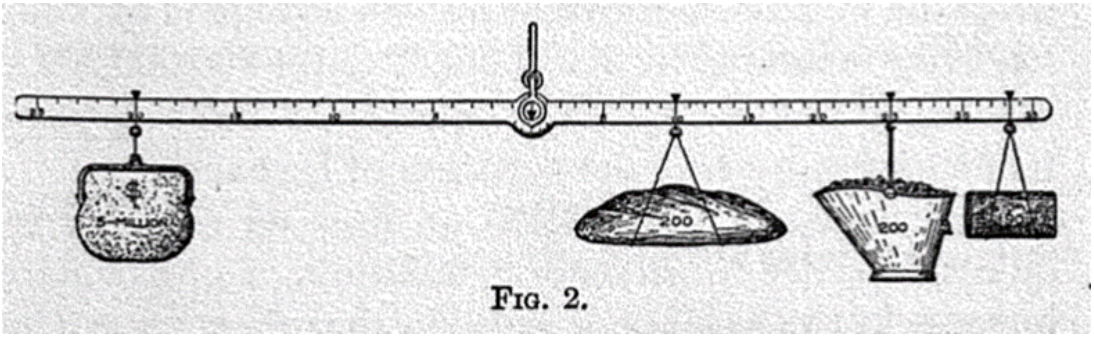


Figure 1.2: Fisher's mechanical balance of exchange, used to represent the quantity theory of money (Fisher 1922, 21).

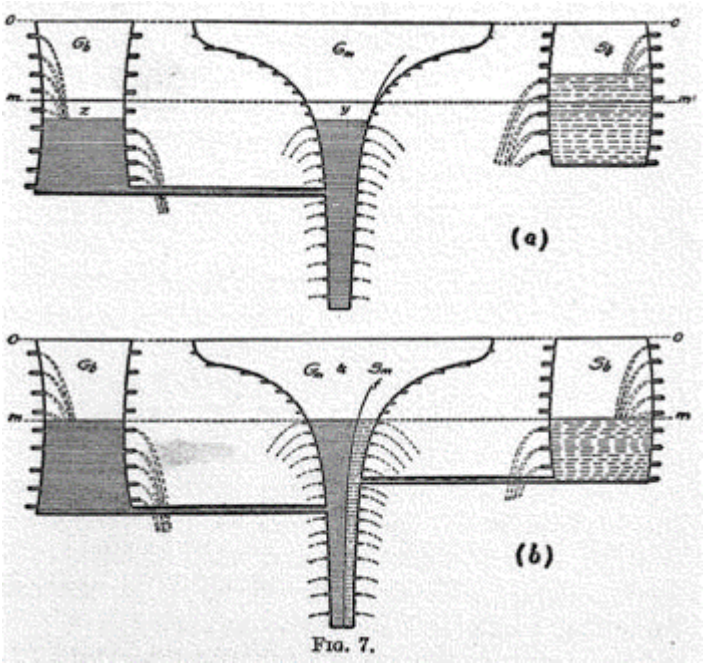


Figure 1.3: Fisher uses a hydraulic model to represent purchasing power in a bimetallism system (Fisher 1922, 119).

The idea of equilibrium itself is a concept borrowed from physics. This fixation on physics is also described as *physics envy*, and Mirowski shows that this orientation towards physics was a fundamental driver in the development of the history of theory (Mirowski 1989, 396). The explicit analogy to physics in economics persists well into the 20th century. A good illustrative example of this, and of economists' basic toolbox, is the machine built by William Phillips (1914-1975) in the 1940s to represent the British economy. The development of homo oeconomicus is accordingly also exemplary of the mechanistic worldview that emerged in neoclassical economics, as Walter Ötsch describes it in his lecture.
Among other things, the complexity of the *economic man* stood in the way of applying mathematics to economics. Jevons therefore attempted to reduce this complexity. Shaped by the

utilitarianism of Jeremy Bentham (1748-1832), Jevons reduced the seven criteria Bentham had formulated, which are central to calculating utility and thus influence human action, to just two, on the grounds that only these were economically relevant (Jevons 1879, 17). In this way, Jevons was able to work mathematically with the economic man (Reiss 2000). This abstraction and objectification of economics and of the economic man is most vividly illustrated by the conceptions of Francis Edgeworth and, later, Frank Knight. Edgeworth speaks of the economic man as a "pleasure machine" and Knight of the "slot machine" (Morgan 2006). Pareto's use of his "homo oeconomicus" also grounds an analysis of the human from a mechanical viewpoint. Pareto writes: "Just as analytical mechanics treats material points and rigid bodies, mathematical economics considers an abstract human being, a homo oeconomicus" (Pareto and Meyer 1902, 1100).

With the marginalist revolution, homo oeconomicus became the focus of economics, which stands in strong contrast to the classical economists (such as Smith, Malthus, Ricardo, and Marx), who focused predominantly on class analysis. Moreover, the abstraction now enabled modelling and mathematical applications, which was meant to bring economics closer to the natural sciences. The rational, utility-maximizing agent became the foundation of economics. The rational choice theory behind it was subsequently developed further and adapted (for example through the concept of bounded rationality) but remains in its fundamentals to this day.

The history behind this development shows that it was not a natural one but that a particular orientation of economics was deliberately pushed forward. The neoclassical orientation, for example, makes it impossible to recognize and analyse structural problems, since the focus is on marginal changes in certain indicators (for example inflation or interest rates). The work of the 2019 winners of the Nobel Memorial Prize in Economic Sciences, Abhijit Banerjee, Esther Duflo, and Michael Kremer, seems exemplary of this. In their research on poverty reduction, they use experiments (so-called *randomized controlled trials*) to examine the effects of specific individual measures to reduce poverty. The analysis (and thus also a change) of the structural causes of these people's poverty falls out of view here.

This development of economics, as outlined here, was strongly criticized right from its emergence. Criticism of the orientation, the fixation on analytical models, and the strong abstractions from the real world continues to this day. In view of the weaknesses of neoclassical economics that have increasingly been exposed in the years since the 2008 financial crisis, Alfred Marshall's concerns about this development have become topical again. In 1881, in a commentary on Edgeworth's "Mathematical Psychics", Marshall wondered whether Edgeworth would succeed in preventing mathematics from running away with him and driving him out of sight of the actual facts of the economy (Vazquez 1995, 251).

[2] In this text, the English term economic man is used when the concern is, in general, the idea of an economic agent. This is because the term homo oeconomicus stands for the currently dominant conception, and economic man is used in this sense in the (predominantly English-language) literature. The term is problematic in that it implies the exclusion of an economic woman. If we look at the orientation of economics today and in history, however, such a portrayal of economics is unfortunately not that far-fetched, as feminist economics shows us.

Further reading

If you are interested, you can also watch the entire lecture by Walter Ötsch (in German).
Lisa Herzog (2014) in her book *Der Wert des Marktes* [The Value of the Market]: The defence of the market from the 18th century to the present (pp. 13-27) (in German).
Lecture by Lisa Herzog on the history of ideas as an interdisciplinary approach to economics (in German).
Hans Christoph Binswanger (2006) in his book *Die Wachstumsspirale* [The Growth Spiral]: From Aristotle to Walras and back – Aristotle as a precursor of modern economics (pp. 377-387) (in German).
The following further reading is for students who want to dive somewhat deeper and engage more intensively with the history of theory:

- Ambler, Lucy, Joe Earle, and Nicola Scott. 2022. "Whitewashes History". In *Reclaiming Economics for Future Generations*, Manchester Capitalism Ser, Manchester: Manchester University Press, 118–60.
- McCloskey, Deirdre N. 1998. *The Rhetoric of Economics*. 2nd ed. Rhetoric of the Human Sciences. Madison, Wis: University of Wisconsin Press.
- Mirowski, Philip. 1989. *More Heat than Light: Economics as Social Physics, Physics as Nature's Economics*. Cambridge: Cambridge University Press. https://doi.org/10.1017/CBO9780511559990.

1.3 How Do These Different Understandings of Economics Come About?

Philosophy of science distinguishes between two basic questions: **Ontology** asks what reality actually is. **Epistemology** asks how we can gain knowledge about this reality. These questions seem abstract but have important consequences for our understanding of the economy.

Many economists, especially in neoclassical economics and the economic mainstream, hold, for example, that economic theory must be independent of images of the human and of society, and thus also independent of social philosophies and values. According to this view, economic theory should be neutral and free of values and able to formulate universally valid laws. This view understands economics as a "positive" science. **Positivism** results from an ontological and an epistemological position, even if these are not always made explicit. Ontologically, positivism assumes that there is a social and physical nature or reality that exists independently of the historical context and is subject to universal laws. These laws exist independently of our perception. Epistemologically, positivism also assumes that laws can be objectively recognized and described. Scientific methods serve this purpose and can bring this reality to light. Since positivism is mainly associated with the natural sciences (although this view is not shared by everyone there either), a positivist conception of economics often draws strongly on the natural sciences. It claims to describe the world "as it is" without having previously made a value judgement or taken a particular perspective. Accordingly, this view distinguishes between positive statements (how the world is) and normative statements (how the world should be) and argues that these spheres can be clearly separated. Economics should focus on positive statements. For economics, this position consequently means that there cannot

necessarily be different schools of thought that reach different conclusions, since there is one objective reality that we can also describe as such.

Critics argue that such a clear separation between positive and normative statements, between facts and values, is not possible for social sciences such as economics and accordingly makes an objective and value-free analysis impossible (see, for example, DeMartino et al. 2016). Hilary Putnam argues that facts and values are often not clearly separable but intertwined (Putnam 2002). Here, too, we find ontological and epistemological basic assumptions that lead to such conclusions. The term **"constructivism"** can serve as a collective term for many epistemological currents. All of these currents assume, in one way or another, a socially constructed reality. Although some currents assume an objective reality ontologically, they agree that the way we generate knowledge and what we recognize as knowledge is historically and socially specific. Accordingly, constructivism supports the position that scientific insights are always also shaped by social contexts, historical developments, and normative presuppositions. Economic concepts and theories are therefore always already shaped in their construction by values and worldviews. Which questions are asked and which answers seem plausible thus also depends on the respective perspectives of the researchers. For economics, this means that different theoretical approaches can legitimately arrive at different interpretations of economic reality or realities.

Our position: We consider a strict separation of the positive and the normative impossible and orient ourselves towards a constructivist understanding of science. The different views on how economics can be defined already show that value judgements are always made implicitly. One perspective can focus on the efficient allocation of scarce resources in a society, another on overcoming poverty and meeting the basic needs of all people in an economy. As soon as we adopt one perspective at the expense of another, we have made a first value judgement.

This does not mean, however, that there are no facts or that all approaches are equally valid, nor that there is no reality independent of our interpretation – we can only approach it more or less closely. Like any other science, economics has to orient itself towards scientific principles and methods. In addition, assumptions and value judgements should be made explicit and transparent.

The dispute over whether economics is free of any value judgements or not has run from the emergence of economics as its own field of science to the present and has not been conclusively resolved. Two quotes from two well-known economists of the first half of the 20th century illustrate this:

> "Economics deals with ascertainable facts; ethics with valuations and obligations. The two fields of enquiry are not the same plane of discourse" Lionel Robbins in An Essay on the Nature and Significance of Economic Science (Robbins 1932, 132).
>
> "As against Robbins, Economics is essentially a moral science. That is to say, it employs introspection and judgement of value" John M. Keynes in a letter to Sir Roy Harrod, 4 July 1938, in (Atkinson 2009, 791)

Excursus: Do assumptions have to be realistic?

Besides the criticism of the value judgements implied in neoclassical economics, its explicit framework of assumptions also leads to controversial discussions, in particular over the extent to which assumptions must represent reality realistically. Do the initial assumptions,

such as the existence of perfect competition, have to be realistic in order to gain valuable insights? At first glance, the finding seems clear: unrealistic or incomplete assumptions would also have to lead to false conclusions. According to the American Paul Samuelson (1915-2009), one of the best-known economists of the 20th century, it is impossible to infer true insights from demonstrably false assumptions. Proponents of so-called **instrumentalism**, however, argue differently. In view of the complexity of human behaviour, they say, one must positively abstract in the analysis and leave out what is inessential. The American economist Milton Friedman (1912-2006), also one of the best-known economists of the 20th century, in search of an efficient method for discovering far-reaching insights and policy recommendations, even establishes an extreme anti-realist position. Analysing within a make-believe world is declared an unproblematic necessity:
"Truly important and significant hypothesis will be found to have assumptions that are widely inaccurate descriptive representations of reality, and in general the more significant the theory, the more unrealistic the assumption. […] To be important, therefore, a hypothesis must be descriptively false in its assumptions" Milton Friedman (1953) [p. 14].

1.4 Styles of Thought and Thought Collectives

As discussed in the previous section, economic theories are always shaped by value judgements and worldviews. There is therefore no clearly objective economic theory but a multitude of perspectives that make it possible to analyse and describe economic phenomena. There is no neutral "view from nowhere" or purely objective perspective, since our perception is shaped by numerous influences, such as our education, our social and cultural environment, our language, our material living conditions, and our personal experiences. These influences shape our perspective on looking at the world.

Perspectives can also be regarded as "glasses": some glasses allow a view of the big picture, while others serve to identify details and subtleties. These different glasses mean that we can see the world in different colours and shades. Nevertheless, this does not mean that the world can be interpreted arbitrarily. There are limits to multiperspectivity, which require economics to orient itself towards scientific principles and methods.

In the philosophy of science, perspectives are also referred to as **paradigms or styles of thought**. Ludwik Fleck (1896-1961), a Polish physician and philosopher of science, developed the concept of the style of thought decades before Thomas Kuhn (1922-1996), who created the better-known concept of the paradigm. In 1944, Fleck was deported to the Buchenwald concentration camp, where he was made to develop a typhus vaccine. However, he supplied the SS only with a placebo and distributed the real medicine to fellow prisoners.

In his scientific work, Fleck emphasized the **dependence of thinking on values and contexts**. He objected to the idea that knowledge is objective, neutral, and universal. He argued that the production and use of knowledge take place in specific environments, which influences their meaning and effect. Knowledge is used differently in concentration camps than in NGOs or corporate research departments. Knowledge is always embedded in institutions and power structures, and scientists are influenced by their environment and experiences.

Thought collectives are groups of people who share a common style of thought and use particular concepts and methods. Thought collectives thus use a particular pair of glasses to look at the world. These thought collectives often resist change and resist further developments of their way of thinking. Since, as Fleck emphasized, people can think, argue, and understand in fundamentally different ways, people who belong to different thought collectives often have difficulty following the trains of thought of other styles of thought.

Fleck's insights show that science and thinking are by no means independent of social, cultural, and historical contexts. They make clear the **diversity of perspectives** from which scientific problems can be viewed and underline that no single theory or way of thinking can capture the whole of reality completely. Instead, different styles of thought and thought collectives are able to provide diverse insights and enrich the understanding of complex phenomena. This leads to different economic policy recommendations and offers a broad range of approaches to solving complex challenges.

1.4.1 Thought collectives in economics

As already indicated several times, economics too consists of a multitude of different thought collectives that take particular perspectives, or styles of thought, based on value judgements and worldviews. This diversity of styles of thought is described by the term "pluralist economics". At present, economics is strongly dominated by one style of thought, neoclassical economics, which is often described as the economic mainstream. Nevertheless, or perhaps precisely for this reason, it is central that students have a rough overview of the different styles of thought in order to gain a comprehensive understanding of the economy. Accordingly, the next section briefly introduces pluralist economics, in which students can then explore the different styles of thought according to their interests. We believe, however, that students should not only get to know the style of thought of the dominant thought collective but also gain insight into how the styles of thought in economics have developed and which other styles of thought exist in economics. In the next section, you will get an overview of the different styles of thought in economics. In what follows, we will mostly call them schools of thought rather than styles of thought, since this is what they are usually called.

Further reading

Yuval Harari and the Peugeot legend in his book *Sapiens: A Brief History of Humankind* (in German)
His thesis: the truly sufficient difference between us and the other animals is the ability to create cooperative networks in which sometimes millions of complete strangers work together towards shared goals. We can cooperate with one another on a large scale because we follow shared intersubjective fictions. We work successfully with strangers because, like them, we believe in things such as gods, nations, money, or human rights. And yet none of these things exists outside the stories that people invent and tell one another. There are no gods in the universe, no nations, no money, and no human rights – except in the shared imagination of human beings.
Podcast with Yuval Harari on "trickle-down economics": What is the trick in trickle down?
How do wealthy elites and their neoliberal mouthpieces manage to convince us that what benefits them – tax cuts, deregulation, and the like – also benefits all of us? And why, of all

things, are the minimum wage, paid overtime, or sick days considered a danger to the economy? Economics is ultimately a story we tell ourselves to explain who gets what – and why. The podcast explores the question: What might a better story look like?

1.5 Pluralist Economics

As has been indicated several times, there are different schools of thought based on different understandings and perspectives. Some of the differences arise from fundamental philosophical questions, from methodological approaches, or from differences in content (for example different foci). Despite its diversity, present-day economics is dominated by the neoclassical school of thought. The dominance was particularly strong towards the end of the 20th century and in the 2000s. The great financial crisis of 2008 boosted the rise of theories that had been marginalized until then, since neoclassical economics was evidently inadequate to fully explain this crisis. Nevertheless, neoclassical economics remains dominant in the economic sciences to this day.

To be able to meet the complexity of present-day challenges, we consider it essential to be able to draw on this diversity of theories and tools. The dominance of a single school of thought carries the risk that we perceive certain phenomena from only one perspective and can accordingly offer only solutions that follow from that perspective, as the following quote illustrates:

> "I suppose it is tempting, if the only tool you have is a hammer, to treat everything as if it were a nail" Abraham Maslow in Psychology of Science (Maslow 1966, 15).

This short section is therefore meant to give you the opportunity to engage with the diversity of the economic sciences, that is, pluralist economics. Engaging with different schools of thought should enable students to perceive and reflect on economic questions in a more differentiated way. It will not be possible, however, to cover all schools of thought in depth. In the self-study phase, you will find opportunities to familiarize yourself independently with the different schools of thought if you are interested. We are also happy to provide further material if you would like to engage more intensively with pluralist economics or individual schools of thought. In addition, for students who want to engage more intensively with different schools of thought, we offer the course "Introduction to pluralist economics" in the spring semester.

Before you set out to explore in the "scavenger hunt through pluralist economics", this video by Professor Ha-Joon Chang of SOAS University gives you a rough overview of different schools of economic thought and how they can (or could) lead to different economic policy programmes.

1.5.1 Scavenger hunt through pluralist economics

Exploring Economics is an open project of the Network for Pluralist Economics, which cooperates with various international actors in designing the e-learning platform. The network consists of many local groups and individuals. There are numerous ways to take part and contribute. We believe that this open platform offers a great many good contributions. We have therefore designed a scavenger hunt through the platform so that you can get to know it in a playful way.

Interactive exercise – Pluralist economics

In this learning section, you solve the crossword puzzle "A journey through economic theories" below using the Exploring Economics platform. Under the heading "Orientation" there, you will find solutions to the terms sought in the puzzle. This gives you an overview of the different schools of thought in economics and of how these perspectives influence research and its results. This exercise is voluntary and is meant to allow you to get to know pluralist economics in a playful way. You can check your answers and display the solution directly in the crossword. You are welcome to contact us to discuss any questions or clarify any uncertainties.
Crossword: A journey through economic theories: available in the online version of this script.

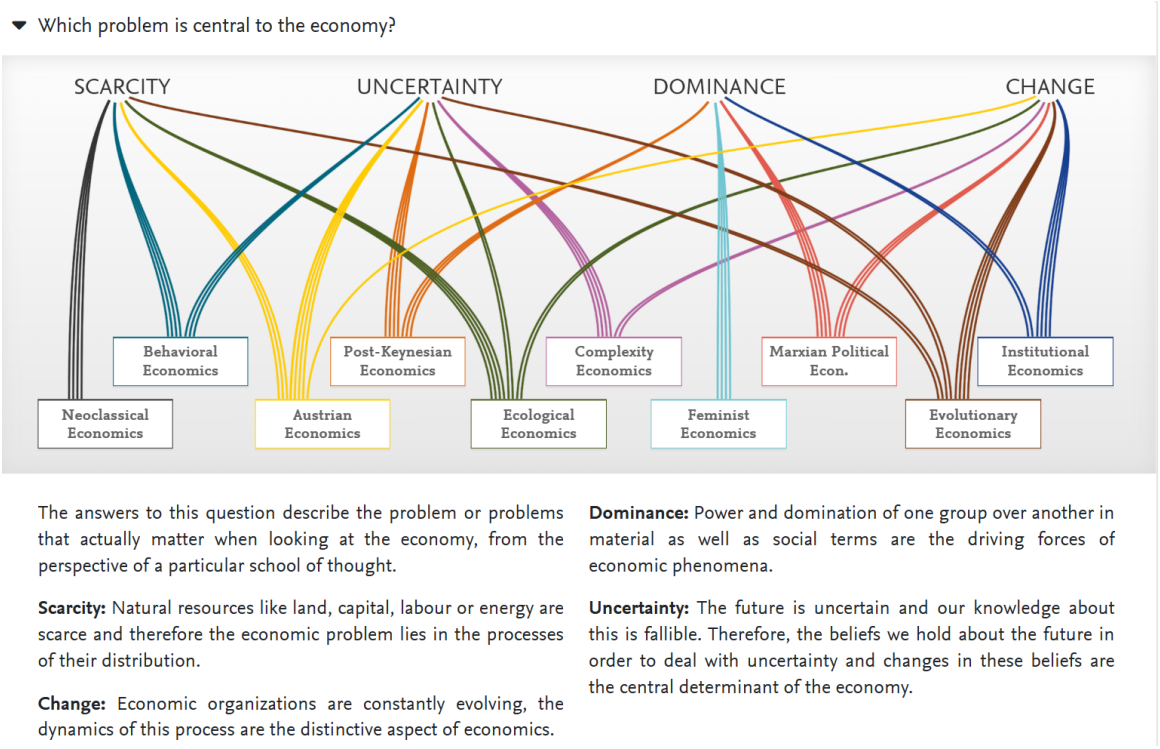


Figure 1.4: Excerpt from the Exploring Economics website, categorizing the different schools of thought according to the question. Source: Exploring Economics.

In this course, we explicitly orient ourselves in what follows towards an understanding that defines economics through its subject matter and takes no theoretical orientation as the basis of our understanding of economics. Accordingly, this understanding also leaves room to draw on many different approaches and schools of thought to understand economic relationships. In what follows, we therefore first want to approach the subject matter and get to know some basic terms and concepts.

1.6 Sustainable Economics: The Approach of This Script

As we have seen in this chapter, there is no single economics but a multitude of schools of thought with different basic assumptions, questions, and perspectives on economic phenomena. We have also established that scientific perspectives are never entirely neutral but always already contain value judgements. This script follows the approach of **sustainable economics**. The guiding question is how our economy can be designed sustainably. Sustainable economics thus explicitly has a normative character. We will see that answering this question also depends strongly on the specific understanding of sustainability, but also on fundamental questions about the goals of economic activity and on the understanding of the subject matter of economics itself.

A first point of approach is the contrast often drawn in debates about a more sustainable economy, that between **green growth** and **post-growth**[3].

Green growth approaches (e.g. OECD 2011; World Bank 2012) assume that economic growth and declining environmental pressure can be reconciled – the economy is meant to grow but become "green". Post-growth approaches (e.g. D'Alisa et al. 2016; Jackson 2009; Seidl and Zahrnt 2010), by contrast, argue that growth itself – and the growth imperative of capitalist economies associated with it (see Chapter 2.1.3, *Economic growth* and Chapter 2.5.4, *What actually makes an economic system capitalist?*) – is a central cause of ecological problems.

This polarization is didactically helpful because it makes a real debate visible. But it is also misleading if taken too literally: in practice, it is rarely an either-or between two closed camps but a question of how to design an economy that respects ecological limits while enabling a good life for all.

[3] The term degrowth is also often used. Although the two terms are not entirely identical, they largely overlap. For simplicity, we will therefore use the term post-growth throughout. In our view, the strength of the term post-growth lies above all in the fact that it does not emphasize a possible shrinkage of the economy so much as the necessary growth independence.

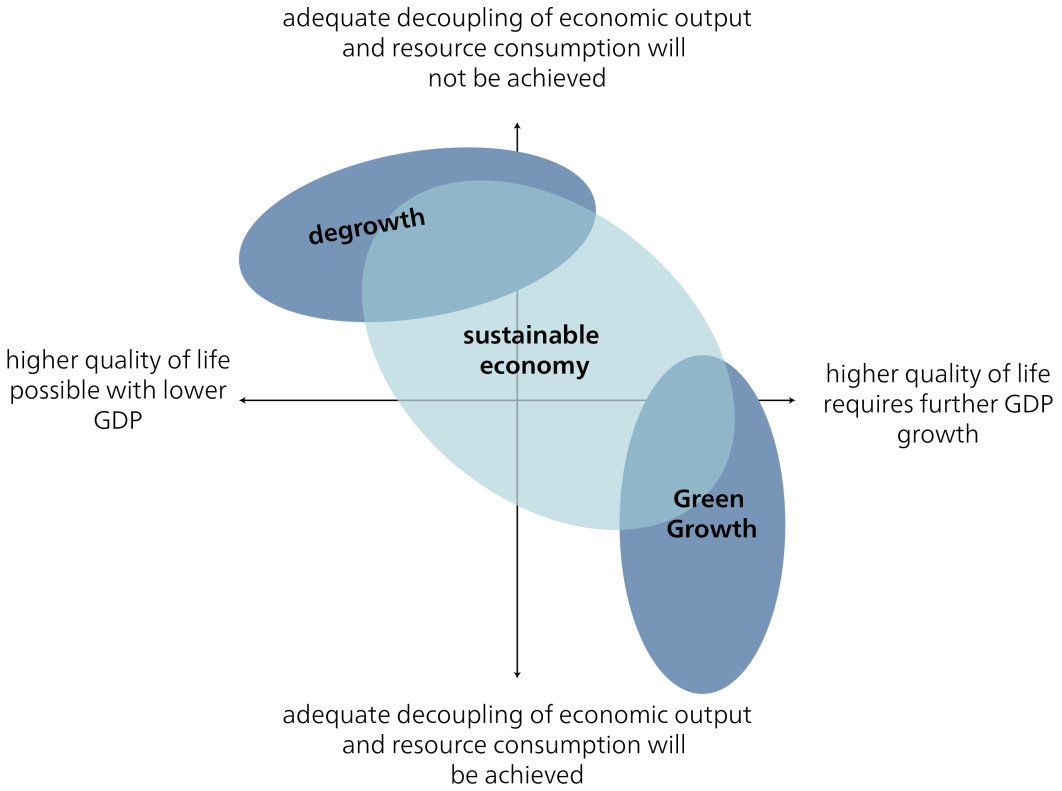


Figure 1.5: Figure 5. Sustainable economics. Source: Own illustration, adapted from Petschow et al. (2018, 22).

This is precisely where sustainable economics comes in: it does not position itself at either of the two poles but asks what ecological and social sustainability require, and only then derives what role growth can or must play. It thus does not start with a fixed answer ("more green growth" or "shrinkage") but asks instead about the goals of economic activity.

We will not yet present a definition of sustainable economics here but will approach an understanding of it over the next chapters. In the next chapter, we approach the subject matter of economics and address, among other things, the question of what the goal of economic activity actually is and what we understand by prosperity. After that, we look at the ecological and social limits and challenges of our current economic system. Sustainable economics aims to develop approaches to overcoming these challenges. Then, in Chapters 4 and 5, we look at possible strategies and roles of the state. In the concluding chapter of this script, we then want to draw a clearer picture of sustainable economics from all this.

Self-test Quiz

Test your understanding of this chapter with 7 short questions. The quiz is voluntary and ungraded, and you can repeat it as often as you like.

Question 1.1 (based on section 1.1)

An economist analyses why people marry and have children as a decision under scarcity of time, money and affection. Which understanding of economics does this reflect?

a) A substantivist understanding, because family life is part of the economy
b) A formal understanding, because economics is defined by its analytical approach, regardless of the subject matter
c) The understanding of classical political economy, because it concerns the reproduction of the working class
d) A feminist understanding, because it makes unpaid care work visible

Question 1.2 (based on section 1.1)

Which of the following questions is typical of a substantivist understanding of economics?

a) How does a student best divide three free hours between studying, sport and friends?
b) How does a household maximize its utility given its budget?
c) How is bread produced and distributed, and who gets how much of societal wealth?
d) How do prices adjust until supply equals demand?

Question 1.3 (based on section 1.1)

What changed with the marginalist revolution at the end of the 19th century?

a) The focus shifted from household management to the wealth of entire nations
b) The focus shifted from social classes, production and distribution to individual decisions on markets
c) The focus shifted from individual markets to economy-wide relationships
d) Economics was re-embedded in a moral and religious order

Question 1.4 (based on section 1.1)

In ancient Greece, chrematistics, the accumulation of money, was regarded as the core of economics.

a) True
b) False

Question 1.5 (based on section 1.3)

Which statement reflects a positivist view of economics?

a) Facts and values are often intertwined and cannot be clearly separated
b) Economics can describe the economy "as it is", and positive statements can be clearly separated from normative ones
c) Different schools of thought can legitimately arrive at different interpretations of economic reality
d) Which questions researchers ask depends on their social and historical context

Question 1.6 (based on section 1.4)

According to Ludwik Fleck, why do members of different thought collectives often have difficulty following each other's arguments?

a) Because one collective usually has access to better data
b) Because they share a style of thought with its own concepts and methods, so they think, argue and understand in fundamentally different ways
c) Because thought collectives are organized along national language borders
d) Because only one collective can be scientifically correct at any time

Question 1.7 (based on section 1.6)

How does sustainable economics, as understood in this script, position itself in the debate between green growth and post-growth?

a) It sides with green growth, because decoupling has been achieved worldwide
b) It sides with post-growth, because the economy must shrink
c) It first asks what ecological and social sustainability require and only then derives what role growth can or must play
d) It treats growth as irrelevant and focuses only on distribution

Answers and explanations

1.1 b). Polanyi distinguishes two understandings. The formal one defines economics by an analytical approach (choice under scarcity, following Robbins) and applies it to any subject, including marriage. Gary Becker received the Nobel Memorial Prize for exactly this extension. A substantivist understanding would instead start from the economy as a subject matter: how people provide their means of subsistence.

1.2 c). The substantivist understanding defines economics through its subject matter: how a society, in relation to its environment, provides the means of subsistence and distributes prosperity. The other three questions start from the analytical approach of choice under scarcity or the price mechanism, which is the formal understanding of the mainstream.

1.3 b). Classical political economy (Smith, Ricardo, Marx) looked at whole economies, social classes and the distribution of the societal surplus. With Jevons, Menger and Walras, the starting point became the rational individual who maximizes utility or profit on markets. The shift from household to national wealth happened earlier (classical political economy), and the return to economy-wide questions came later with Keynes.

1.4 False. False. Economics (oikos + nemein) meant household management embedded in politics and ethics, with the happy life (eudaimonia) as the ultimate goal. Chrematistics, which aimed at accumulating money, stood in sharp contrast to it and was rejected.

1.5 b). Positivism assumes a reality governed by universal laws that can be described objectively, so economics should stick to positive statements (how the world is). The other options describe the critique: Putnam's entanglement of facts and values and the constructivist view that knowledge is shaped by social contexts and perspectives.

1.6 b). A thought collective shares a style of thought, a particular pair of "glasses" with its own concepts and methods. Collectives tend to resist change, and people from different collectives see the world through different glasses. Fleck stressed that knowledge always depends on values and contexts, which is why the script argues for pluralist economics.

1.7 c). The polarization is didactically useful because it makes a real debate visible, but taken literally it is misleading. Sustainable economics does not start from a fixed answer ("more green growth" or "shrinkage") but from the goals of economic activity: respecting ecological limits while enabling a good life for all.

2 Scope of Economics

As we discussed in the previous chapter, economics can be defined in two ways: through an analytical approach (as the mainstream does) or through a subject matter – that is, through *what* economics deals with in terms of content. In this chapter, we approach the subject matter and ask concretely: What is actually the object that economics studies?

You will find that even basic questions – such as "What is work?" – are answered differently. The reason is that, depending on one's understanding of the economy, one also understands its individual aspects differently. As introduced above, we therefore distinguish different schools of thought in economics. Each of these schools of thought brings its own understanding of the economy and accordingly looks at economic questions differently.

To approach the scope of economics, we follow a simple means-end logic: we begin with the *goal* of economic activity – prosperity – and then ask *with what* (resources, production) and *how* (market or economic system) this goal is achieved. This chapter is accordingly structured as follows:

- **Prosperity in a changing world:** What is the goal of economic activity – what do we understand by prosperity?
- **Forms of provision in the economy:** What is needed for this prosperity to arise in the first place?
  - (Re)production factors: What is needed for goods and services to come into being?
  - Land – Can we replace nature with technology at will?
  - Capital – Why do some people own capital and others do not – and how does it differ from money?
  - Labour – Why do I earn as much as I do – and why is some work, such as care work, often not paid at all?
  - Types of goods: Is the sun a public good – and which properties determine how a good is treated economically?
  - External effects: Why does a flight often cost less than a train ride – even though it harms the environment much more?
- **Abstract markets or capitalist production system:** How is it coordinated who uses these resources and how, and who receives the result?
  - The neoclassical perspective: Does anyone need to steer the economy at all – or does it regulate itself?

- The political-economy perspective: Who actually determines the rules of the market – and in whose interest?
- What is capitalism? – What actually makes an economic system capitalist?
- Why do companies (and entire economies) actually have to grow constantly?
- Are financial crises accidents – or built into the system?

Overarching guiding question of this chapter

How does a society ensure at all that the needs of its members are met?

We are aware that, for students without a background in economics, there are many new terms. The aim of this chapter is therefore not that you know all topic areas in detail afterwards but rather that you gain a solid overview of the scope of economics. For students of economics, the chapter serves at the same time as a review and as a broader contextualization of existing knowledge.

In-depth questions

Several in-depth questions are listed for each subsection. On the one hand, these serve to check your own knowledge of the respective area; on the other hand, they are also meant to prompt thought and reflection. If you have any questions about the content or would like to explore a topic in greater depth, you can contact us (see information document).

2.1 Prosperity in a Changing World

Guiding question

What is the goal of economic activity – what do we understand by prosperity?

When we ask about the scope of economics, we inevitably encounter the question of its goal: What is the purpose of economic activity in the first place? An obvious but incomplete answer is: to create prosperity. But what exactly is meant by "prosperity"?

At least in Western societies, prosperity is often understood materially, that is, as how much money we have and what things we can afford with it. But prosperity could just as well include life expectancy, education, social relationships, leisure time, an intact environment, or subjective well-being. Depending on how prosperity is defined, what counts as "good" economic development also changes. This question is thus anything but neutral: it implicitly determines what a society should strive for in the first place.

Because prosperity is so multifaceted, its measurement is correspondingly difficult and contested. The most widespread index for measuring prosperity to this day is gross domestic product (GDP). It is omnipresent in everyday life, in politics, and in the media and is regarded in many places as

the most important economic indicator of all. In what follows, we therefore first look at what GDP actually measures – and what it does not.

2.1.1 Gross domestic product (GDP) as an indicator of (material) prosperity

GDP is regarded as the main indicator of a nation's prosperity. It measures the total value of all goods and services produced in a country within a year, after deduction of all intermediate inputs. The concept of GDP in its current form was developed only in the 1930s – first attempts to measure a country's prosperity go back to the work of William Petty in the 17th century – to combat the Great Depression and to plan armaments spending in the Second World War in the USA and England (Schmelzer and Vetter 2019, 57–59). A precondition for this was also the invention of the "economy" as a separate sphere of social life that can be statistically recorded and measured (Schmelzer and Vetter 2019, 57). Despite its present widespread use, GDP is also constantly criticized. Even economists who advanced the development of GDP warned against using GDP as a general measure of prosperity. Simon Kuznets (1901-1985), one of the most prominent economists in this field, said: "*However one may use them, such figures [...] seem very useful; they seem to measure and make comparable something clearly defined and significant. But on closer inspection, it turns out that the impression that such estimates are clear and unambiguous is deceptive.*" (our translation of Lepenies 2013, 90)

This warning can be made concrete through several central points of criticism – which ultimately all show how far GDP is from the broader notion of prosperity we outlined at the outset:

- GDP records only what is traded for money. Unpaid care work or undeclared work remains invisible.
- GDP says little about actual well-being, since it is reduced purely to material possessions.
- Social and ecological consequences of economic activity are ignored – environmental pollution does not appear as a cost in GDP. Paradoxically, a natural disaster can even raise GDP if its reconstruction has to be financed. Personal accidents, too, raise GDP through medical treatment, although they clearly worsen the well-being of the person affected.
- The distribution of income remains unaccounted for: a high GDP says nothing about whether this prosperity is distributed fairly.
- GDP is based on annual growth rates, which favours short-term thinking – at the expense of long-term sustainability goals.

> Interactive exercise – Does it count in GDP?
>
> *Does it count in GDP?: available in the online version of this script.*

Figure 2.1 shows how GDP per capita in Switzerland has developed over the last 150 years. It becomes apparent that GDP in Switzerland rose sharply above all in the post-war period. Figure 2.2 shows the development of Switzerland's GDP in comparison with selected other countries over the last 70 years.

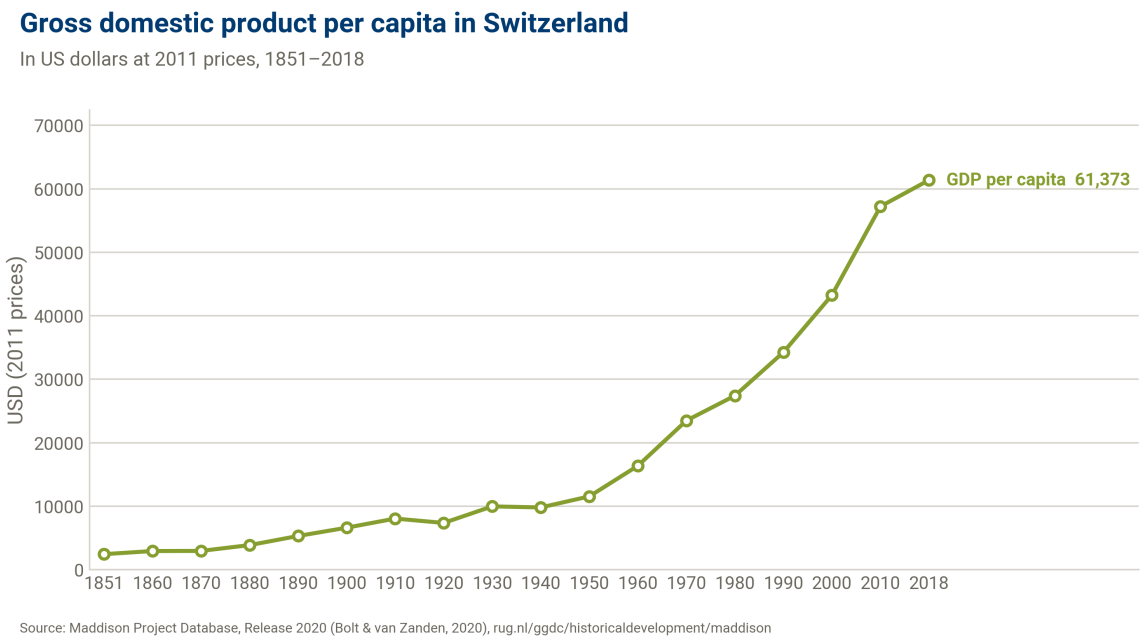


Figure 2.1: Development of GDP per capita in Switzerland at 2011 prices in USD. Source: Own illustration with data from The Maddison Project (2020).

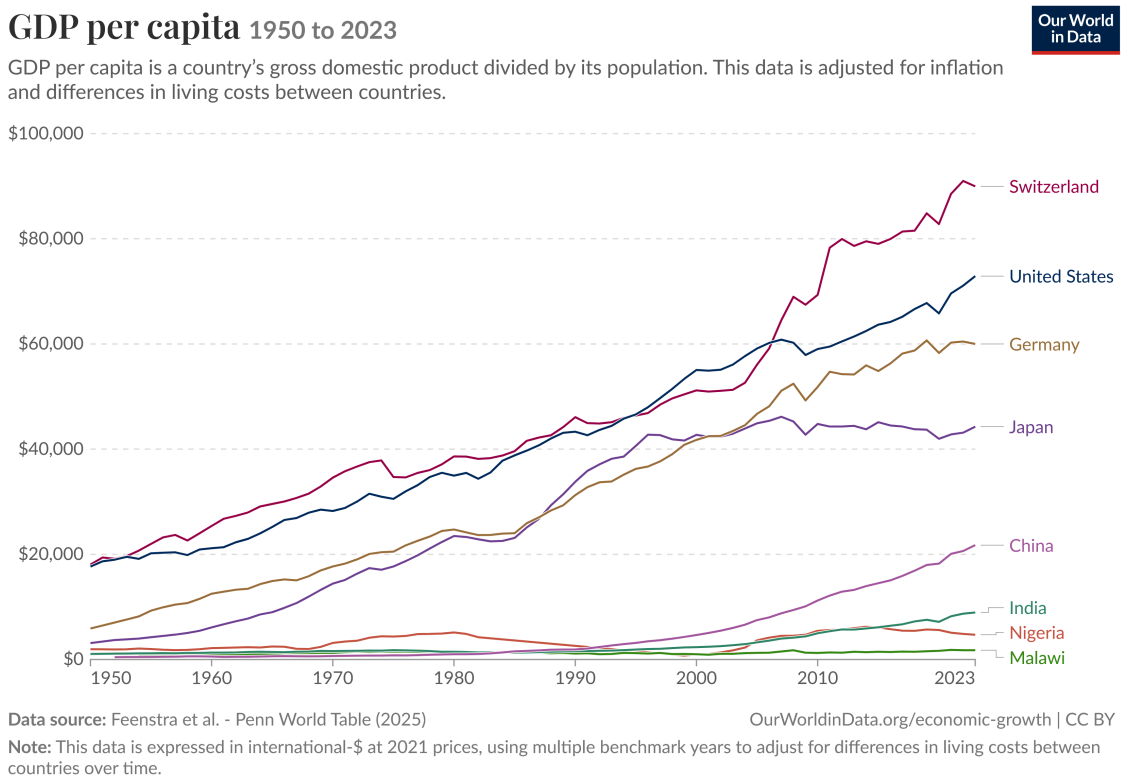


Figure 2.2: GDP per capita of selected countries, 1950-2023. Source: Feenstra et al. – Penn World Table (2025) – with extensive data processing by Our World in Data.

2.1.2 Alternative indicators of prosperity

If GDP screens out so many facets of prosperity – how could prosperity be captured instead? In view of the limitations mentioned, several alternative indicators have been developed that aim to paint a more comprehensive picture:

The **Human Development Index (HDI)** supplements GDP with life expectancy and level of education. The HDI thus follows a concept of prosperity and human well-being according to which people should be enabled to lead a life worth living according to their own ideas. In Chapter 3.2.2 (*Justice and inequality*), we go into this approach in more detail.

The **Genuine Progress Indicator (GPI)** goes further: besides positive factors, it also includes negative ones such as environmental pollution and social inequality – in total about 26 indicators from the economic, social, and ecological dimensions, including the value of unpaid work.

Since it is fundamentally difficult or even impossible to measure prosperity, these indicators, too, are criticized for their limitations, and accordingly many further indicators exist for measuring a country's prosperity. Besides GDP, however, none has so far been able to establish itself.



Figure 2.3: Aspects of the GPI. Source: GNHUSA.

To illustrate how far different indicators can diverge, Figure 2.4 shows the calculations by Kubiszewski et al. (2013) for the development of GDP and GPI per capita since the 1950s for a group of 17 countries.

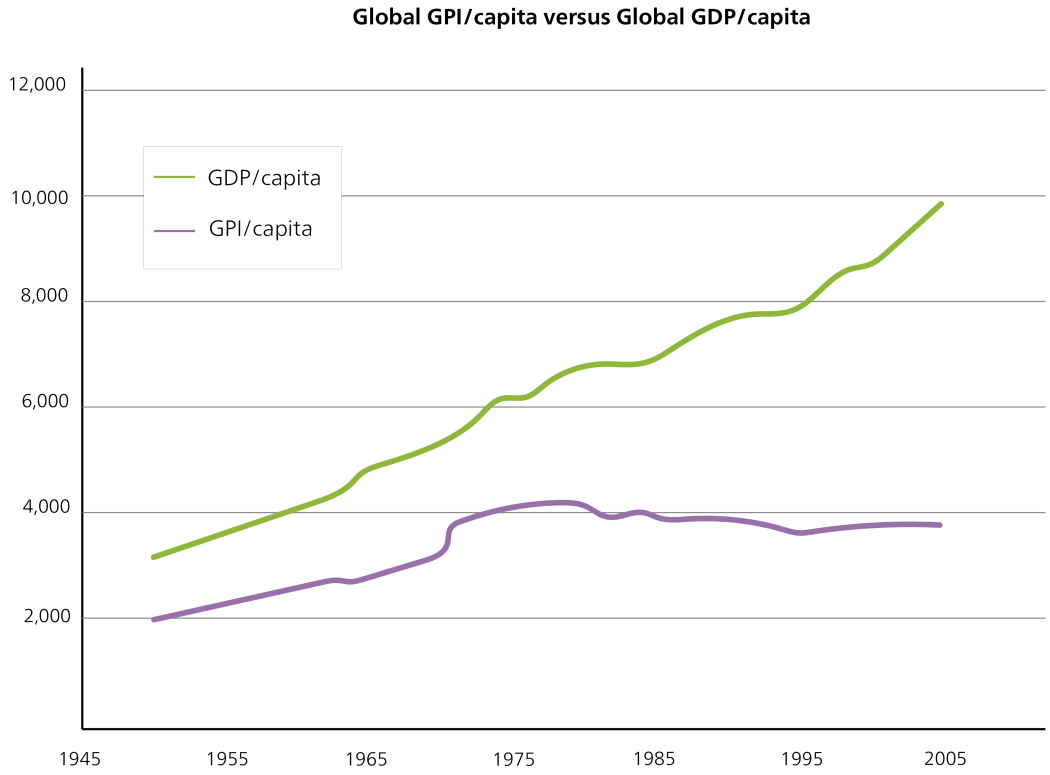


Figure 2.4: GPI vs. GDP per capita. Source: Own illustration with data from Kubiszewski et al. (2013, 63).

2.1.3 Economic growth

If prosperity is mostly measured by GDP, then the *increase* in this prosperity over time – economic growth – almost automatically becomes the central political goal. Economic growth refers to the increase in a country's economic output over time, usually measured by the development of GDP. It is the central target variable for politics, society, and the economy – so central that Schmelzer and Vetter (2019, 62–68) speak of the **growth paradigm**.

Given this omnipresence of economic growth, it is astonishing that economic growth is a very young phenomenon. Only with the Industrial Revolution in England in the 19th century could sustained economic growth (measured by GDP) be observed in the world for the first time. For the longest part of human history, people lived without long-lasting economic growth. Meanwhile, economic growth is not only a central cornerstone of our current economic and social system but has developed into a compulsion (Binswanger 2019). We will go into this systemic growth imperative and the growth paradigm in more detail later in the course. The significance of economic growth is intensively discussed within sustainable economics, with positions that are far apart. While some want to hold on to economic growth (in an adapted form), others advocate overcoming economic growth as a compulsion (green growth vs. post-growth). At a later point, we will engage with these discussions in depth.

Figure 2.5 shows the development of GDP per capita for different countries or regions over the last 1,000 years. The Industrial Revolution in the 19th century is clearly visible.

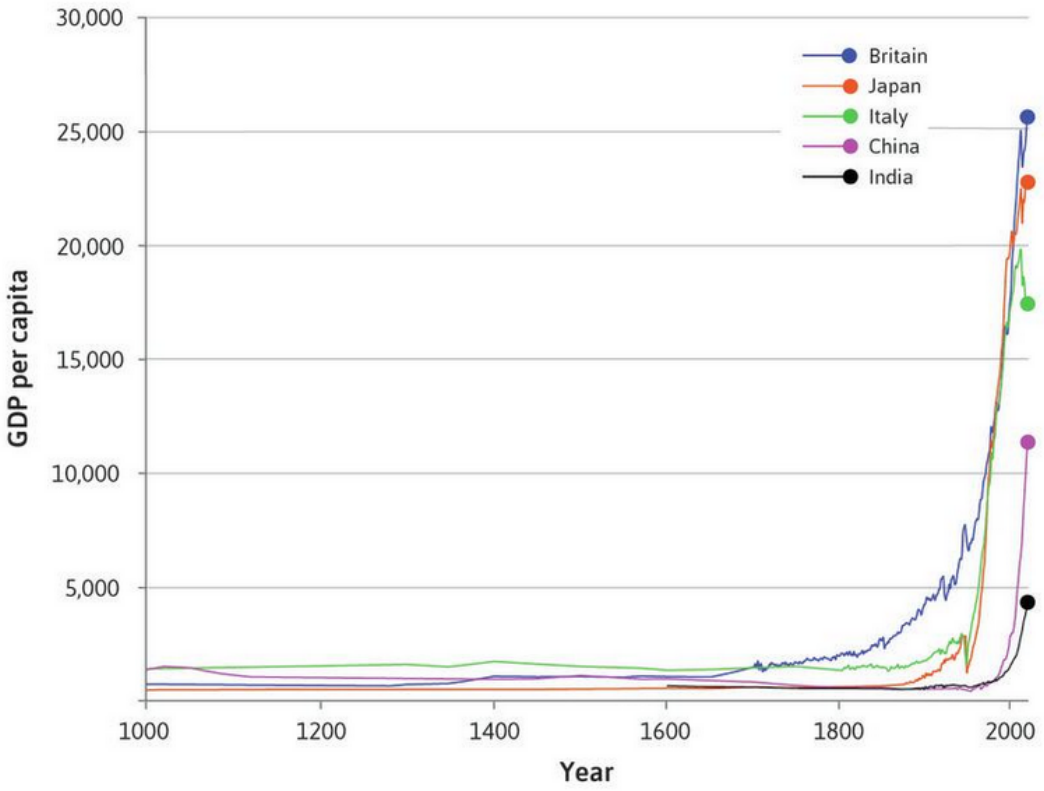


Figure 2.5: History's hockey stick: Gross domestic product per capita in five countries (1000-2018). Per capita income stagnated for centuries. Only with the Industrial Revolution and fossil capitalism did incomes and production volumes rise exponentially – albeit very unequally from region to region. Source: https://books.core-econ.org/the-economy/microeconomics/01-prosperity-inequality-02-historys-hockey-stick.html#figure-1-1.

In-depth questions

- Does GDP seem to you a sensible measure of a nation's prosperity? What speaks for it, and what against?
- Think of examples from everyday life in which the growth paradigm can be recognized.
- What could be the reasons that GDP and GPI diverge so strongly? And what could be the reason that the trends in the figure diverge only from the 1970s onwards?

Further reading

- Binswanger, Mathias. 2019. *Der Wachstumszwang: warum die Volkswirtschaft immer weiterwachsen muss, selbst wenn wir genug haben* [The growth imperative: why the economy has to keep growing even when we have enough]. Wiley-VCH Verlag GmbH & Co. KGaA.
- Lepenies, Philipp. 2013. *Die Macht der einen Zahl: eine politische Geschichte des Bruttoinlandsprodukts* [The power of a single number: a political history of gross domestic product]. Erste Auflage, Originalausgabe. Suhrkamp.
- Schmelzer, Matthias. 2016. *The hegemony of growth: the OECD and the making of the economic growth paradigm*. Cambridge University Press.
- Sen, Amartya. 2011. *Development As Freedom*. Knopf Doubleday Publishing Group.

2.2 Forms of Provision in the Economy

Guiding question

What is needed for prosperity – as we discussed it in the last section – to arise in the first place?

For a society to achieve prosperity, it must first ensure that goods and services are provided at all. Through the division of labour, an economic system always faces fundamental questions of organization: How is societal production distributed? Who (re)produces, maintains, and repairs what? Who consumes what? In other words: How is what is necessary for prosperity produced in the first place and distributed among the members of a society?

As became clear from the preceding chapters, the answers to these questions depend strongly on which understanding of the economy one takes as a basis. Here, we follow Karl Polanyi's (1957) substantivist understanding, which defines the scope of the economy beyond the market and proposes a broad understanding of the economy.

For our starting question – what is needed for prosperity to arise – Polanyi's broad substantivist understanding is particularly illuminating. It shows that real economies consist of very different institutions and logics. A housing cooperative works with different business models than plumbers and steel companies, the public hospital differs from the industrial firm, and unpaid care work is organized differently from assembly-line work. Karl Polanyi therefore distinguishes four socioeconomic organizing principles, or forms of provision, that can be found in actually existing economies and interact in different ways: householding, reciprocity, redistribution, and market exchange (Polanyi 2017):

(1) **Householding** refers to forms of self-sufficiency and is rooted in families and households. In ancient Greece, the oikos, that is, the whole house, was an autarkic, that is, self-supplying, economic unit. But even today, a large part of the economy still takes place in the household, in particular as unpaid care, nursing, and childcare work as well as housework (for example cooking, cleaning, gardening).

(2) **Reciprocity** is based on the principle of give-and-take and defines an exchange of goods and services between persons beyond the market and the state. On the one hand, it takes place in local communities and between persons who know one another, for example among friends, in the neighbourhood, or in associations, and includes, among other things, neighbourly help and community work. On the other hand, there are also many historical examples of reciprocal exchange over long distances, in which peoples exchange goods, with ceremonial motives and motives of maintaining relationships often playing a role alongside economic ones. To endure, however, this form of provision requires that giving and taking do not get too far out of balance or are underpinned by a corresponding ideal of justice (see Chapter 3.2.2, *Justice and inequality*).

(3) **Redistribution** defines a systematic flow of resources towards an administrative centre and their subsequent redistribution. Examples are public education, health, and pension systems financed by taxes or levies. Redistribution allocates resources to members of a society (who are often unknown to one another). It takes place within political territories, in particular the nation-state, and therefore goes beyond local communities.

(4) Finally, **market exchange** defines the exchange of goods and services at market prices. This is the commodified[1] area of economic activity. Depending on the type of goods and services traded and on their reach and structure, these markets can differ. The logic of individual profit and ability to pay predominates in market relations.

Interactive exercise – Polanyi's four forms of provision

Polanyi's four forms of provision: available in the online version of this script.

In-depth questions

- Find further examples from your everyday life for each form of provision.
- Find an example of a good or service that is provided through all four forms of provision. How do the four forms of provision change the situation and the character of the transaction or provision?

Back to the guiding question: prosperity thus does not arise through market exchange alone. Real economies are always **mixed economies**. They combine all four forms of provision, even if in everyday life only market exchange is often perceived as "the economy". Not every aspect of life and economic activity can – or should – be turned into a marketable, commodified good. Which of these forms of provision contribute how strongly to a society's prosperity (and which of them become visible in classical prosperity measures such as GDP) is thus itself a central economic question.

[1] Commodification means that something becomes a commodity and is traded on markets. That is, something is produced and distributed according to a market logic that previously did not work according to this logic (for example food produced for self-sufficiency). In principle, almost anything can be affected by commodification, for example labour, goods, natural resources, CO_2, and so on. With the development of capitalism, more and more areas of the economy were commodified, which is why this appears self-evident to us in many areas.

2.3 (Re)production Factors

> Guiding question
>
> What is needed for goods and services to come into being?

To answer this question, economics needs a concept for the "ingredients" of economic activity: the **factors of production**. Factors of production are all the inputs that are necessary to produce an output – a good or a service. The aggregated factors of production of an economy are called **factor endowment**. The origins of the factors of production lie in the physiocratic thinking of the 18th century; as a concept, they became established in classical economics. According to classical economics, the factors of production of economic activity can be divided into three categories: **land, capital, and labour.** Over time, this trinity has repeatedly been criticized, extended, and adapted. Further factors such as entrepreneurship, knowledge and technology, and nature itself, for example, have been brought into play. Often, however, these aspects are also subsumed under one of the three existing factors.

But the question "What is needed for something to come into being?" is thereby only half answered. Many schools of thought (such as neoclassical economics, post-Keynesian economics, and in part also Marxist economics) focus above all on the **production** of goods and services for sale on the market. What is often screened out is **reproduction** – that is, everything that is necessary for something to continue to be usable. A bicycle, for example, has to be pumped up regularly and its chain oiled so that it rides reliably. A cup has to be washed thousands of times so that it can be used again. But people, and thus workers, also have to be "reproduced" (this is often described as social reproduction). We all need food, care, and so on in order to live and to show up for work.

Whoever focuses only on production thereby screens out a central part of what makes goods and services possible in the first place: unpaid reproductive work such as cooking, cleaning, or childcare is often not analysed but simply assumed as given. Feminist economics shows, however, that this work is an integral part of economic life. It forms the basis on which market-mediated activities can take place in the first place. The separation into "productive" market-mediated work and "unproductive" housework, and its gender-specific connotations, developed only with the emergence of capitalism and is by no means natural. Questions of production and reproduction are thus also strongly connected to the capitalist production system (see Chapter 2.5, *Abstract Markets or Capitalist Production System*).

In what follows, we turn to the three classical factors of production – land, capital, and labour – which, depending on the perspective, play a central role not only for production but also for reproduction.

2.3.1 Land – can we replace nature with technology at will?

The factor of production land often covers not only the area or ground itself but also the natural environment. The question is often how land and natural resources are handled and how they are used. These questions affect us very directly in everyday life. The design of living space, for example, is of central importance for all of us. But how we deal with agricultural land, forests, lakes, and the

living beings within them also plays a central role in our everyday lives. In addition, land and natural resources are an elementary factor for the production of goods and services. Land is indispensable for production, whether for factories or agriculture. But natural resources such as water and rare earths are also central as inputs for production.

What is special about land is that, on the one hand, it has unchangeable natural characteristics (for example land is immobile and cannot be multiplied), while, on the other hand, how it is handled is very strongly socially determined (for example rights of ownership, access, and use). This starting point means that the handling of land has a great deal of conflict potential, is necessarily linked to questions of distribution, and can also be an important aspect of inequality. Accordingly, questions about the handling of land are deeply political.

2.3.1.1 Perspective 1: Nature is replaceable in principle

Despite their vital importance, land and natural resources receive little attention in economics. In many schools of thought such as neoclassical economics or, in part, post-Keynesian theory, land is either screened out, assumed, or treated as a pure asset (for example housing) and thus subsumed under capital. In these theories, land very rarely finds its way into production functions. And when land and natural resources are explicitly considered, they are often understood as "ecological services" (*ecosystem services*). Through pricing, they are then conceptualized as an input factor and commodity and embedded in a market logic. In such understandings, land and natural resources are thus implicitly or explicitly understood as "pure input factors" and/or assumed as self-evident. A forest, for example, is reduced to its purely economically exploitable dimension, such as the sale of timber and game. What is often neglected is that nature is never available in the form of goods but must always be made usable and exploitable by humans through labour (Schaupp 2024). In these understandings, it is also implicitly assumed that, for example, the loss of biodiversity can be compensated by technological solutions. In a cost-benefit analysis, all costs and benefits are then summed up, and all the different aspects are thus equated through monetary valuation. For sustainable economics, such fundamental questions are of central importance. In addition, through the reference to market logic, questions of land are depoliticized, and questions of power and distribution are implicitly excluded.

2.3.1.2 Perspective 2: Nature has limits that technology cannot lift

In contrast, there is a fundamentally different understanding of land and natural resources as non-replaceable goods that form the basis of social life. In this understanding, the loss of biodiversity or the destruction of fertile soil cannot be compensated. These losses are final. Such understandings often come from ecological economics, but some currents of feminist and Marxist economics also build on this understanding. The natural limitedness and the need for social organization in dealing with land are taken as the starting point, and questions of power and distribution are often made explicit.

Whether nature can be replaced by technology at will is thus not a purely technical question but a theoretically and politically charged one. In Chapter 3.1, we go deeper into different understandings of nature and their consequences. What is important to take away here above all is that land,

and its significance for economic life, can be understood and conceptualized in different ways. In addition, land and natural resources have some unchangeable, natural characteristics. Who has access to them, however, is not subject to any natural economic law but is the result of economic, political, and social structures.

2.3.2 Capital – why do some people own capital and others do not?

We encounter capital every day in various forms, yet the term mostly hides in rather technical vocabulary. With a certain amount of start-up capital, for example, we can found a company, invest money in a capital investment (for example buying securities), or invest in our own education, which is referred to as human capital. Not least, our economic system – capitalism – is named after the (company) capital that empowers its owners to make the decisions that matter for the company. Accordingly, it must play an important role. But what exactly is it? The examples given already show that an important aspect is that one invests (in) something that is later supposed to yield a return (a *capital* return): by investing start-up capital, I want to build a company that later makes profits, the purchase of securities is supposed to yield regular interest, and the investment in education is supposed to pay off through a higher wage in working life. But do these examples all refer to capital to the same extent, or are there differences? Again, this question is answered differently depending on the perspective.

2.3.2.1 Capital as a thing – inequality as the result of different investments

Neoclassical economics sees capital first and foremost as a factor of production, that is, mainly machines, tools, factory buildings, and so on. With this (physical) capital, goods are then produced with the help of labour as the second factor of production. On the capital market (or the market for financial capital), the rate of return is then determined, which in equilibrium corresponds to the productivity of physical capital. Capital is understood here primarily as a commodity that is traded on a market and valued and distributed according to market mechanisms. The main thought here is of the production of goods, but in this understanding human capital can very well also be understood as such. I invest time and forgo wages during my training and then expect a return in the form of a higher wage, which is regulated through the market mechanism.

This understanding fits our guiding question as follows: whoever owns more capital has invested (or inherited) more in the past. Inequality therefore appears here primarily as the result of different investment decisions and opportunities, not as a structural problem. To distinguish it from money: **money** can also be capital in this understanding, provided it is invested in order to achieve a return – money that merely lies in a drawer would, strictly speaking, not be capital.

2.3.2.2 Capital as a social relation

Post-Keynesian theory, too, understands the concept of capital primarily as a factor of production but connects entirely different dynamics with it (see Chapter 2.5). In particular, a concept of capitalism and the distinction of classes are connected with it: workers who have no capital and therefore

have to work for their wages, and capitalists who have capital and can deploy it to make profits. Questions of power and distribution thus also increasingly come into view.

Post-Keynesian economics took this view of the relationship between classes from Marxist economics. Marxist economics makes this relational aspect of capital the central point: capital here is not a thing but a **social relation**. The deployment of capital is based on the separation of the means of production from the workers, who have to work for a wage – that is, sell their labour power, since they themselves have no capital. In this understanding, we do not speak of capital, for example, in the case of machines used for agricultural self-sufficiency. But if an entrepreneur buys machines and pays wage workers to till the land for them, and the harvest is sold on the market to make a profit, we speak of capital in this understanding. Capital is thus tied to the use of wage labour to make a profit, and this class relation is the central aspect of capital. The Marxist concept of capital is accordingly tied to the specific context of capitalism and cannot be extended at will to other times or contexts. The concept of human capital, too, makes little sense in this understanding. The capitalist dynamic that capital must constantly multiply (accumulation) is likewise already inherent in this concept as a social relation. In our example, the entrepreneur would then, for example, invest the profit made in further machines to expand production. This dynamic would then continue steadily.

To distinguish it from money: mere money is not yet capital in this understanding. Only when it is deployed to mobilize wage labour and make a profit from it does it become capital. Hodgson (2014) therefore also finds this understanding insufficient and understands capital as everything that can potentially be used to make a profit. Concretely, this means that all wealth is understood as capital, provided it can also serve as collateral for taking out loans. This understanding of capital also underlies, for example, Piketty's analysis in his major work *Capital in the Twenty-First Century* and can be connected with various schools of thought (Piketty and Goldhammer 2014).

Although public debates often act as if it were entirely clear what is meant by "capital", quite different understandings appear here – and with them different answers to our guiding question. Whether inequality in capital ownership is understood as the result of individual decisions, as a structural power relation, or as a self-reinforcing wealth advantage depends on which concept of capital we take as a basis. It is therefore central to be aware of which concept of capital we use and for what purpose.

2.3.3 Labour – why do I earn as much as I do – and why is some work, such as care work, often not paid at all?

Labour is the third factor of production and is necessary so that goods and services can be obtained from raw materials. Coordinating, mental, and executing activities are necessary for the purposeful use of capital in order to be able to produce. Labour is thus a central factor for value creation. In many cases, labour also determines people's everyday lives and takes a central role in economic policy reporting, for example through discussions of unemployment, employment figures, and minimum wages. This often concerns a specific form of labour: **paid employment.**[2]

[2] Paid employment means activities for which people receive a direct return in the form of money.

This is precisely where our guiding question comes in: Why do some people earn a lot for their work and others little – and why does some work count as worthy of payment at all, while other work, such as care and nursing work, often remains unpaid? As we will see, the answer depends strongly on which economic understanding of labour one follows.

2.3.3.1 Centrality of paid employment – what counts as work at all

In our everyday lives, too, much revolves around paid employment. But this centrality of paid employment developed only in the course of 19th-century industrial capitalism and became the central societal instance of integration and orientation. While in premodern household economies paid and unpaid activities were closely connected and the household was the central economic unit, now only market-mediated, remunerated labour was increasingly recognized as productive, while care and housework were devalued. With the development of capitalism, more and more people became dependent on paid employment to make a living. And the capitalist system, too, depends on paid employment, since excessively high unemployment can weaken economic development and lead to a societal crisis (post-Keynesians emphasize that with low employment, aggregate demand also falls away). In addition, social security and tax revenues were built strongly on paid employment, and societal participation and social recognition also orient themselves towards paid employment. In the second half of the 20th century, this development was consolidated by the idea of the *standard employment relationship* and the male breadwinner as a societal guiding model.[3]

Since the 1970s, this model has come under increasing pressure, in Switzerland too, through the flexibilization and precarization of employment (for example the gig economy). Since the 1970s, unemployment and underemployment have risen in Switzerland, as have temporary work, part-time employment, and precarious and informal employment (Benelli and Streckeisen 2014; Halbeisen et al. 2012; Sheldon 2007; Scherschel et al. 2012; Baumgartner et al. 2019). Over the last 50 years, women have also increasingly entered the labour market but mainly work part-time.

In line with the centrality of paid employment, many schools of thought – such as neoclassical and post-Keynesian economics – focus strongly on paid employment, while unpaid reproductive work is mostly tacitly assumed. Feminist economics, by contrast, works with a considerably broader concept of labour that also includes unpaid work. It often draws on the so-called **third-person principle**: any activity that could in principle also be performed by a third, paid person counts as work. Cooking, cleaning, or childcare, for example, thus fall under "work" just as much as factory work, regardless of whether they are remunerated or not.

Feminist economics also points out that person-related services (for example nursing, care) follow a different logic than, say, factory work. Unlike factory work, the productivity or efficiency of person-related services can hardly be increased, because the quality of the work depends directly on time and human interaction and can therefore hardly be standardized or planned. As a result, wage costs in these areas rise relative to other sectors, since they cannot (for example through new technologies) show ever higher labour productivity, while wages rise with societal development. The

[3] Following Bernet (2021, 128), the term "standard employment relationship" is understood as a "permanent, socially insured, and collectively protected full-time position with a secure monthly income, clearly regulated working hours, protection against dismissal, and certain instruments of co-determination" [our translation]. Despite its name, this employment relationship was never really normal. There were always many people who did not work under such conditions, very often migrants and women.

economist William Baumol described this under the term *cost disease* and shows how, for example, costs in technological sectors have fallen massively while they continue to rise in person-related services. In Switzerland, this is clearly visible, for example, in the health-care system, where costs rise, calls for efficiency improvements arise but cannot be implemented, which puts staff under great pressure.

2.3.3.2 But how are working time and wages determined?

This brings us to the core of the guiding question: Why do *I* earn as much as I do? Here, too, the schools of thought diverge. Neoclassical economics sees the central explanation in the market mechanism of the labour market: in equilibrium, the wage corresponds to labour productivity. People maximize their personal utility by deciding, according to their preferences, on an optimal ratio between leisure and consumption, which is made possible by selling labour (time). Whoever wants to work finds work in this model too: in market equilibrium, there is no (involuntary) unemployment. From this perspective, one thus earns what corresponds to one's own productivity.

Feminist, post-Keynesian, and Marxist economists contradict this: because of the societal centrality of paid employment, people are ultimately compelled to work and cannot freely determine their working time. Power imbalances in the labour market and institutions such as trade unions play a central role in this process. In addition, feminist economics emphasizes that these questions are also often shaped by societal norms, so that the distribution of work, working time, and wages is also co-determined by questions of gender.

2.3.3.3 Unemployment

In line with the centrality of paid employment, often only figures on paid work are recorded and included in analyses. The most important indicator used is usually the **registered unemployment rate**. This is mostly calculated as follows:

$$\text{Unemployment rate} = \frac{\text{Number of registered unemployed}}{\text{Labour force}}$$

This indicator is only one of several indicators that are used, however. The International Labour Organization (ILO), for example, relies on unemployment in the ILO sense and, accordingly, the ILO unemployment rate (*Erwerbslosenquote*) as an indicator. All persons of the permanent resident population in Switzerland who are without work, are looking for a job, and could start an activity within a short time count as **unemployed according to the ILO**. In 2025, the registered unemployment rate in Switzerland averaged 2.8%, and the seasonally adjusted ILO unemployment rate in the fourth quarter of 2025 was 5.0%. Unlike the registered unemployment rate, the ILO unemployment rate also includes, for example, self-employed persons, persons affected by long-term unemployment such as those whose benefits have run out, and young people who are not yet in gainful activity directly after their training. Accordingly, the ILO unemployment rate is higher than the registered unemployment rate. Such indicators say nothing about the type of employment and employment relationships, however, so they are not useful for such an assessment. More fundamental questions

about the meaning and purpose of work and of a high-employment goal also cannot be discussed with such indicators.

As with questions about working time and wages, the schools of thought also see different reasons for unemployment. In neoclassical economics, unemployment is interpreted above all as the result of market disturbances or market failure, that is, when certain factors disturb the frictionless functioning of the market mechanism. A classic example of such "market disturbances" is minimum wages above the market equilibrium, which raise costs for companies and thus lead to unemployment. If there are no disturbances in the market mechanism, there is in principle no (involuntary) unemployment. Post-Keynesian economics emphasizes above all that unemployment does not arise from the labour market but depends on economic development and aggregate demand. If the latter falls, no more people are hired or employees are dismissed, and unemployment rises. It also emphasizes, however, that full employment is not necessarily desired by employers, since a certain level of unemployment also strengthens companies' bargaining power (Kalecki 1943). Marxist economics, too, emphasizes that unemployment is a structural part of capitalism and strengthens the bargaining power of capital vis-à-vis employees (Magdoff and Magdoff 2004). Feminist economics emphasizes that unemployment is not gender-neutral and that women are more often part of the hidden reserve who would like to do more paid work, or any paid work at all, but cannot, for example because of unpaid care work.

The following figure shows the development of the unemployment rate in Switzerland since 1920. Unemployment was and is very low in Switzerland in international comparison. Especially during the post-war economic boom, there was almost no unemployment in Switzerland, which is why Switzerland brought in many workers from abroad – who often worked under precarious conditions (Halbeisen et al. 2012). Only with the crisis of the 1970s did unemployment return to Switzerland, although Switzerland prevented a sharp rise by exporting unemployment abroad – by 1977, almost 250,000 foreign workers had to leave the country (Halbeisen et al. 2012, 910).

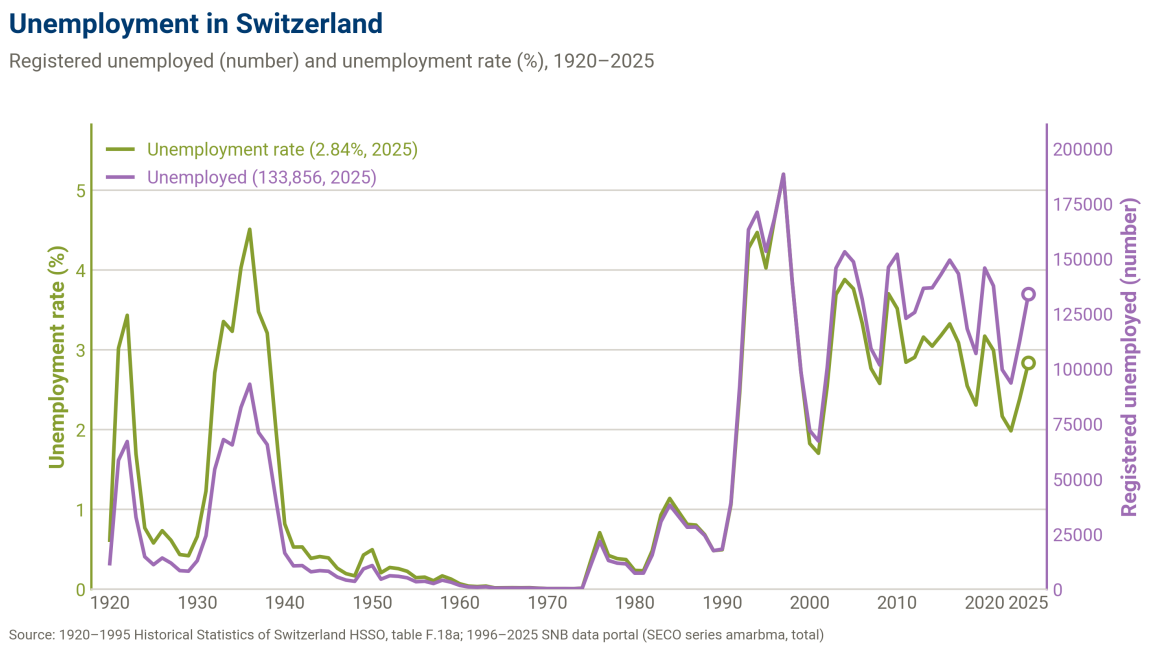


Figure 2.6: Unemployment in Switzerland 1920-2025. Source: Own illustration based on data from the Historical Statistics of Switzerland HSSO (1920-1995) and from SECO via the SNB data portal (1996-2025).

2.3.3.4 Unpaid work

> Interactive exercise – Guess first: paid or unpaid?
>
> *Guess first: paid or unpaid?: available in the online version of this script.*

Since 1997, the Federal Statistical Office (FSO) has recorded the monetary value and the quantity of unpaid work in Switzerland every four years with the household production satellite account (SHHP). In Switzerland in 2020, 9.8 billion hours were worked unpaid (mostly by women), more than were spent on paid work (7.6 billion hours). All unpaid work in 2020 is estimated at a value of CHF 434 billion. If GDP is extended by total household production, the latter corresponds to 41.4% of the value added of the extended total economy. Figure 2.7 shows how paid and unpaid work are distributed between the genders. Overall, women perform somewhat more working hours per week than men; the big difference is that the majority of these are not paid.

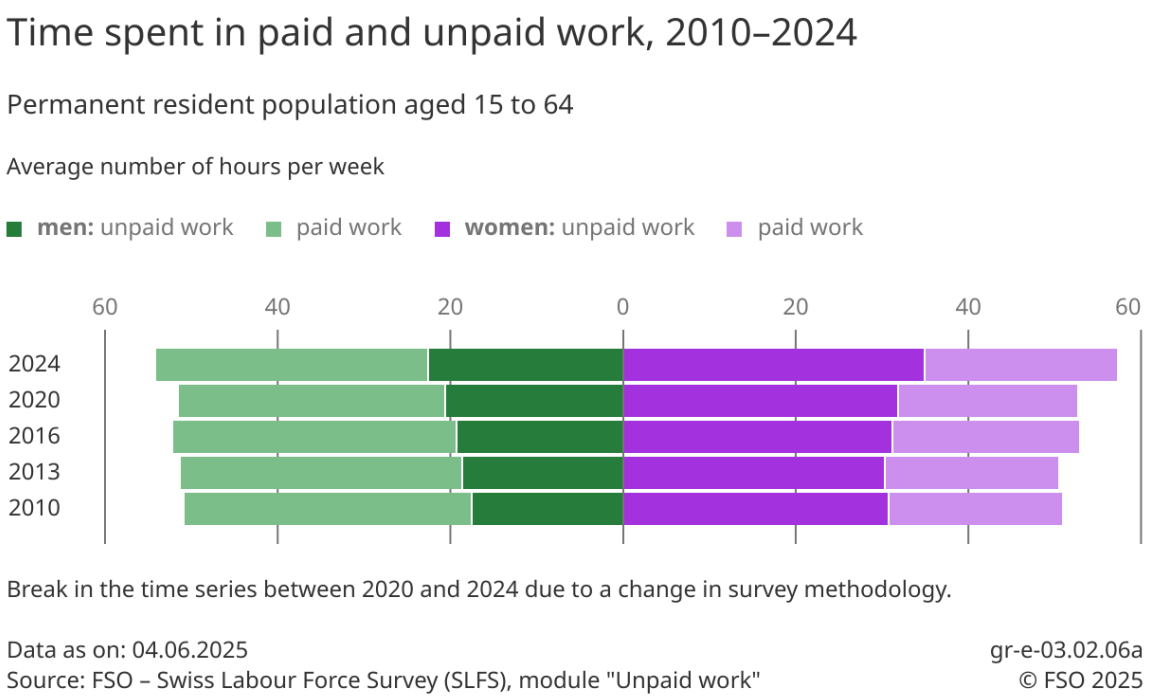


Figure 2.7: Time spent in paid and unpaid work, 2010-2024. Source: FSO, Swiss Labour Force Survey (SLFS), module "Unpaid work" (https://www.bfs.admin.ch/bfs/en/home/statistics/work-income/employment-working-hours/balancing-unpaid-work.html).

The question "Why do I earn as much as I do?" thus cannot be answered by individual productivity alone. It depends on which activities count as "work" at all, on how bargaining power and institutions shape the labour market, and on which historically grown norms determine whose work is visible and remunerated. Care work often remains unpaid – not because it would be less necessary, but because our present notion of "productive" work is historically tied to the rise of paid employment in capitalism.

In-depth questions

- Collect two examples for each of the three factors of production.
- Collect examples of (re)production factors that are insufficiently captured in the three classical factors of production.
- Think about, using examples, how it affects production if there is more labour than capital, if there is more capital than labour, if the factor land is not available, or if all factors of production are given in balanced proportion.

Further material

The short documentary "Did we toil more in the past?" (in German) gives a rough overview of the history of work.

Table 2.1: Types of goods by excludability and rivalry in consumption.

	Rivalry in consumption	
Excludability	**Yes**	**No**
Yes	Private goods: bread, flat, clothes	Toll goods: cable television, streaming services, national parks
No	Common-pool goods: deep-sea fishing grounds, the Gotthard tunnel on Easter weekend	Public goods: legal order, river corrections, fireworks

2.4 Types of Goods

> Guiding question
>
> Is the sun a public good – and which properties determine how a good is treated economically?

Production creates things and services that we humans can use and draw on. In production, we turn nature into usable things and services that serve people in satisfying their needs. But how can we describe these things and services as a category? The things and services produced are often referred to as goods, and attempts are made to describe them by their properties. It becomes apparent that these properties are often not simply inherent in the goods but are strongly shaped by societal structures. Accordingly, central questions about how goods are handled, such as access to them, cannot be reduced to the properties of goods but are always also shaped by societal negotiation.

2.4.1 Properties of goods

Many schools of economic thought distinguish goods according to two central properties. *Rivalry* describes whether a good can be consumed by different people at the same time without loss of utility. The *excludability principle* describes whether it is possible to exclude persons from using a good. On the basis of the two properties, rivalry and excludability, all goods can be divided into one of four types of goods. Table 2.1 gives an overview of this distinction:

Private goods are both excludable and rival in consumption. This means that it is possible to keep persons from consuming through measures and that the consumption of a good cannot be repeated by any number of persons. Because of these properties, private goods are characterized as well suited for trade on markets.

Toll goods (or club goods) are excludable but not rival in consumption. Cable television or streaming platforms, for example, can exclude persons from consumption through a subscription fee, but the consumption of the subscribers does not suffer when the number of users rises. Apart from digital goods, the separation between private goods and toll goods is more difficult. Membership in a fitness club and the use of its premises, for example, is non-rival only until everyone wants to pursue their New Year's resolutions in January and all the machines are occupied.

Common-pool goods (also called commons) are not excludable and yet rival in consumption. These often include natural resources such as fish stocks or the availability of water for irrigating fields. To prevent overuse of these goods, coordination between the different claims to use is indispensable.

Public goods are neither excludable nor rival in consumption. Because they are not excludable, such goods can hardly be traded on markets. The use of public goods without paying for them is described as the free-rider problem. Through state financing, such necessary goods can nevertheless be provided.

This puts the question from our guiding question in a new light: Is the sun – whose light we all enjoy at the same time and without exclusion – really necessarily a public good? Some schools of economic thought, such as ecological economics, and work from social anthropology emphasize that the classification of these goods is not a natural one. Excludability in particular is not a property inherent in the good but depends above all on technical possibilities, societal negotiation, and societal power relations. Especially in the handling of natural resources, the question of excludability is primarily a social and/or technical one. Water resources, for example, can technically be made easily accessible but also excludable. Even sunlight could theoretically be made excludable – as Mr. Burns plans in the series *The Simpsons*, for whom the sun is literally the enemy. In this respect, the question is not so much whether the sun is a public good but whether the sun is treated as a public good. Concretely, this means that whether the sun *remains* a public good is ultimately a question of technology, power, and societal negotiation.

In categorizing goods, it should also be noted that this involves a reification. Goods are reduced to their purely economic value. This can be very problematic, especially for resources that are managed as common-pool goods. What is often screened out is how common-pool goods structure social relationships and communities, how participation takes place, and how a relationship to nature and the environment is maintained. Ugo Mattei therefore argues that it is too narrow to say that we "have" a commons. Rather, the point is to explore to what extent we "are" commons, insofar as we are also part of the environment (Mattei 2014, 76). For this reason, social anthropology, for example, often speaks of the verb "commoning" instead of the noun "commons".

The tragedy of the commons and how to overcome it

Actors cannot be excluded from the consumption of public and common-pool goods. This can result in overuse of these goods, which Garrett Hardin (1968) described as the tragedy of the commons. Privatization of these goods (so-called *enclosure*) is often put forward as a solution: through the allocation of clear property rights, actors can be excluded from use.
Elinor Ostrom researched the same topic and observed how overuse can be counteracted through self-organized cooperation within clear boundaries even without the allocation of property rights. Working solutions against the overuse of non-excludable goods are, however, strongly context-dependent and thus hard to generalize. For her research on commons, Ostrom was awarded the Nobel Memorial Prize in Economic Sciences in 2009 (Ostrom 2010). Commons research, which by now is rich, knows many studied examples of commons that were not overused and did not fall victim to the tragedy. There are examples of commons that have been managed collectively and sustainably for centuries (see, for example, De Keyzer

Table 2.2: Examples of positive and negative external effects.

	Effect on	
Type of effect	**Production possibilities of third parties**	**Consumption possibilities of third parties**
positive	Apple growers and beekeepers benefit from each other through pollination and apple blossoms, respectively.	Parents have their children vaccinated; society benefits from a lower infection rate and lower health costs.
negative	Industry emits pollutants into waterways; third parties can no longer fish.	Industry emits pollutants into waterways; third parties can no longer swim in the river.

2018; De Moor 2015). The commons research of Ostrom and others accordingly shows that there are solutions beyond privatization and nationalization for preventing the tragedy of the commons. This also fundamentally breaks up the dichotomy of market and state and opens up points of connection beyond these two spheres. In Chapter 5, we turn in more detail to the relationship between market and state.

2.4.2 External effects – why does a flight often cost less than a train ride?

The price for the use of goods does not always correspond to the true societal costs. If the good is nevertheless used at this price, the costs are shifted onto third parties, that is, neither onto producers nor onto consumers but, for example, onto taxpayers, subsequent generations, or nature. If this is the case, one speaks of **external effects**.

This is exactly what answers our guiding question: a flight ticket is often cheaper than a train ride because many of the actual costs, such as climate damage from CO_2 emissions, are not included in the flight price at all. These costs are borne instead by third parties. The market price thus does not reflect the full societal costs. External effects need not be only negative, as in this example, however; they can also be positive. Table 2.2 gives some examples of negative and positive external effects.

How problematic such external effects are judged to be differs depending on the school of thought. In neoclassical economics, external costs are usually understood as an exception in an otherwise well-functioning market. In contrast, William Kapp, in the tradition of ecological economics, pointed out as early as 1950 that the more a system is based on maximizing private profit, the stronger the incentives to shift costs onto people, society, and nature (Kapp 1950). Accordingly, the current capitalist system contains inherent tendencies towards the externalization of costs. Stephan Lessenich accordingly describes the affluent societies of the Global North as externalization societies that outsource costs to the Global South (Lessenich 2020).

Interactive exercise – External effects

External effects: available in the online version of this script.

In-depth questions

- Collect two examples each for the four types of goods that you encounter in everyday life.
- What are the advantages and disadvantages of privatizing common-pool goods? Think about it using a concrete example.
- Research two examples of external effects and how they were or are being addressed politically.

2.5 Abstract Markets or Capitalist Production System

Guiding question

How is it coordinated who uses resources and how, and who receives the result?

Markets are omnipresent in our everyday lives. We encounter them, for example, on the weekly shop at the shopping centre, when shopping online, or when visiting a restaurant. In our imagination, markets often also form the central area of what we understand by the economy. But as we saw in Chapter 2.2, markets are only one of four different forms of provision. Nevertheless, markets take a central role in economics. Some schools of thought, for example, base their understanding of the economy almost exclusively on markets (for example neoclassical and post-Keynesian economics). Accordingly, the economic circuit is usually reduced to households and firms as market actors. The former make their labour power available to the latter in return for an income. By means of labour power and other factors of production, firms produce goods and services, which households consume.

In a simplified economic circuit, expenditure at the aggregate level is always matched by equal revenue (see Figure 2.8). Households receive wages (W_{tot}) and can spend them again as consumption expenditure (C_{tot}). Firms receive profits (P_{tot}) and spend them again in the form of investments (I_{tot}). If this circuit is extended by further actors (state, banks, foreign countries), more complex relationships can also be depicted (see Figure 2.9).

$$\text{Revenue} = \text{Expenditure}$$

$$P_{tot} + W_{tot} = I_{tot} + C_{tot}$$

Other schools of thought take other forms of provision into consideration more explicitly. Feminist economics, for example, emphasizes the importance of unpaid work as the basis of the paid economy. Accordingly, it takes an important role in the depiction of economic activities. For illustration, Figure 2.10 shows the circular flow model by Diane Elson, which makes unpaid work visible. Ecological economics also includes nature in such depictions, since nature too provides the basis for all economic activity. In Chapter 3.1, we will look at this in more detail.

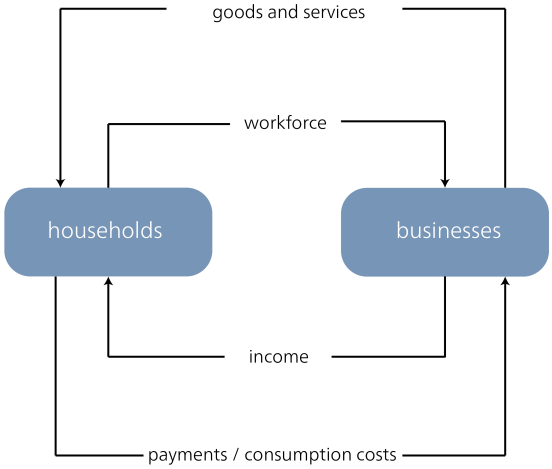


Figure 2.8: **Simple economic circuit** – Flows of goods and money between households and firms: labour in exchange for income, consumption expenditure in exchange for goods.

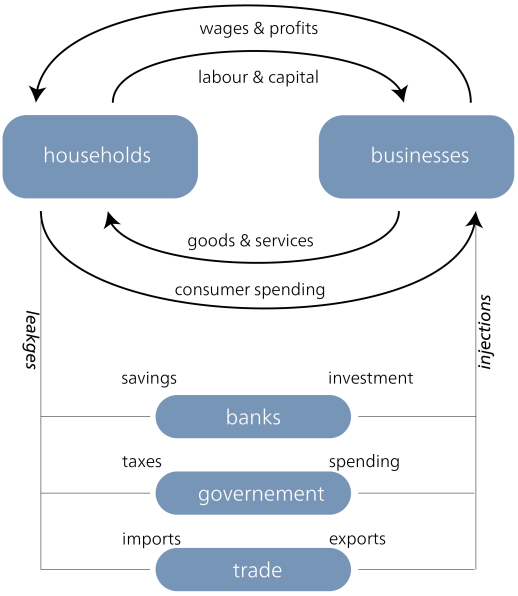


Figure 2.9: **Extended economic circuit** – Includes banks, the state, and foreign trade and shows how savings, taxes, and imports act as *outflows*, while credit, government spending, and exports act as *inflows*.

Source: Own illustrations after Rogall (2013) [p. 32] and Bäuerle et al. (2026, 29).

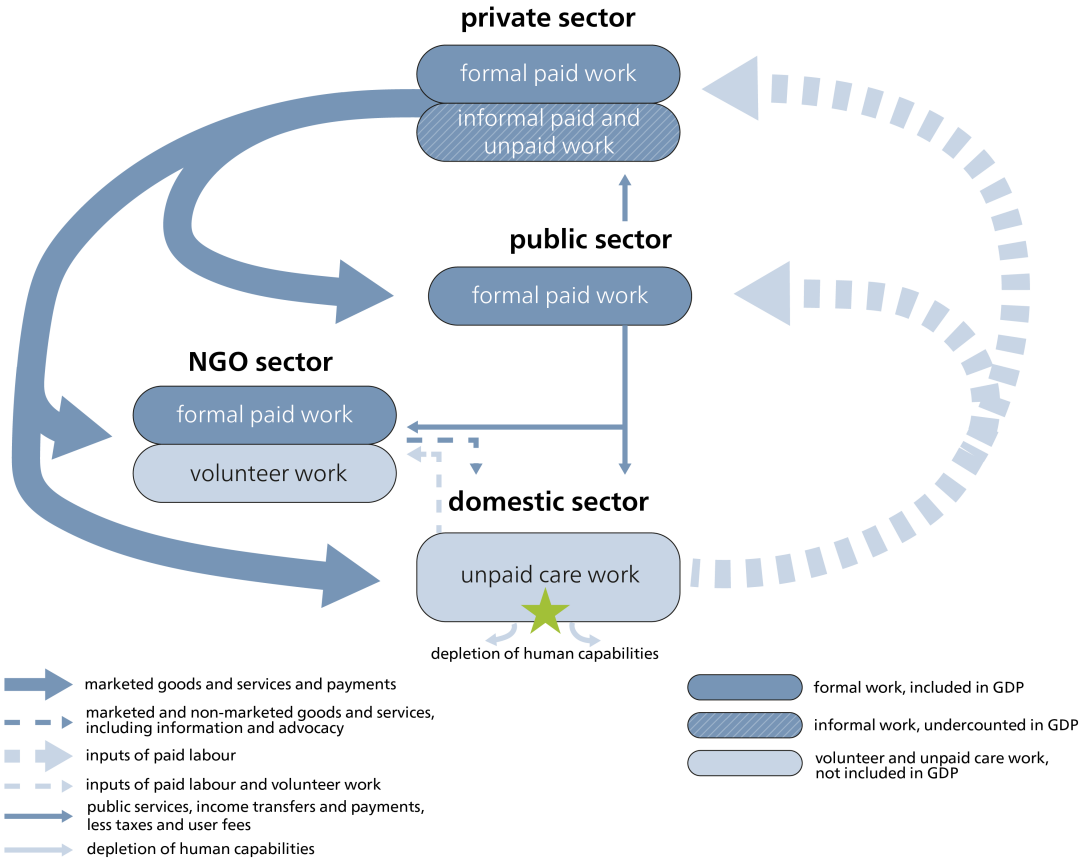


Figure 2.10: Circular flow model including unpaid work. Source: Own illustration based on Elson (2000).

The schools of thought agree on the central importance of markets – but not on *what* markets actually are and how they work. In what follows, we compare two fundamentally different answers to our guiding question: the abstract understanding of markets in neoclassical economics and the political-economy understanding of capitalist markets. A third perspective, institutional economics, which emphasizes the active shaping of markets, will be taken up again later in the discussion of the relationship between market and state (Chapter 5, *Role of the State and Policy Paradigm for Sustainability*).

2.5.1 How the market works in neoclassical economics

Neoclassical economics regards the **market as the central, self-regulating institution** for providing goods and services. What is central here is that markets are conceptualized abstractly, that is, they are understood as independent of historical and societal circumstances. This understanding implies that we are dealing here with (quasi-)natural mechanisms or laws that apply universally. This is the central difference from the political-economy perspective, which we consider afterwards.

2.5.2 The market as central coordinating mechanism

In the neoclassical understanding, markets coordinate economic activity through the interplay of supply (willingness to offer something) and demand (willingness to acquire something). The profit and utility functions of the actors, which can be calculated on the basis of the marginal principle (see the excursus further below), form the basis for supply and demand. Total demand results from the aggregation of the individual functions. Firms try to maximize their profit, and individuals try to maximize their utility (according to their preferences) through consumption. In this starting position, competition and the price mechanism subsequently determine how scarce resources are distributed (as presented in Chapter x, this is the central economic problem in neoclassical economics). If individuals can thus freely decide what they want to buy and sell, the market mechanism ensures that goods go where they create the highest utility, that is, to an efficient or optimal market outcome. Market actors and structures are shaped merely by these maximizations; there are no other influences, such as the favouring or discrimination of persons or groups. No political or moral goals are pursued.

In economics, markets are mostly represented by means of a price-quantity diagram. The axes show quantity and price (or wage in labour markets and interest rate in capital markets), and supply and demand are depicted as curves. The slope of the curves changes with the elasticity of supply and demand and shows how strongly these change when quantity or price changes. Neoclassical economics assumes, on the one hand, that the market tends towards an equilibrium and, on the other, that this equilibrium represents the most efficient outcome. The so-called Pareto efficiency (named after the economist Vilfredo Pareto) serves as the central reference for assessing efficiency. A situation is understood as Pareto-efficient, or Pareto-optimal, if no person can be made better off (in terms of subjective utility) without something having to be taken away from another person. As long as one person can be made better off without our having to take something away from another person, the situation counts as inefficient. As discussed in Chapter 3.2, this also contains specific ideas of justice.

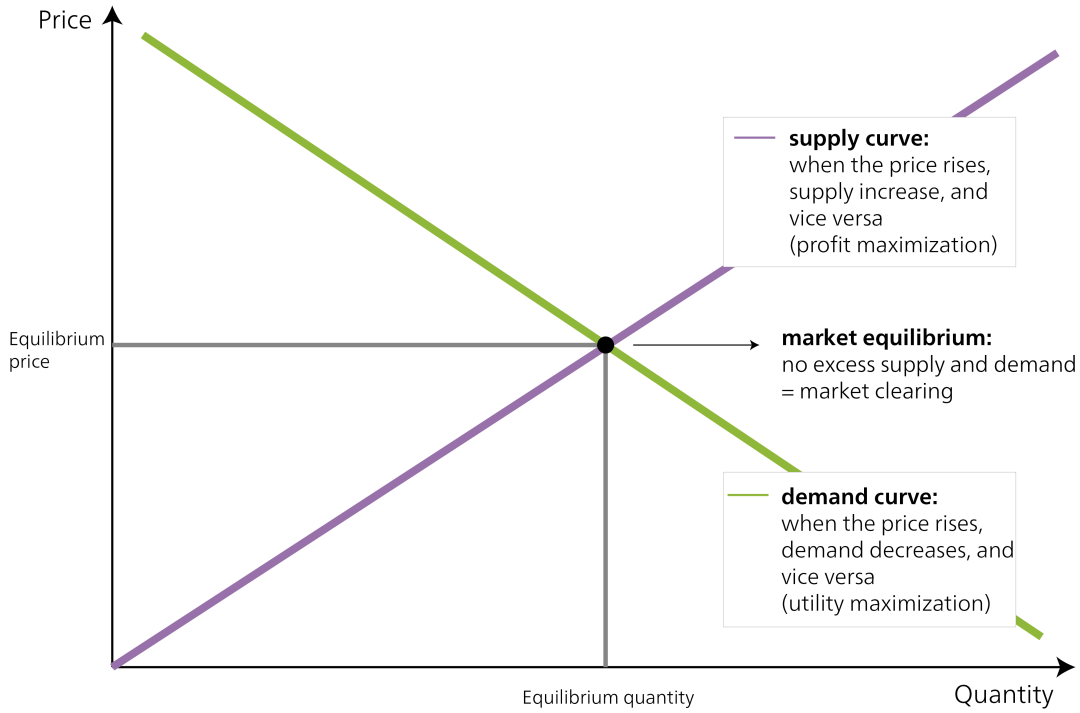


Figure 2.11: Graphical representation and determination of a market equilibrium. Source: Own illustration.

If the supply of a good is too high, the price falls – if demand is too high, it rises. In market equilibrium, demand equals supply and the price stabilizes. In this way, the price mechanism decides who produces, what and how much is produced, and for whom it is produced. The price is therefore regarded as the central allocation instrument on markets. A factory that produces with outdated machines, for example, cannot produce at the same price as another factory with a more modern setup. Because of its excessive production costs, it drops out of the market and has to produce other goods or file for bankruptcy. Competition ensures that this mechanism works and that inefficient firms exit the market. A central assumption for the supply curve to slope upwards is that, beyond a certain size, the marginal costs of firms rise again (even in the long run), for example through increased coordination effort, so that there is, to a certain extent, a limit to growth for firms. This assumption is contested, however (see, for example, Keen 2010; Pirgmaier 2017). If this assumption is not met, the supply curve could also have a different slope, and there would be no guaranteed market equilibrium with many competitive firms; there could also be a tendency towards monopoly. Even Alfred Marshall, the founder of this price-quantity diagram, repeatedly wrestled with the question of whether there might not also be a tendency towards monopoly (Marshall 1919, 316; 2013, 324).

The central basic assumption for this way markets work is that of perfect competition. Under **perfect competition**, individual suppliers have no influence on the price, since if the price rises, buyers purchase from other suppliers. Firms have to take the market price as a given quantity and can de-

Table 2.3: Market forms matrix: market forms by number of suppliers and buyers. The polypoly is a condition for the assumption of perfect competition. Source: Own illustration after studyflix.

	Buyers		
Suppliers	**one buyer**	**few buyers**	**many buyers**
one supplier	Bilateral monopoly	Limited monopoly	Monopoly
few suppliers	Limited monopsony	Bilateral oligopoly	Oligopoly
many suppliers	Monopsony	Oligopsony	Polypoly

cide only on the quantity they offer. They are accordingly in the role of price takers and quantity adjusters.

For perfect competition to be possible, several preconditions are necessary:

- There is a large number of suppliers and buyers in the market. This state is called a **polypoly**. If there are only a few suppliers or buyers, one speaks of an oligopoly or oligopsony, respectively. If there is even only one supplier or buyer, one speaks of a monopoly or monopsony, respectively (for an overview, see Table 2.3).
- All actors have **complete information**, that is, there are no hidden agreements among certain market participants or firms that offer inferior goods without disclosing it.
- Consumers have **no preferences at all for individual suppliers**. Markets work fully only if consumers can choose between the different suppliers without obstacles. Otherwise, a supplier could raise the price above the market price and still have customers.

If the above preconditions are not met, market failure can occur, and the market no longer works optimally. In a neoclassical understanding, when market failure arises, the state should intervene with regulations. As long as no market failure occurs, however, the state should keep out as far as possible and only guarantee the framework conditions. In Chapter 5, we then look more closely at how the relationship between the market and the state can be understood.

2.5.2.1 Types of markets

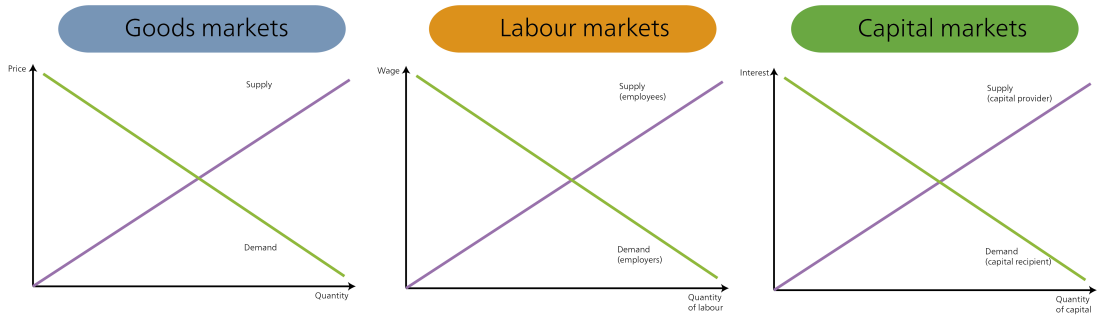


Figure 2.12: Different types of markets. Source: Own illustration.

This market mechanism of coordinating supply and demand is applied to various markets. A key distinction is made between goods markets and factor markets. On **goods markets**, goods and services are traded. On **factor markets**, the factors of production (land, capital, labour) are offered and demanded. Households, for example, offer their labour power on the labour market, and instead of a price, the level of the wage paid arises through the price mechanism from the demand of firms. In the neoclassical understanding, all these markets fundamentally work according to the same logic. The labour market works on the same principle as the market for potatoes. Other schools of thought, for example post-Keynesian economics, emphasize that labour markets differ strongly from goods markets.

In-depth questions

- For each of the three preconditions of perfect competition, think of examples from today's economy where they are met and where they are not.
- Are there differences in the demand for different goods? What influence does this have on markets for food, cars, and leisure activities?
- Think about the advantages and disadvantages of conceptualizing the market as an abstract concept, that is, independently of the historically specific context.

Excursus: Price level

Given the great importance of the price mechanism for the functioning of markets, the price level in an economy is one of the most important variables that is observed and analysed. Price-level stability is measured by the inflation rate of goods and services. With low inflation of up to 2%, one speaks of a stable price level. Statistical offices use a standardized "basket of goods" to record how the price index of the cost of living changes over time. The composition of the basket differs between countries and mostly covers only goods markets, not capital markets.
In Switzerland, the Federal Statistical Office (FSO) compiles the Swiss Consumer Price Index (CPI). The table below shows which consumer goods make up this index and in what proportions. The basket contains "the most important goods and services consumed by private households" [our translation]. The weighting is based on annual household surveys and is meant to reflect their actual consumption behaviour as accurately as possible.

Table 2.4: Composition of the basket of goods for the Swiss Consumer Price Index (CPI) and weights, 2026. Source: BFS (2026) (original in German, own translation).

Category	Weight in %
Total	**100.000**
Food and non-alcoholic beverages	10.307
Alcoholic beverages and tobacco	3.468
Clothing and footwear	2.420
Housing and energy	25.595
Household goods and housekeeping	3.302
Health	17.379

Transport	10.715
Information and communication	3.279
Leisure, sport, and culture	7.501
Education	0.851
Restaurants and hotels	9.568
Insurance and financial services	2.252
Other goods and services	3.363

Excursus: The invisible hand of the market

The term invisible hand is often used to describe the **self-regulation of markets**. The term goes back to Adam Smith. Adam Smith used the term twice in his earlier work *The Theory of Moral Sentiments* (1759) and once in his work *The Wealth of Nations* (1776). In *The Wealth of Nations*, Smith uses the term in the chapter on "restraints upon the importation from foreign countries of such goods as can be produced at home". In this context, Smith writes that if someone prefers to support domestic rather than foreign industry, this person is led by an invisible hand. Noam Chomsky therefore interprets Smith as describing a "home bias" in this passage. Different economists have interpreted the passage in different ways, however. The most popular interpretation sees in this passage the justification of the self-regulating forces of the market. Market activity, it holds, is a regulating force that leads individuals to satisfy their individual needs as well as possible, which at the same time serves society by distributing goods as well as possible. Such an interpretation is contested in view of the surrounding context and is repeatedly criticized by experts in the history of theory.

> "He generally, indeed, neither intends to promote the public interest, nor knows how much he is promoting it. By preferring the support of domestic to that of foreign industry, he intends only his own security; and by directing that industry in such a manner as its produce may be of the greatest value, he intends only his own gain, and he is in this, as in many other cases, led by an invisible hand to promote an end which was no part of his intention." (Smith and Recktenwald 2013, 371)

Excursus: The marginal principle – marginal utility and marginal cost

The marginal principle is central to the conceptualization of the market from a neoclassical perspective, since it takes on the explanation of the emergence of value and utility and thus of demand. The marginal principle has its origins around 1870 in the marginalist revolution and to this day forms the basis of neoclassical economics. It deals with the question of the value of commodities, which also influences the allocation of goods, services, and factors of production. In contrast to classical economics, the marginal principle of neoclassical economics detaches itself from the objective theory of value and formulates the subjective theory of value.
The objective theory of value of classical economics postulates that all commodities have an objective value. The labour put into the production of the commodity was commonly used as

the basis for determining this value, which founded the labour theory of value. In the subjective theory of value, by contrast, no absolute values are considered but the subjective values that people attribute to commodities and their marginal changes relative to the starting position. This focus on marginal changes founds the marginal principle. The marginal principle made it possible to transfer differential calculus to cost, utility, and yield functions, as a result of which the use of mathematical models in economics experienced a lasting boom.

Yield function of land and marginal yield

The marginal principle was originally observed in the yield function of land. Studies of agriculture in the 18th century revealed the relationship between factor input and yield. The marginal yield is the increase in yield through one additional unit of a factor of production.

> "It lies in the nature of agriculture – and this is a very noteworthy circumstance – that the additional product does not rise in direct proportion to the number of additional workers employed; rather, each worker employed later delivers a smaller product than the one before." Johann Heinrich von Thünen 1850, secondary quotation after Winfried Reiss in *Mikroökonomische Theorie: Historisch fundierte Einführung* (2007, pp. 90f.) [our translation].

What von Thünen observed with the factor labour, A. R. J. Turgot also found with the use of the factor capital in the form of fertilizer. The principle of marginal yield can equally be transferred to industrial production and, for example, to the number of workers in a factory.

Gossen's laws

In 1854, Hermann Heinrich Gossen developed his laws of human intercourse and the rules for human action flowing from them. In them, he formulated the lawfulness of individual preferences and their satisfaction, which is oriented towards the respective utility.

Gossen's first law: The law of diminishing marginal utility

The law of diminishing marginal utility, also called the law of satiation, reads:

> "The magnitude of one and the same pleasure decreases continuously if we continue to prepare that pleasure without interruption, until satiety finally occurs." Hermann Heinrich Gossen in *Entwicklung der Gesetze des menschlichen Verkehrs, und der daraus fliessenden Regeln für menschliches Handeln* (1854, p. 4) [our translation].

The utility of the first cup of coffee in the morning is greater than that of the third or fourth. It is similar with living space, where, for example, the first bathroom in a flat is of greater use to the residents than the second. According to the theory, this law can in principle be transferred to all goods.

Gossen's second law: The law of equalization of marginal utilities

The law of equalization of marginal utilities, also called the law of equalization of pleasures, states:

> "A human being who is free to choose between several pleasures but whose time is not sufficient to prepare all of them fully must, however different the absolute magnitude of the individual pleasures may be, in order to bring the sum of his pleasure to the greatest, before he fully prepares even the greatest one, prepare

them all partially, and in such a proportion that the magnitude of each pleasure, at the moment when its preparation is broken off, remains the same for all" Hermann Heinrich Gossen in *Entwicklung der Gesetze des menschlichen Verkehrs, und der daraus fliessenden Regeln für menschliches Handeln* (1854, p. 12) [our translation].

This law aims at the optimal use of limited means for different goods. An individual's optimal utility is reached when the marginal utility (that is, the utility of the next additional unit) is the same for all goods. Mathematically, this means that the marginal utility functions of the different goods are set equal. The marginal utility corresponds to the maximum price that the person concerned is willing to pay. From the aggregation of the marginal utility (functions) of all individuals, the total demand (function) can be determined. The marginal principle accordingly forms the core of neoclassical consumer theory. For interested students (who are familiar with the basics of neoclassical economics), the critical discussion of neoclassical consumer theory by Ben Fine (in German) is recommended.

Marginal cost and marginal yield in firms

As derived above, marginal yield describes the yield of an additional unit. Analogously, the term **marginal cost** describes the cost of an additional unit. Marginal costs can be determined by differentiating the mathematical cost function. The costs [C] of a firm result from the sum of the fixed costs [C_f] (for example rent, machines) and the variable costs [C_v] (labour power, raw materials, intermediate products).

$$C = C_f + C_v$$

According to neoclassical theory, firms orient themselves, besides the cost function, on the revenue function. This describes how much revenue [R] arises from the sale of a good depending on quantity [X] and price [P].

$$R = P * X$$

Differentiating the revenue function yields marginal revenue, the revenue of one additional unit sold. A firm produces until marginal cost exceeds marginal revenue. This can be calculated by setting the differentiated cost and revenue functions equal. In market equilibrium (under perfect competition), the price theoretically equals marginal cost. Although neoclassical theory focuses strongly on price, it has engaged little with how firms set prices. In his dissertation, Nubbemeyer (2010) shows that until the middle of the 20th century it was controversially discussed whether firms actually orient themselves towards the marginal principle or towards alternative concepts (for example mark-up pricing, as in some post-Keynesian models) when setting prices. Although the debate has not been conclusively resolved, the idea of the marginal principle dominates economics.

2.5.3 Markets as arenas of power and distribution – capitalist production system

While neoclassical economics regards markets as universal institutions, other schools of thought examine capitalist markets in their historical and societal specificity. That is, these analyses do not

claim universal validity but want to understand the dynamics on capitalist markets in a specific context.

From this perspective, markets are not a natural state but are actively **shaped and regulated** by actors, institutions, property rights, and laws. This immediately raises our guiding question: Who determines the "rules of the game", and whose interests do they serve? This question opens up the view to **power and distribution**, which play a central role in political-economy perspectives. A political-economy perspective means that economic, political, and societal aspects and structures are understood in relation to one another, so that power, institutions, and distribution, for example, receive central attention. Political economy is thus a collective term, and different schools of thought can take a political-economy perspective (there is also a strand of political economy in neoclassical economics). Marxist and post-Keynesian economics, as well as parts of feminist and ecological economics, understand themselves in principle as political-economic. In contrast to neoclassical economics, capitalist markets in this perspective are moreover regarded as inherently **unstable**. Class conflicts and power dynamics are thus not an exception but structurally built in.

To make this perspective more concrete, we ask in the next section: What exactly makes an economic system *capitalist*?

2.5.4 What actually makes an economic system capitalist?

There are many different definitions of what constitutes the essence of a capitalist economic system. Some central aspects can nevertheless be identified (Fraser 2022):

- The means of production are predominantly in private ownership.
- The majority of employees are separated from the means of production. They thus do not own the means of production. They therefore have to sell their labour power for a wage.
- Economic decisions and the distribution of goods are predominantly coordinated through markets.
- Firms produce predominantly with the aim of making profits. Competitive pressure means that a large part of these profits is reinvested, so that constant capital accumulation becomes a central feature of the capitalist economy. This leads to a compulsion for constant growth.

These conditions may seem very ordinary to us. Many of us do wage work, we encounter markets everywhere, and competition and the profit motive seem so self-evident in economic policy news that they need no further explanation. In fact, however, this way of doing business became dominant only in the course of the 19th century with the rise of industrialization in England. Before that, in Western Europe, peasants were bound to the land and the landlord through the feudal system, and the economy was largely shaped by agriculture. Production for one's own needs and local supply dominated in many places. In addition, the communal use of land and resources, alongside seigneurial and private ownership, played an important role in supplying the population. Markets (for example weekly markets, annual fairs, and so on) also played a role but were of less importance in everyday life and for supplying the population than they are today. Moreover, markets were strongly embedded in society and in a moral system – ideas of a just price, for example, played a central role.

An influential, predominantly Marxist interpretation sees a decisive step in the emergence of capitalism in the fact that people were separated from their access to land and the means of production. Over centuries, more and more land in common ownership, that is, commons, was privatized. Through these enclosures and the privatization of communally used land, many peasant households lost their livelihood and increasingly became dependent on wage labour (Polanyi 2017). This was in part a very violent process and illustrates that markets do not arise independently of societal and political processes. They are actively created and shaped by various actors, state interventions, property rights, laws, and so on. It becomes clear that power is also a central factor.

But how and when exactly capitalism emerged is still contested in scholarship today. While some authors see its origin above all in changed class relations and conflicts in England (for example Wood 2015), others emphasize the central role of colonialism, slavery, and global trade (for example Beckert and Dierlamm 2026). Many current works try to combine these two perspectives. And while some locate the origin of capitalism as early as the Middle Ages, others see the decisive steps of its emergence only in the industrial capitalism of the 19th century. Here too, it is a question of perspective and of the understanding of what exactly capitalism is (for example, whether we can already locate capitalist elements in long-distance trade from the 16th century onwards or not).

2.5.4.1 Why do companies (and entire economies) actually have to grow constantly?

A central dynamic of the capitalist economy is the accumulation of capital. Firms do not produce only to provide goods but to make profits and to invest these profits again in the production process. Marx describes this dynamic as a movement from money (M) through the production of commodities (C) back to more money (M') – (often summarized in this notation: M-C-M'): capital is used to produce commodities whose sale generates a larger sum of money. Capital must thus multiply continuously. This dynamic is not determined solely by the individual profit interests of individual firms but also arises from competitive pressure between firms. Firms that do not invest, bring forth innovations, or raise their productivity risk losing market share. Competition thereby creates systemic pressure towards growth, expansion, and constant productivity increases. Several tensions arise from this growth dynamic:

- **Capital vs. labour:** Since the majority of people do not own means of production, they depend on selling their labour power for a wage. At the same time, firms strive to make profits by raising labour productivity and reducing costs. Profits and wages thus stand in an inverse relationship, since higher wages usually mean lower profits and vice versa. Marx describes this relationship as the fundamental class conflict between capital and labour. In doing so, Marx describes a process of exploitation. Firms do not pay employees the full value they create through their work and skim off the rest as profit (this remainder is also called surplus value). Since workers do not receive the full value of their work, Marx calls this process exploitation. In Marxist theory, exploitation describes not primarily a moral judgement of individual firms but a social relation (see also Chapter 2.3.2, *Capital – why do some people own capital and others do not?*).
- **Distribution of income and demand:** While firms may have an interest in reducing wage costs, wages are at the same time the most important source of income for a large part of the population and thus a central basis for consumer demand. Strong pressure on wages

can therefore raise the profitability of individual firms in the short term but contribute to weaker aggregate demand in the long term. Marxist approaches describe this as a possible tendency towards crises of overproduction: firms can produce more than can be sold given the available purchasing power. Post-Keynesian approaches also take up this problem but emphasize in particular the importance of demand for economic development. If demand falls (rises), firms tend to invest less (more), which in turn can have a negative (positive) effect on economic development.

- **Competition and concentration:** On the one hand, competition is regarded as an important mechanism for innovation and efficiency gains. On the other hand, competition can lead in the long term to a concentration of capital, since successful firms can gain market share and displace smaller firms. As a result, dominant firms and monopoly-like structures can emerge.
- **Societal and ecological limits:** Feminist economists point out that the capitalist economy depends on labour that often takes place outside markets, in particular unpaid care and reproductive work. These activities create the preconditions for the paid economy to be possible at all. Ecological economists also emphasize that capitalist accumulation is based on the use of natural resources. The continuous expansion of production and consumption can thereby come into conflict with the limited ecological capacities of the planet.

The growth dynamic described and its tensions mean that capitalist economic systems are particularly dependent on growth. Matthias Binswanger (2019) even argues that capitalist economic systems are subject to a **growth imperative**. Firms must make profits to finance investments and to survive in the long term. If firms on average do not earn profits, this can lead to a downward spiral: investment declines, employment falls, and aggregate demand decreases. Growth thus stabilizes not only individual firms but the entire economic system. Schmelzer and Vetter (2019) argue in turn that **growth dependency** does not necessarily mean that every single economic activity must grow. Rather, a longer-term absence of growth can create considerable economic and societal tensions. A phase of stagnation or shrinkage can lead to rising unemployment, falling incomes, corporate crises, and a burden on public budgets (Schmelzer and Vetter 2019). If growth fails to materialize, the system thus falls into crisis.

This compulsion or dependency is reinforced by institutional factors. Modern, credit-based financial systems, for example, presuppose future growth so that loans can be repaid. In addition, in many countries – including Switzerland – social insurance systems are closely tied to paid employment. High employment and rising incomes thereby become important preconditions of societal stability. Growth is thus not only an economic goal but deeply embedded in societal institutions. According to Hartmut Rosa (2003), growth has even become a central cultural expectation.

From the perspective of ecological economics, this orientation towards growth is fundamentally problematic, since economic growth has historically been, and continues to be, closely coupled with rising energy and resource consumption. Whether growth can be permanently decoupled from environmental pressure is the central point of contention between green growth and post-growth approaches – more on this in Chapter 7.

Interactive exercise – Why growth becomes a compulsion

Why growth becomes a compulsion: available in the online version of this script.

2.5.5 Are financial crises accidents – or built into the system?

Money and credit are an important precondition for capitalist dynamics. Through loans, firms can make investments, expand production capacities, and finance innovations. Since the 1980s, there has been increasing liberalization and deregulation of financial markets in many countries. In the period after the Great Depression of 1929, banks and financial markets were still strongly regulated. With the deregulation of the 1980s, the role of the financial sector within capitalist economic systems also changed. While financial markets originally had above all the task of enabling productive investment, financial returns and short-term increases in value gained increasing importance. This development is called *financialization*. It refers to the growing importance of financial markets, financial institutions, and financial motives for firms, households, and states. Firms, for example, orient themselves more strongly towards short-term return targets and increasing share value, while financial institutions increasingly make profits by trading financial assets such as shares, bonds, commodities, or derivatives. With the importance of the financial market for the economy as a whole, the power of the financial industry also grows.

Deregulation relied above all on theories from neoclassical economics, which argued that deregulated financial markets would function efficiently in principle because of their information processing and the rationality of individual market participants (Fama 1970). The great financial crisis of 2008 fundamentally called these assumptions into question, however. While the financial crisis represents, from a neoclassical perspective, an exogenous shock, that is, an external shock that disturbed the equilibrium, other schools of thought, above all post-Keynesian economics, emphasize that financial crises arise from the system itself and that the system is thus inherently unstable. Post-Keynesian economics emphasizes that investment depends primarily on expectations about future profits. Firms invest when they expect future revenue to exceed the cost of the investment. These expectations are fundamentally uncertain, however. As a result, investment and economic activity can be subject to strong fluctuations. With his financial instability hypothesis, Hyman Minsky (1992) shows that phases of economic stability in particular can generate new instabilities. In times of rising confidence, firms and households increasingly take on credit, and financial institutions are willing to take on higher risks. A dynamic can thereby develop in which rising asset prices and increasing indebtedness initially reinforce economic expansion. At the same time, however, the system becomes more prone to crises. If expectations are disappointed, lending, investment, and consumption can fall abruptly. The great financial crisis of 2008 was an impressive example of this. From this perspective, financial crises are therefore not simply external disturbances of an otherwise stable market system but can arise from the normal dynamics of capitalist financial systems.

The tensions described, as well as the instability of the financial system, do not mean that capitalist societies inevitably collapse. But they show that capitalist markets do not find a stable equilibrium on their own. Instead, they are unstable and prone to crises. Historically, capitalist societies have developed various mechanisms to limit these conflicts and to establish stability. These include, for

example, state regulation, social insurance, labour rights, public investment, or the opening up of new markets. The concrete shape of capitalism therefore differs strongly between historical periods and also between different regions. An often-discussed example of stabilization is the Fordist accumulation regime after the Second World War, when employees largely accepted the capitalist order of ownership and economy and firms accepted rising wages and the expansion of the welfare state. Productivity growth then translated strongly into rising wages. This was a development that, however, stalled from the 1970s onwards. This then had strong effects on the lives of many people. In Chapter 3, we discuss the social limits and will also go into these developments in depth.

In-depth questions

- Think about where in your everyday life you encounter tensions of the capitalist mode of production and how they affect your everyday life.
- Look for examples of interactions between the economy, politics, and society.
- Are you aware of current processes of enclosure? How are they connected to the capitalist mode of production?

Further material – Oeconomia: Economic growth, debt, inequality, and profits

In 2020, Carmen Losmann made a documentary film (in German) on the relationship between economic growth, debt, inequality, and profits. The film deals with these complex relationships in a very accessible way and stimulates reflection. Even though the film is in part somewhat imprecise, it is well worth watching as an introduction.

Self-test Quiz

Test your understanding of this chapter with 9 short questions. The quiz is voluntary and ungraded, and you can repeat it as often as you like.

Question 2.1 (based on section 2.1.1)

A storm destroys houses in a village, and the reconstruction is financed and carried out within the same year. What happens to GDP?

a) GDP falls, because wealth was destroyed
b) GDP can rise, because the reconstruction is recorded as production, even though well-being has worsened
c) GDP stays the same, because destruction and reconstruction cancel each other out
d) GDP is not affected, because natural disasters are excluded from national accounts

Question 2.2 (based on section 2.1.2)

Which indicator includes negative factors such as environmental pollution and social inequality as well as the value of unpaid work?

a) Gross domestic product (GDP)

b) Human Development Index (HDI)
c) Genuine Progress Indicator (GPI)
d) Consumer Price Index (CPI)

Question 2.3 (based on section 2.2)

Your neighbour helps you move house, and a few weeks later you look after their cat while they are on holiday. Which of Polanyi's forms of provision is this?

a) Householding
b) Reciprocity
c) Redistribution
d) Market exchange

Question 2.4 (based on section 2.3.3)

Feminist economics often uses the third-person principle to define work. Which activity counts as work under this principle?

a) Jogging to stay fit
b) Sleeping
c) Cooking dinner for your family
d) Reading a novel for pleasure

Question 2.5 (based on section 2.4.1)

Deep-sea fishing grounds are hard to exclude anyone from, but every fish caught is no longer available to others. Which type of good are they?

a) Private good
b) Toll (club) good
c) Common-pool good
d) Public good

Question 2.6 (based on section 2.4.1)

Whether a good is excludable is a fixed, natural property of the good itself.

a) True
b) False

Question 2.7 (based on section 2.4.2)

Why is a flight often cheaper than a train ride, even though it harms the environment much more?

a) Because airlines are more efficient than railways
b) Because many of the real costs, such as climate damage, are not included in the ticket price and are borne by third parties
c) Because train tickets include a tax that subsidizes air travel
d) Because demand for flights is lower than demand for trains

Question 2.8 (based on section 2.5.2)

A small town has only one supermarket, and all residents shop there. Which precondition of perfect competition is violated?

a) Complete information
b) A large number of suppliers and buyers (polypoly)
c) No preferences for individual suppliers
d) None, the market is still perfectly competitive

Question 2.9 (based on section 2.5.4)

According to the political-economy perspective in the script, what drives the compulsion of capitalist economies to grow?

a) Population growth, which constantly increases demand
b) The individual greed of managers
c) Competitive pressure that forces firms to reinvest profits, so that capital must continuously multiply (M–C–M')
d) Central banks, which set growth targets

Answers and explanations

2.1 b). GDP measures the market value of goods and services produced, not the losses. Reconstruction counts as output, so a disaster can paradoxically raise GDP. This is one of the main criticisms: GDP ignores social and ecological consequences and says little about actual well-being.

2.2 c). The GPI combines about 26 economic, social and ecological indicators, including negative ones and the value of unpaid work. The HDI supplements GDP with life expectancy and education. The CPI measures the price level, not prosperity.

2.3 b). Reciprocity is give-and-take between people beyond market and state, for example among neighbours, friends or in associations. To endure, giving and taking must not get too far out of balance. Householding is self-provision within the household, redistribution flows through an administrative centre (for example a tax-financed pension system), and market exchange happens at market prices.

2.4 c). Under the third-person principle, any activity that could in principle be done by a paid third person counts as work, whether it is paid or not. Someone else could cook for your family, but nobody can jog, sleep or enjoy a novel on your behalf.

2.5 c). Common-pool goods are non-excludable but rival in consumption. This combination creates the risk of overuse that Hardin called the tragedy of the commons. Elinor Ostrom showed that self-organized cooperation within clear boundaries can prevent overuse without privatization or nationalization.

2.6 False. False. Excludability depends above all on technical possibilities, societal negotiation and power relations. Water can be made accessible or excludable, and even sunlight could in theory be

made excludable (Mr Burns in The Simpsons). So the question is less whether the sun is a public good than whether it is treated as one.

2.7 b). This is a negative external effect: costs are shifted onto third parties such as other people, future generations or nature. Neoclassical economics treats such externalities as exceptions. William Kapp argued that the more a system is based on maximizing private profit, the stronger the incentive to shift costs, so externalization is built into capitalism.

2.8 b). With only one supplier and many buyers, the market is a monopoly, not a polypoly. Under perfect competition, individual suppliers have no influence on the price because buyers could switch to other suppliers. If the preconditions are not met, neoclassical economics speaks of market failure, which justifies state intervention.

2.9 c). Firms produce to make profits and reinvest them. Firms that do not invest, innovate or raise productivity risk losing market share, so competition creates systemic growth pressure. Institutions reinforce this dependency: credit-based finance presupposes growth, and social insurance is tied to paid employment.

3 Problem Analysis: What Is Going Wrong?

> Guiding question
>
> If prosperity for all – within ecological limits – is the goal we took as our starting point in Chapter 2: Why do we not achieve this goal, and what exactly stands in its way?

The current way of doing business leads to major challenges, which manifest themselves in various crises. The current crises are referred to as a multiple crisis and, more recently, increasingly as a **polycrisis** (see, for example, Lawrence et al. 2024 or this explainer video by Adam Tooze), because they touch people's working and living worlds in many ways and are interwoven with one another. Some of the individual crises are strongly connected and share common causes. Others appear rather independent of one another and do not necessarily have the same causes. Taken as a whole, however, all these crises culminate in this polycrisis, which often seems overwhelming. In this chapter, we devote ourselves to the **analysis of these crises** before turning to solutions for a sustainable economy. In what follows, we address the ecological and social problems separately. It is important to stress, however, that these dimensions are closely linked and can reinforce one another. The separation of these areas is meant to make the introduction easier.

As became apparent in the preceding chapters, the analysis itself is also shaped by the perspective a person takes. The problem analysis would turn out differently depending on the school of thought. Neoclassical economics, for example, does not necessarily see rising income and wealth inequality as a problem as long as it comes about through market mechanisms. A school of thought that includes aspects of power in its analysis, such as the feminist or the Marxist one, sees this inequality as a central problem for the economy and society. In line with the orientation of the course, we will frame the problem analysis as broadly as possible, across as many schools of thought as possible. Later in the course, we will also see more precisely which analyses and conclusions come from which school of thought. In the first part of this chapter, we address the ecological limits of current economic activity and the relationship between nature and the economy. Then, in the second part, we look at the social limits of the current system and at how the economy and society are related.

3.1 Ecological Limits – How Much Earth Do We Actually Have – and How Much Do We Use?

The ecological limits of our economic activity have been concretely tangible for some time and are not just abstract discussions about climate models. We can feel them in our everyday lives. While this script was being written in the summer of 2026, the fourth heatwave arrived in Switzerland, wildfires raged in France and Spain, and the consequences of heat and drought were noticeable

everywhere in Europe. In the context of a sustainable economy, the question arises of how we connect these developments with economic activities or, more generally, how we can understand the relationship between the economy and nature. How we understand this relationship influences what we perceive as a problem and also forms the basis of the range of possibilities within which we can think about and develop solutions. Here, we will focus mainly on neoclassical and ecological economics, since they have two fundamentally different understandings of this relationship. While ecological economics emphasizes that the economy is embedded in the natural environment and necessarily depends on nature as its foundation, neoclassical economics understands nature and the economy as two separable spheres. Before we go into the basic understandings of this relationship in more detail (3.1.2) and look at what they mean for the problem analysis, we briefly look at some basic aspects of the ecological crises.

3.1.1 Planetary boundaries and climate crisis

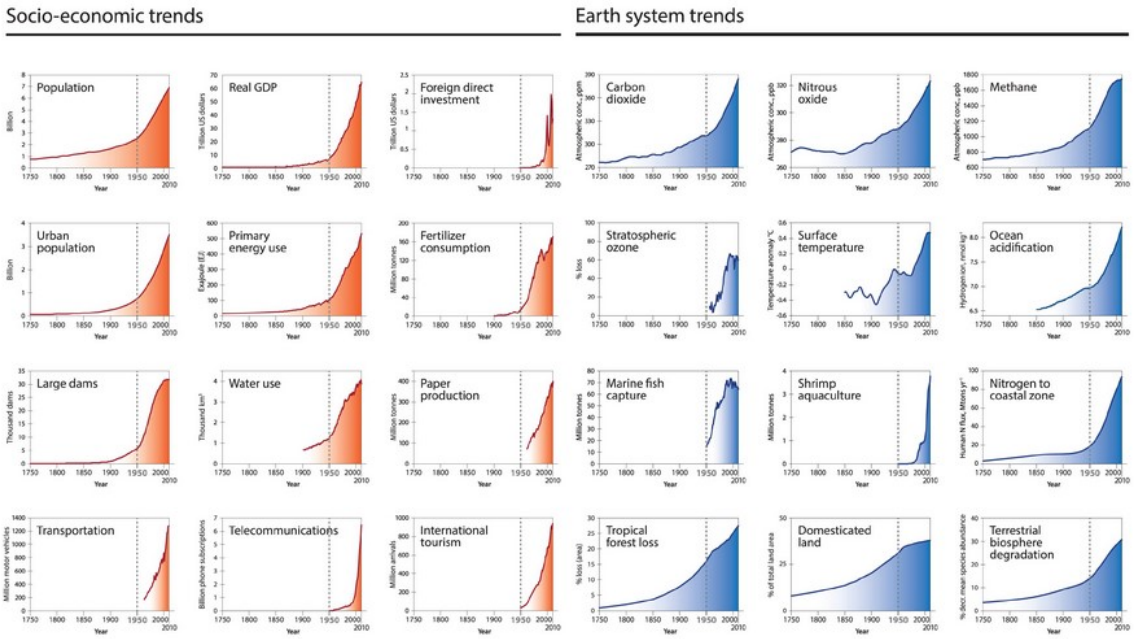


Figure 3.1: Trends of various socioeconomic and Earth system indicators between 1750 and 2010. The rapid increase is called the "Great Acceleration", and its effects are regarded as an indicator of the new Anthropocene epoch. Source: https://futureearth.org/the-great-acceleration/.

In the wake of the Industrial Revolution, a uniquely productive mode of production emerged, which enabled a massive increase in material prosperity. At the same time, but far less noted, there was also a massive increase in the consumption of natural resources, environmental pollution, and emissions (Jarrige et al. 2020). Socioeconomic growth went hand in hand with the acceleration of biophysical trends. The description of exponential growth dynamics is called "the Great Acceleration" (Steffen et al. 2015). The graphic above shows, by way of example, some important biophysical and socioeconomic indicators, all of which begin to rise with the Industrial Revolution. From the middle of the 20th century, the tendency towards exponential growth becomes obvious.

Excursus: Exponential growth

The exponential function is used to describe the size of things that are subject to accelerating growth. A decisive step is the realization that even moderate growth of, for example, five per cent means exponential growth, so the increase keeps getting larger (this applies, for example, to economic growth or to case numbers during the COVID-19 pandemic). Exponential growth is hard for us humans to grasp and is commonly underestimated, since the slope is quite moderate at the beginning. The growth rate, however, keeps increasing, because the moderate increase in the following year refers to the value that rose in the previous year – and at some point it becomes enormous. A revealing figure that quickly exposes growth processes is the doubling time, which can easily be calculated in one's head: 70 divided by the percentage growth per unit of time. If something grows by seven per cent per year, the doubling time is ten years; at five per cent, it is fourteen years; and at one per cent growth, it is seventy years. The number 70 is the result of multiplying 100 (the percentage value of a doubling) by the natural logarithm of 2. You do not need to understand how this number is derived. All that matters is: 70 divided by the percentage growth.
For those who are interested, there is a documentary on exponential growth: "Understanding exponential growth: The water lily principle" (in German). The following **thought experiment** also helps with understanding: A client offers you two different fees for a 64-day assignment. Which fee do you choose? A: You receive CHF 10,000 every day. B: You receive CHF 0.01 on the first day, and on every following day the amount is doubled.
Table 3.1 shows the payoff matrix for this thought experiment.

Table 3.1: Thought experiment: Payoff matrix under linear versus exponential growth.

Number of days	Payout A	Payout B
8 days	CHF 80,000	CHF 1.28
21 days	CHF 210,000	CHF 10,485
26 days	CHF 260,000	CHF 335,544
30 days	CHF 300,000	CHF 5,368,709
64 days	CHF 640,000	CHF 92,233,720,368,547,758

The payout of B after 64 days is more than 92 quadrillion CHF, which corresponds to about 95,000 times the wealth of Elon Musk (according to Forbes, the richest person on Earth as of June 2026).

Since the late 18th century, when the Briton Thomas Robert Malthus (1766-1834) first put forward a theory of overpopulation in his *Essay on the Principle of Population*, the discourse of a planetary carrying capacity has persisted in certain scientific theories. Even though Malthus's ideas have meanwhile turned out to be outdated – he calculated the maximum population depending on a fixed quantity of producible food – the basic idea returns in the form of neo-Malthusianism. In the study *The Limits to Growth*, published by the Club of Rome in 1972, a neo-Malthusian approach calculated a population limit also by means of available food, but additionally including available resources, environmental pollution, and industrial output. Although the predicted developments did not materialize, the study still carries weight today in an ecological perspective that problematizes limitless growth.

The latest theoretical development in this regard is the natural-science concept of **planetary boundaries**. Unlike before, these do not refer to a maximum population but to indicators from Earth system science. As a starting point, the researchers choose the state of the Earth system in the Holocene (the current and latest epoch of Earth's history, which began about 11,700 years ago). In this epoch, they argue, the planet had largely ideal conditions for giving rise to human civilizations. Deviations from this bring humanity into an uncertain zone in which tipping points can be reached. Crossing them can either halt previous developments, change their direction, or accelerate them strongly. One example is the extinction of many large mammals at the end of the last ice age following human migration to the American continent. The concept of planetary boundaries advises applying the precautionary principle here in order to reduce possible harm to humans and the environment. The current Planetary Health Check takes the planetary boundaries as a fundamental framework for evaluating the health of our planet (Ceasar et al. 2024; Sakschewski et al. 2025). The recently founded initiative "Planetary Health Check" offers, besides the report, exciting opportunities on its website to engage with the planetary boundaries and their development over the last 70 years.

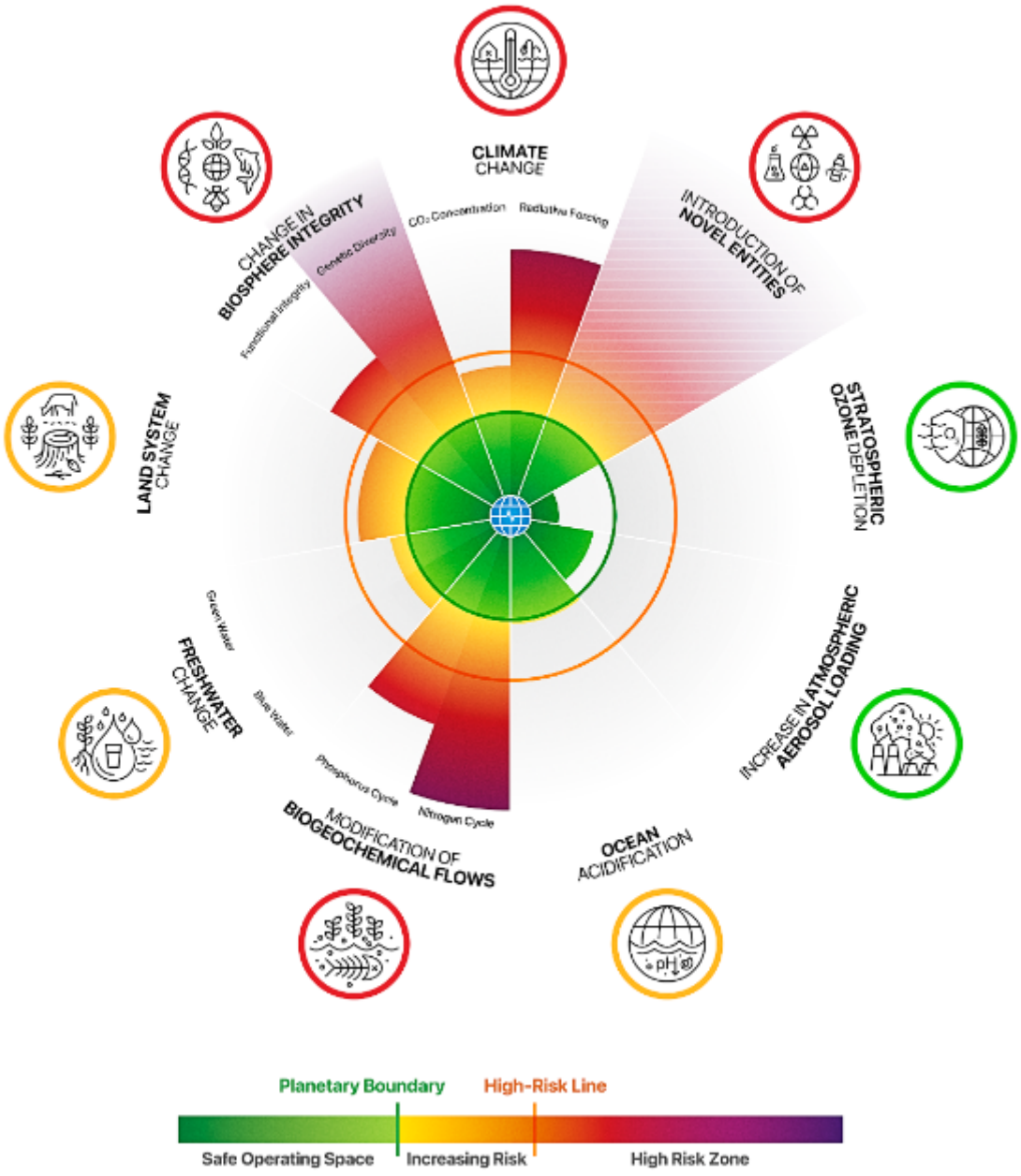


Figure 3.2: Planetary boundaries of the various subsystems and the respective risk assessment of the current state. Green denotes the safe operating space, yellow an increased uncertainty. Beyond yellow indicates that the risk of fatal consequences is assessed as high. Source: Sakschewski et al. (2025, 29).

The planetary boundaries refer to nine different subsystems, such as land-use change or ocean acidification. For each of them, an inner circle (safe operating space) and an outer circle (increased uncertainty) are defined. Seven of the nine planetary boundaries have already been crossed (Sakschewski et al. 2025). For ocean acidification, the boundary has recently been crossed for the first

time, while aerosol loading is declining somewhat again. While the trend for stratospheric ozone levels does not show a clear direction, the degree of transgression is increasing for all boundaries previously identified as crossed (Sakschewski et al. 2025).

Within these nine subsystems, there are two so-called core boundaries: biosphere integrity and climate change. These two systems combine the processes of many other subsystems and act at a supraregional level. Reaching tipping points in these two systems can therefore push the entire Earth system into a new state. The rapid changes in the climate in particular pose a major challenge to humanity in the 21st century. Because the energy system, the transport infrastructure, and industrial agriculture are based on fossil fuels such as oil and gas, excessive amounts of greenhouse gases such as carbon dioxide and nitrogen oxides are continually emitted. These accumulate in the atmosphere and prevent the heat that reaches the atmosphere through solar radiation from escaping again (greenhouse effect). As a result, glaciers and ice sheets have shrunk, the oceans have become warmer, and sea levels have risen over the last 30 years. Extreme temperature anomalies and precipitation events are also continually increasing. Today, the concentration of greenhouse gases in the Earth's atmosphere is the highest in the last 800,000 years, and the global average temperature has risen by more than one degree Celsius since the Industrial Revolution.

To avoid reaching tipping points in the climate, climate policy set the two-degree target with the Paris Agreement in 2015. The rise in average temperature since the Industrial Revolution is to be kept "well below 2 degrees Celsius". Even at two degrees of warming, massive biophysical changes are likely. And to reach this target, energy supply must forgo fossil fuels entirely from 2050. Such a shift poses major challenges for society, in particular because fossil energies were a central factor in the development of the current economic system and strongly structure it (see, for example, Huber 2013; Kallis and Sager 2017; Malm 2016).

3.1.2 Different understandings of the nature-economy relationship

As already discussed in Chapter 2 on the scope of economics, some schools of thought understand the economy mainly as a separate sphere in which actors interact on markets. Others emphasize the embeddedness of economic activities in society and nature. We already encountered this distinction in Chapter 2.3.1, where we dealt with land as a factor of production. Neoclassical economics focuses very strongly in its understanding on a separate economic sphere. These efforts to study a separate economic sphere in isolation go back, among other things, to the marginalist revolution, when attempts were made, among other things, to separate economic motives from non-economic motives in humans. Accordingly, in this understanding, the economy stands to some extent alongside nature, and the two are of equal standing (see Figure 3.3).

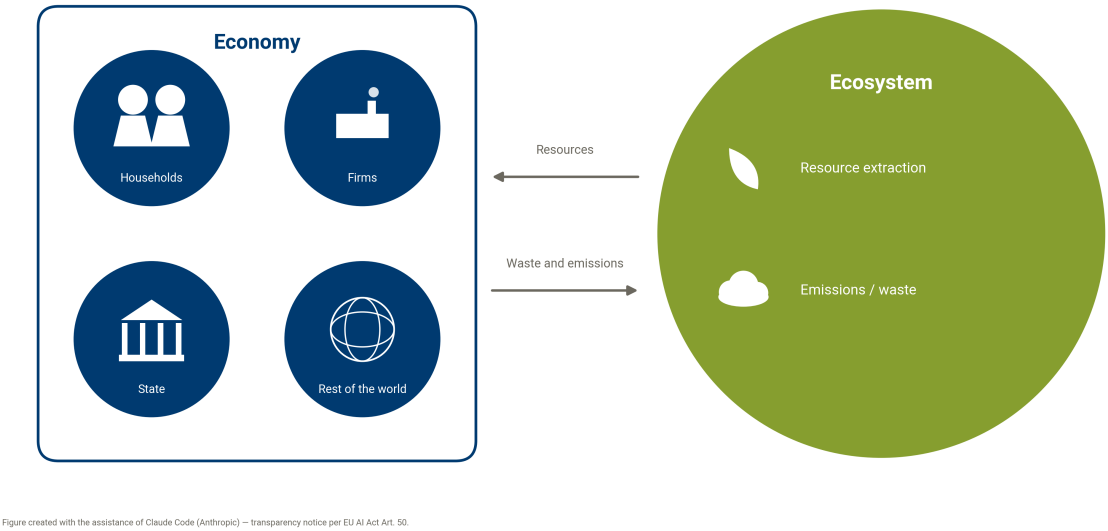


Figure created with the assistance of Claude Code (Anthropic) — transparency notice per EU AI Act Art. 50.

Figure 3.3: Economy and nature as two separate spheres. Source: Own illustration

A fundamentally different understanding emerges for ecological economics. Here, the economy and nature are not conceived side by side, but the economy is understood as embedded in an ecosystem. The economy is a subsystem of the global ecosystem and completely dependent on its functions. All economic activities rely on the extraction of energy and raw materials from nature and inevitably end in the form of waste and emissions that are released back into the environment. The economy therefore cannot be understood independently of the biophysical limits of the planet. While capital, labour, or knowledge can be increased, the Earth's resources and its capacity to absorb pollutants are limited. Figure 3.4 conceptualizes the embedding of the economic subsystem in a larger finite global ecosystem. The economy is thus embedded in a finite global ecosystem, draws energy and resources from it (source function), and also releases matter and energy back into this ecosystem (sink function). Only solar energy newly enters the system from outside, while waste heat leaves the Earth at the same time.

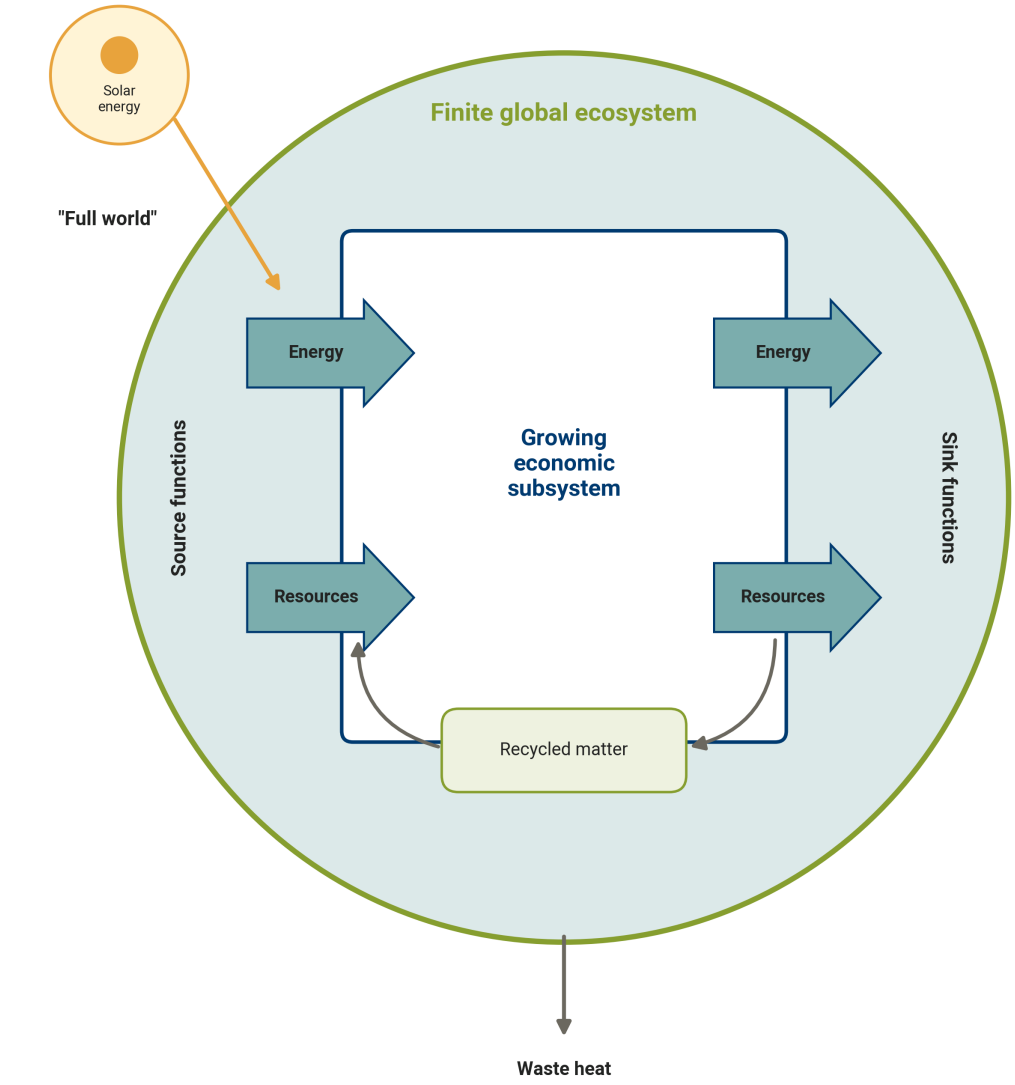


Figure created with the assistance of Claude Code (Anthropic) — transparency notice per EU AI Act Art. 50.

Figure 3.4: Economy embedded in a finite ecosystem. Source: Own illustration based on Herman Daly's concept of Economics for a full world.

These fundamentally different understandings have important consequences for how we can understand sustainability. The concept of sustainability can be placed on a continuum from weak to strong (see Figure 3.5). While the neoclassical understanding entails a weak understanding of sustainability, ecological economics has a strong understanding of sustainability. In what follows, we go into the two understandings.

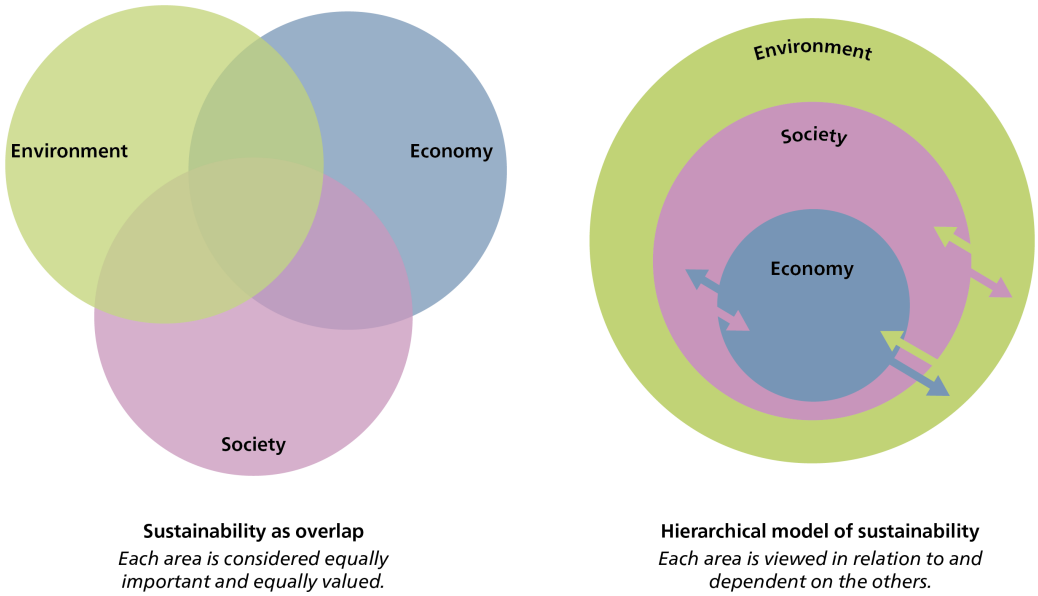


Figure 3.5: Weak (left) and strong (right) sustainability. Source: Own illustration.

3.1.2.1 Weak sustainability – the neoclassical position[1]

Weak sustainability sets a rather "lenient" requirement for sustainability. For the proponents of this view, an action is sustainable if it offers an advantage to the overall system or at least does not diminish its quality. Weak sustainability can therefore be regarded as a kind of basic requirement for sustainability, with the aim of maintaining a certain level of well-being or quality of life or of ensuring equality. The concept of weak sustainability "assumes the far-reaching and, at least in principle, unlimited [...] substitutability of all kinds of capital" (our translation of Ott and Döring 2004, 41) and is thus based on the premise of neoclassical economics, which conceives the different spheres as standing side by side with equal standing. Natural capital, for example, can be replaced by other capital; the loss of biodiversity, for instance, could be replaced by technological capital. It is assumed that it is ultimately irrelevant in what physical composition the inherited capital stock is passed on to the next generation – what matters is only that total capital and total utility, and thus the overall level of welfare, are maintained. The concept of weak sustainability builds on neoclassical utility theory, according to which it is irrelevant how utility is generated. Environmental economics, which emerged from neoclassical economics, fundamentally draws on the concept of weak sustainability.

Within the concept of weak sustainability, a measure can nevertheless be regarded as sustainable even if it comes at the expense of natural capital. This is the case, for example, if the loss is compensated by an increase in human or manufactured capital. The extraction of raw materials such as coal or excessive crude oil production could accordingly be quite defensible. This understanding

[1] Excerpt from Pufé (2017, 105ff.) and https://thesustainablepeople.com/starke-und-schwache-nachhaltigkeit/ (in German).

tends to recognize the strong role of technological progress for sustainable development. Assessing measures on the basis of such a concept is naturally complex, since it is always tied to many assumptions. Apart from fundamental questions about sustainability such as "What is a good life? How is the level of prosperity measured?", this approach requires that the value of the different types of capital be determined. In practice, this means that a monetary value is attributed to everything. As we saw in Chapter 2.3.1 on land, this is an important prerequisite for substitutability. It makes the different types of capital comparable and interchangeable. Even where there is no market price, a monetary value must therefore be set. Whether such monetary valuation is possible or sensible is discussed again and again. On the one hand, there is the question of whether we can give nature a price; on the other, monetary valuation necessarily implies the substitutability of the natural environment. As early as 1974, William Kapp doubted whether this monetary valuation manages to capture the substantial social value that is actually at stake (Kapp 1974). In addition, fundamental and normative questions arise in operationalization. Which factors are then taken into account? What is the value of the beauty of nature? What is the value of a rare bird or a whale? This short video gives insights into such questions. Monetary valuation then forms the basis for cost-benefit analyses, in which, for example, the costs of a climate measure are weighed against its benefits. In the excursus further below, you will find, if you are interested, the opportunity to dive somewhat deeper into this topic.

3.1.2.2 Strong sustainability – the ecological economics position

In contrast to weak sustainability, the strong sustainability position regards the value of natural capital as irreplaceable. Characteristic of the proponents of the strong concept of sustainability is that their optimism regarding the individual possibilities of substitution is considerably lower. They therefore emphasize the importance of an intact "natural capital stock". That the natural capital stock could be drastically reduced in favour of another type of capital is unthinkable to them. Human-made capital and natural capital are interchangeable only to a limited extent. Proponents of strong sustainability thus start from an ecocentric view, in contrast to the anthropocentric view of weak sustainability.

In contrast to the confidence of neoclassical economists regarding substitutability, advocates of strong sustainability do not believe in solutions based on remediation and reaction but rely on prevention and anticipation. For example, the growth of the ozone hole can be compensated only to a limited extent by measures such as sunscreen, protective clothing, and medical aftercare. Even technological approaches such as geoengineering are viewed critically, since they treat the problem superficially instead of addressing its causes. Geoengineering refers to technical interventions in geochemical or biogeochemical cycles in order, for example, to slow climate change or ocean acidification.

So far, the reduction of environmental pressures has mainly concentrated on end-of-pipe technologies, such as flue-gas cleaning systems on factory chimneys or catalytic converters in cars. At the same time, long-term environmental problems were pushed into the background by such measures as long as their effects were not perceived clearly enough, as in the case of climate change or the loss of biodiversity. This apparent protection of the environment is explained by the concept of **externalization**, in which environmental and social costs are shifted outwards, as described, for example, by Lessenich (2018). This means that pricing takes into account only macroeconomic and

business-economic costs, partly because these are easier to quantify, while the social and ecological costs of providing goods and services are left out or shifted outwards. This leads to distortions in market prices.

> Interactive exercise – Weak or strong sustainability?
>
> *Weak or strong sustainability?: available in the online version of this script.*

3.1.3 Climate crisis – market failure or systemic problem

As explained above, climate change concerns one of the central planetary boundaries and is probably one of the greatest societal challenges of this century. Although all schools of economic thought recognize the climate crisis as a central problem, they differ greatly in how they understand and explain it and in what policy measures can be derived from it. While neoclassical economics sees the climate crisis above all as a major market failure, which can be solved by creating a functioning market, ecological economics, for example, sees the problem more fundamentally in the capitalist mode of production, that is, in the way we do business. In what follows, we look at the two understandings using climate change as an example. Such a juxtaposition could be applied in a similar way to many ecological crises, for example biodiversity loss, since the underlying differences lie in the understanding of the fundamental nature-economy relationship.

3.1.3.1 Climate change as market failure

From the perspective of neoclassical economics, the climate crisis is first and foremost a market failure. As explained in Chapter 2.5.2 (*The market as central coordinating mechanism*), markets are fundamentally regarded as efficient institutions for coordinating economic activities. Under certain conditions (perfect competition), they ensure that resources are used where they create the greatest societal benefit. Since the price mechanism is central to the functioning of markets, an important prerequisite for this is that all costs and benefits of an economic action are reflected in market prices. If this is not the case, the price mechanism cannot function optimally.

And this is exactly the case with the climate crisis. Through the emission of greenhouse gases, companies and households cause costs that are borne not by themselves but by society. These costs are not automatically included in the market price of goods. Such consequences not included in the market price are called negative externalities (see Chapter 2.4.2, *External effects – why does a flight often cost less than a train ride?*). The costs of climate change – such as extreme weather events, crop failures, or biodiversity losses – are therefore not fully reflected in the prices of fossil fuels. Fossil energy appears artificially cheap, which produces more emissions than would be desirable for society. This is shown in Figure 3.6. The dashed purple line would correspond to the social marginal costs, and the solid purple line corresponds to the private marginal costs. Since only the private costs are reflected in the price, the price of goods based on fossil fuels is too low, and accordingly the quantity at market equilibrium A is too high. Concretely, this means that the quantity of CO_2 emitted is too high. If all social costs were reflected in the price, the optimal market equilibrium

would be at point B with a quantity y^*. The grey triangle denotes the overall economic welfare loss caused by these negative externalities.

As hinted in Chapter 2.5.2 (*The market as central coordinating mechanism*) (and set out in more detail in Chapter 5), market failure is, in neoclassical economics, a reason for the state to intervene. According to this analysis, the central economic policy task is now to internalize these external costs. This means that all costs and benefits are integrated into the price. The polluter-pays principle (that is, that the costs are borne by those who cause them) is to be restored by putting a price on emissions. The most important instruments include CO_2 taxes or emissions trading systems, in which CO_2 emissions receive a price and are traded via a market. Both approaches aim to raise the price of fossil fuels and thereby create incentives for lower-emission technologies, innovation, and behavioural change. Companies and households then decide for themselves how they reduce their emissions at the lowest cost. From a neoclassical perspective, market-based instruments therefore enable efficient climate protection at the lowest possible overall economic cost, since in this way we move to the optimal equilibrium point B. This approach requires the substitutability from weak sustainability and that it is possible to put a price on all costs and benefits (for example the loss of biodiversity), as explained before.

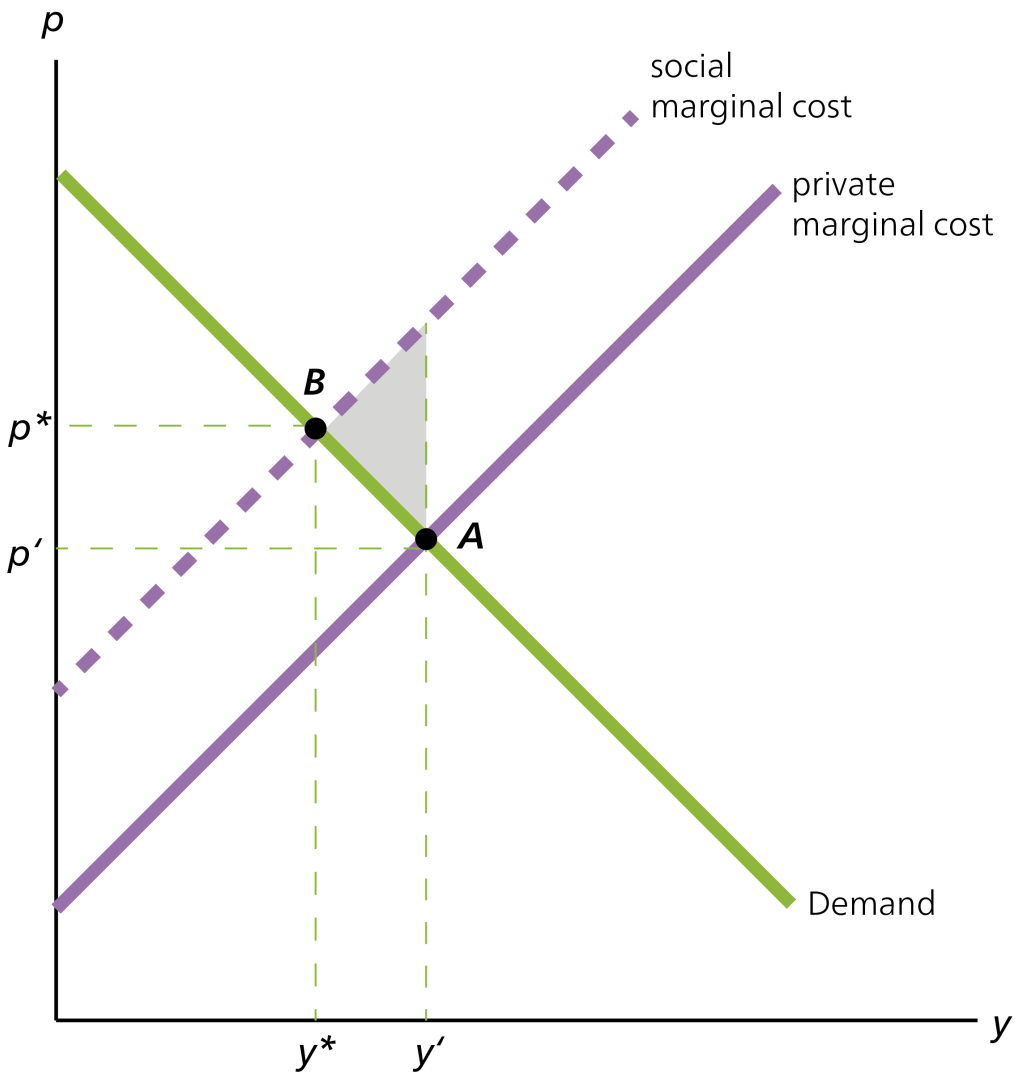


Figure 3.6: Welfare loss (grey triangle) with external costs. Source: Own illustration.

In this approach, fundamental structures are taken as given, and accordingly the fundamental economic order is not called into question. Markets continue to be regarded as a suitable mechanism for allocating resources; they merely have to be corrected where market failure occurs. Economic growth is accordingly not regarded as a problem. Rather, it is assumed that, with functioning markets, price signals, technological progress, and innovation can lead in the long term to a decoupling of economic growth and environmental pressure.

3.1.3.2 Climate change as systemic problem

Ecological economics does share the assessment that environmental costs are often externalized, but it interprets the climate crisis in a fundamentally different way. It regards climate change

not merely as a market failure but as a symptom of an economic system that is based on permanently rising material and energy throughput and thereby exceeds the planetary boundaries of the Earth.

The starting point here is the understanding, discussed several times, that the economy is not an independent system. Against this background, the climate crisis appears not as an isolated environmental problem but as an expression of fundamental tensions between economic development and the ecological carrying-capacity limits of the Earth system. Climate change is thus closely linked to other ecological crises – such as the loss of biodiversity, the overuse of natural resources, or the disruption of biogeochemical cycles. The various environmental problems are understood as interconnected symptoms of an economic system whose material and energy throughput is continually growing.

In contrast to neoclassical economics, ecological economics therefore does not understand environmental problems primarily as the result of missing market prices. The idea of negative externalities is not rejected, however. But these are not regarded as an exception in an otherwise functioning market; they are regarded as a structural feature of capitalist modes of production. Companies compete with one another and strive to keep production costs as low as possible. As long as a system is based on maximizing private profit, there is an incentive to shift costs onto people, society, and nature, that is, to externalize them (Kapp 1950; Saave and Jena 2022). The externalization of ecological costs is therefore not a random market failure but is understood as an inherent consequence of the growth and competition logic of capitalist economic systems.

Beyond this, ecological economics emphasizes the particular significance of economic growth. Historically, the growth of gross domestic product went hand in hand almost everywhere with rising consumption of energy and materials. Although the resource and energy intensity of many production processes has fallen, these efficiency gains were often overcompensated by the growth of production and consumption. For ecological economics, what is decisive is therefore not only how efficiently resources are used but whether absolute resource and energy consumption stays within the planetary boundaries.

In this context, it distinguishes between relative and absolute decoupling from economic growth. Relative decoupling is said to occur when resource consumption or emissions grow more slowly than gross domestic product. Absolute decoupling, by contrast, occurs only when environmental pressure actually falls despite continued economic growth (see Figure 3.7).

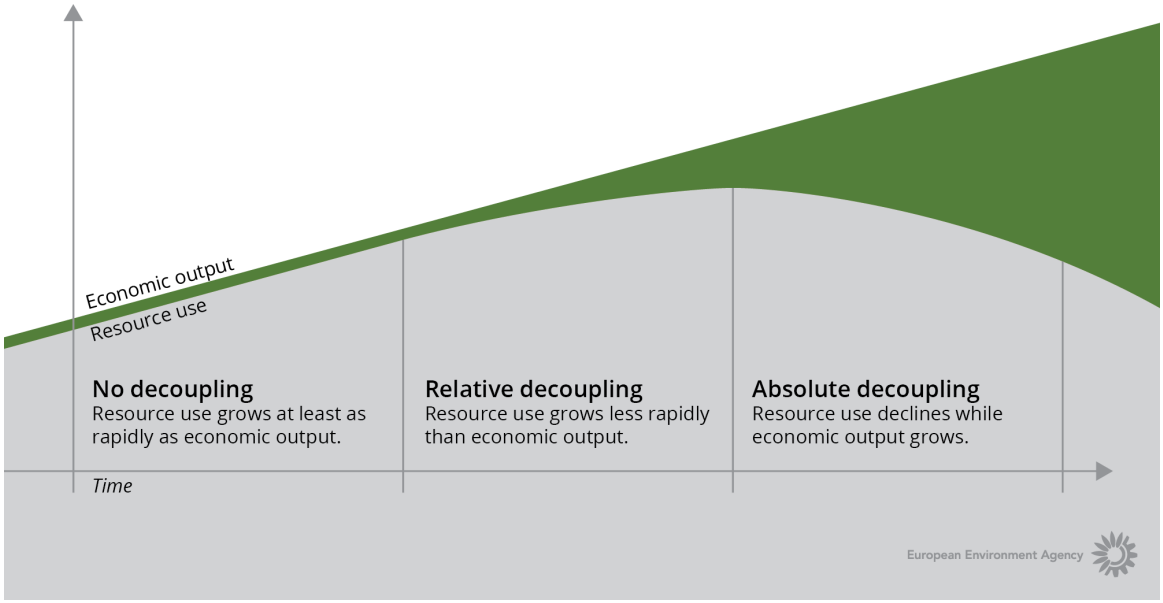


Figure 3.7: Relative and absolute decoupling; Source: EEA (2015).

While relative decoupling can be observed in many countries, proponents of ecological economics are sceptical about the possibility of lasting absolute decoupling on the scale required globally. Part of the observed emission reductions in industrialized countries is also based on the relocation of resource-intensive production to other regions of the world rather than on an actual reduction of global material and energy consumption. As shown in Figure 3.8, absolute decoupling can be observed in some countries of the Global North (although not all emissions are included). This concerns, however, not resource consumption but only CO_2 emissions, and it remains questionable whether this is possible at the global level. At the global level, no absolute decoupling of CO_2 emissions and economic growth can be observed.

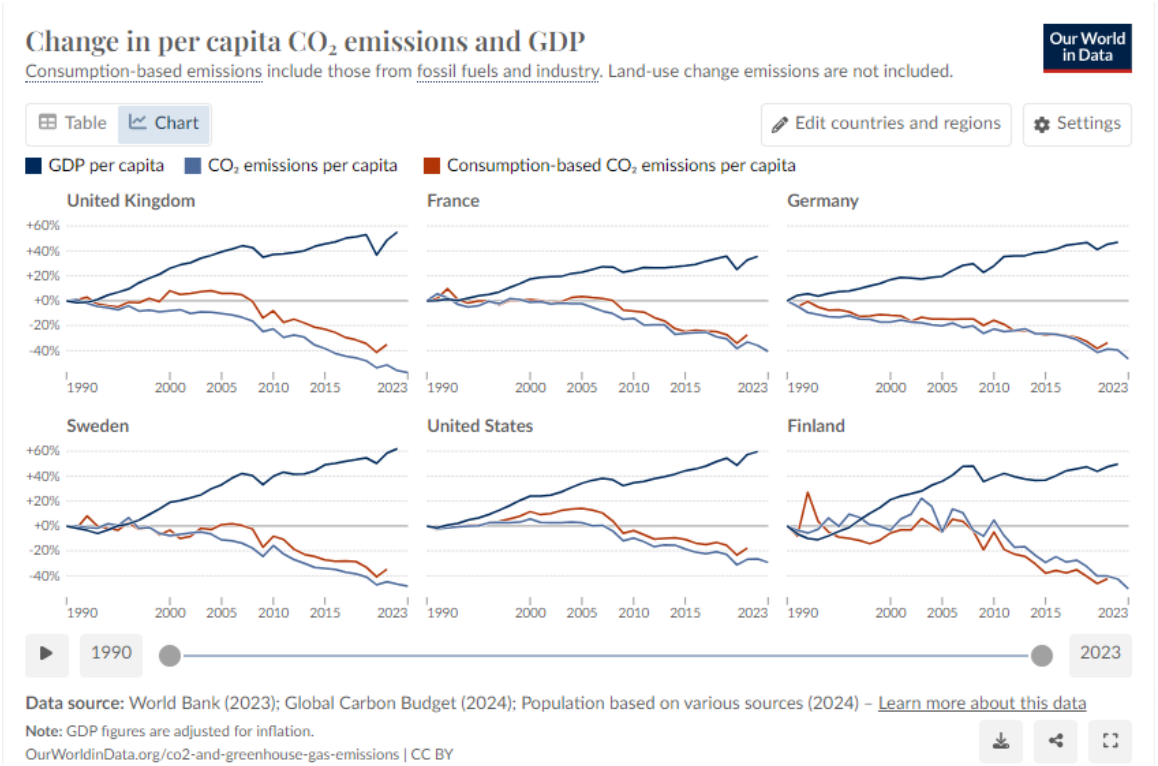


Figure 3.8: Absolute decoupling in selected industrialized countries. Source: Global Carbon Budget (2025); population figures based on various sources (2024) – with extensive data processing by Our World in Data.

Another important point is the discussion on *rebound effects*. Technological innovations often increase the efficiency of resource use but thereby often make products and services cheaper. Falling costs can in turn lead to greater demand, so that part of the originally expected environmental relief is lost again. Modern vehicles, for example, consume less fuel per kilometre than earlier models, but at the same time the lower costs can lead to larger vehicles being bought or more distance being travelled, which constitutes direct rebound effects. Indirect effects would occur here, for example, if the lower costs lead to more air travel on holidays, which can then also lead to higher energy consumption (see Figure 3.9). Energy-efficient household appliances or buildings, too, do not automatically lead to a correspondingly lower energy consumption. From the perspective of ecological economics, efficiency improvements are therefore indispensable, but they alone are not sufficient to reduce absolute resource consumption sufficiently.

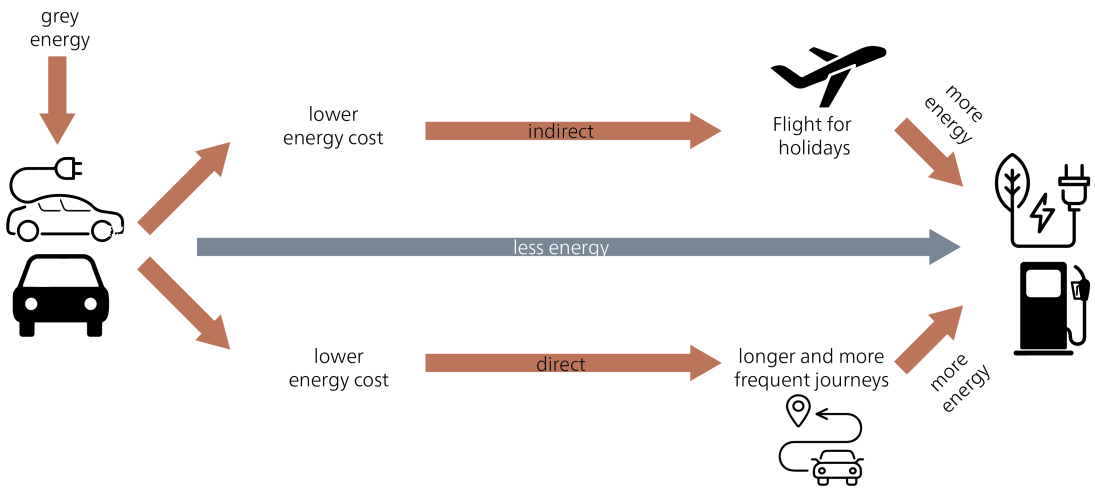


Figure 3.9: Indirect and direct rebound effects of energy-saving passenger cars. Source: Own illustration.

The problem analysis of ecological economics leads to different policy conclusions than in neoclassical economics. It is also important to emphasize, however, that the internalization of external costs through CO_2 prices or emissions trading is not rejected in principle; it is regarded merely as one building block of a more comprehensive transformation. Contrary to the neoclassical perspective, the economic system is fundamentally called into question here, and solutions also include fundamental transformations of the existing system. In Chapter 7, we will look more closely at various possible approaches that follow from an analysis of strong sustainability, but also at those that are more oriented towards a neoclassical understanding. In many sustainability debates, the differences lie at their core in these fundamental differences.

Excursus – Cost-benefit analysis

Cost-benefit analysis makes it possible to weigh costs against benefits. This instrument makes it possible to evaluate the expected consequences of decisions by weighing the associated costs and benefits. The important precondition of this analysis is the monetary valuation of all consequences. Only in this way can an evaluation be made with comparable quantities. The establishment of cost-benefit analysis goes back to the discussion of social costs (external costs) in the 1950s and 1960s. Based on the work of economists such as William Kapp (1950), it was argued that the social costs that arise should be prevented through regulation. Ronald Coase (1960), by contrast, argued that regulations, too, would constitute costs for companies: opportunity costs through forgone profit. For example, the profit of a chemical company that pollutes a river through its activity. This profit of the company would have to be offset against the social costs. According to Coase, such a total calculation is supposed to provide a basis for arguing against regulations and for extending property rights (that is, the market).
Cost-benefit analysis has several problems. One difficulty is that the value of many areas of societal and natural life cannot be clearly expressed in a price. It is not possible, for example, to put a clear and objective price on an endangered animal or plant species, the existence

of a forest, clean air, or a human life. To meet this challenge, researchers have developed various methods intended to make it possible to quantify monetary values. Two main procedures are important here: with **stated preference methods**, actors can be asked about different future scenarios. Previously unknown circumstances can thus be weighed against one another. In **revealed preference methods**, a utility valuation for the future is derived from decisions already made. This prevents actors from giving strategic answers when valuing expected future scenarios. However, conditions that have not occurred before cannot be valued. Although there are established procedures for valuing preferences, they are limited in their possibilities for weighing costs and benefits. As the term "preference" already suggests, these are always subjective views. The monetary valuation derived from them will therefore also always be subjective, that is, not an objective number.

Example:

At the beginning of this millennium, it emerged that the tobacco company Philip Morris had commissioned a cost-benefit analysis in the Czech Republic on the effects of smoking on the country. The authors concluded that smoking made an annual profit of about 5.8 billion crowns (about USD 150 million) for the Czech state budget. Because of the lower life expectancy of smokers, the state has to provide for them for a shorter time and therefore saves money. This report triggered a public outcry, whereupon the tobacco company apologized. But this example also shows the subjective nature of cost-benefit analysis. By reducing the value of human years of life to a number, the company had not only crossed a moral line. As a scientist later showed, the company had also made false assumptions and left out wider-reaching costs in the report. Her cost-benefit analysis of the same scenario concluded that smoking in the Czech Republic contributed to an annual deficit of about 14 billion crowns.

A further problem is that a cost-benefit analysis implies that the different types of capital are substitutable for one another, that is, that the loss of biodiversity, for example, can be technologically substituted. We discussed this problem in detail in this chapter.

Discounting – the time factor

The time factor is included in cost-benefit analysis by means of *discounting*. This means that costs incurred in the future are valued as smaller. The discount rate influences how strongly perceived costs decrease with the passage of time. Various decision bases can be used to determine the discount rate. Intergenerational justice, the decision between strong and weak sustainability, the uncertainty of future events, or the time preference of the decision-makers influence how strongly future costs are discounted.

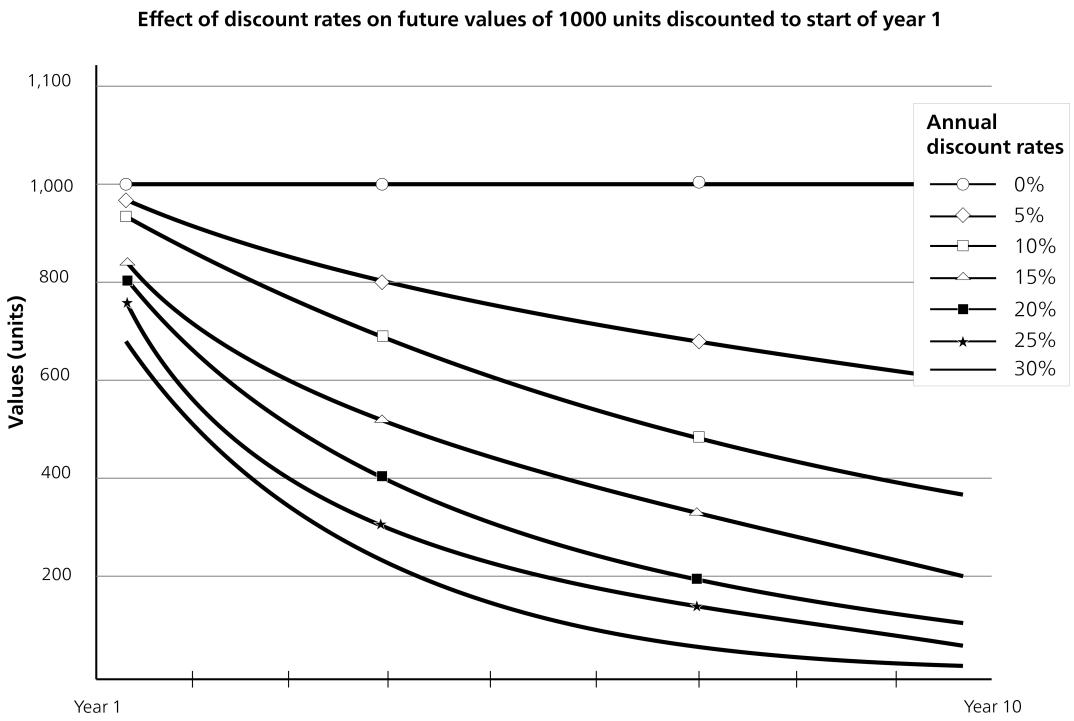


Figure 3.10: Illustration of different discount rates over time.

Figure 3.10 shows different discount rates over the same period. The costs of climate damage of CHF 1,000 occurring in 15 years, for example, are valued at about CHF 800 at a discount rate of 1.5%. At a rate of 5%, only about CHF 460 are included as perceived costs in the cost-benefit calculation.

Example:

In 2006, the British economist Nicholas Stern published the so-called Stern Report on behalf of the British government. In this comprehensive document, he describes climate change as the greatest market failure in history and argues by means of cost-benefit analysis that one per cent (later he argued for two per cent) of global GDP must be invested in emission reduction. Economists reacted to Stern's report in mixed ways. While some supported it, others argued that he calculated the costs of global warming incorrectly. The main argument was his low discount rate of 0.1 per cent, which he used to value future costs. The economist William Nordhaus took over Stern's calculations and changed only the discount rate from 0.1% to 3%. Accordingly, investments against global warming are still necessary but considerably lower than Stern demanded. In the logic of cost-benefit analysis, Nordhaus concludes that global warming in the optimal scenario rises to more than 3 degrees by 2100 (Nordhaus 2019). In this example, the question of the discount rate constitutes a clear ethical positioning. Stern chose the rate of 0.1%. He argues that future generations should not be given less weight because of their later point in time. The discount of 0.1 represents the low risk that humanity is wiped out by a catastrophic event. Nordhaus's discount rate, by contrast, weights the welfare of

coming generations by 3% less with each year. This is on the assumption that the economy grows continuously at this rate and future generations will accordingly be wealthier, and the costs will weigh less in the future.
The calculation of costs and benefits in the future thus depends on many assumptions and ethical questions. These are meant to be represented numerically in the discount rate.

Further reading

Buller, Adrienne. 2022. *The Value of a Whale: On the Illusions of Green Capitalism*. Manchester University Press.

3.2 Social Limits: Is the Pie Growing for Everyone – or Only for a Few?

When industrial production in the cities picked up speed in the early 19th century, industrialists needed workers en masse. They "found" them through increasing internal migration from rural areas. Through the advancing *enclosure* and privatization of the commons, a large part of the rural population increasingly lost its livelihood and had to migrate to the cities to earn a living there as industrial workers. Only in this way could industrial production and the capitalist market economy establish themselves as a new economic and social system. But the economic upswing by no means benefited all those involved equally: while factory owners attained unprecedented wealth, growing destitution prevailed in the new working-class districts. Countless reports testify to an unprecedented extent of suffering and inhumane conditions, both in the factories and in the overcrowded barracks of workers' families. Shifts of 14 to 16 hours without breaks, child labour, and appalling health risks were the bleak norm. But anyone who insisted on rights quickly risked their job and thereby pushing their family to the brink of survival from misery: there was no shortage of labour, and the hardship was great enough that any conditions had to be accepted.

But criticism of the new conditions, and in particular of human destitution, also manifested itself early on. Besides the first self-help associations such as consumer cooperatives, intellectual responses to the misery of industrial capitalism also increasingly formed. Various wings of early socialism named the problem of the dehumanizing conditions (for example, Robert Owen in England). The objections to the market system were mostly formulated from the perspective of the working class. People did not want to simply keep accepting the miserable situation in which wage workers and their families found themselves. A central endeavour, for example, was the organization of the workers in cooperative associations that organized both the production and the distribution of goods according to principles of a fair wage and concrete need.

In the middle of the 19th century, Karl Marx, a German living in industrial England, set about writing the most influential critique of capitalism to this day. Almost a century later, Karl Polanyi (1886-1964) finally described, from a historical perspective, human misery as a necessary consequence of a market society. Polanyi described the *market* of the time as a capitalist market that had slipped away from any social control and was in this sense "disembedded". He explained his observation by the inclusion of money, labour power, and land in the market-mediated process of price formation. In his conviction, only this gave rise to a system of market exchange detached from all normative

restrictions, which tends to destroy all social obligations, reliabilities, and notions of honour in the lifeworld. At the same time, according to Polanyi (2017), the complete *commodification*, and thus the creation of a pure market society, could nowhere fully prevail, since societies at a certain point take countermeasures to counteract, for example, the emerging misery and societal disembedding and alienation (Polanyi 2017). The trade union and cooperative movement establishing itself towards the end of the 19th and the beginning of the 20th century, but above all the emerging welfare state, can, in this reading, be understood as a counter-reaction inherent in simultaneously advancing capitalism. It has, to a certain extent, a double function of *limiting* it: to protect people from its consequences and, at the same time, to protect capitalism from itself.

3.2.1 Social limits in the 21st century

As Figure 3.1 under the ecological limits shows, in the 20th century not only Earth system trends but also socioeconomic ones accelerated strongly. Although social progress (or counter-movements) has noticeably reduced extreme poverty and misery in many industrialized countries, various social limits worldwide are at similar risk of being crossed as ecological ones. *Doughnut Economics*, a well-known book and framework of the same name by the economist Kate Raworth, accordingly supplements the nine "planetary boundaries" with twelve "social foundations" (Raworth 2017). These include, for example, food security, health care, and access to education. In a similar way, the 2030 Agenda, adopted by all UN member states, reflects in its 17 *Sustainable Development Goals* (SDGs) the continuing urgency of social questions as part of sustainable development.

Despite this supposedly broad acceptance of the goals, social limits are often less clear to draw than planetary boundaries and are accordingly discussed more controversially. As we will see, social questions are closely tied to ideas of justice and of a good life. The term "boundary" is therefore less obvious at first glance, especially in comparison with the biophysical limits of planetary ecology. On closer inspection, however, the notion of a social boundary, or a foundation that must not be fallen short of, makes perfect sense if we understand society and humanity as a whole as social systems whose functioning – similar to ecology – can be thrown out of balance. This can happen through social as well as primarily economic changes, which moreover often complement each other. The misery of workers in early capitalism, for example, was at first glance characterized by material poverty, but migration to the cities also meant being torn out of the familiar social environment with all its contacts and rhythms.

Today, too, scientists emphasize social limits that go along with the ever-recurring need to adapt to constantly new laws of capitalist economy and society. The German sociologist Hartmut Rosa describes modernization as a process of "social acceleration" (see, for example, Rosa and Celikates 2024; Rosa 2003). According to Rosa, alongside technological progress, **social change**, for example the change of social relationship patterns, and the **pace of life** also accelerate. These acceleration processes are driven by economic, cultural, and structural factors and mean that, despite constant technological progress and the associated efficiency gains, we increasingly feel short of time and stressed. For example, the enormous advances in communication technology have made it possible for us to communicate over long distances much more efficiently (for example emails instead of sending letters). Where we once wrote at most a few letters, the number of emails sent has increased massively, so that we fill the freed-up time, to a certain extent, with sending additional

emails, which in turn brings further obligations and changes relationship patterns. The possibilities and demands of being reachable have thus increased, for example. With technological progress, the possibilities of experiencing parts of the world also accelerate, for example through advances in transport and communication, but also in completing more and more tasks with the help of artificial intelligence. The potentially experienceable world grows faster than the share we can actually experience. Relatively speaking, the share of the world that we can experience therefore shrinks. Such acceleration dynamics have effects on the social well-being of individuals and societies and therefore cannot be disregarded.

A further social limit that increasingly occupies economists (although by no means all of them) is the growing inequality of income and wealth in many places and globally. But why can income and wealth inequality be described as a social limit? The epidemiologists Kate Pickett and Richard Wilkinson, for example, have shown that social, health, and environmental problems are considerably more pronounced in OECD countries with high inequality than in countries with low income inequality, as Figure 3.11 shows (Wilkinson and Pickett 2011; Wilkinson and Pickett 2024). With increasing inequality, for example, child mortality rises and life expectancy falls. According to Pickett and Wilkinson, the reason could be, for example, the higher stress caused by status competition. In addition, post-Keynesian economics emphasizes that distribution can also have important effects on the economy and that high inequality can have a negative effect on aggregate demand: if most people would like to spend more but cannot, while a few rich people save most of their income as wealth, less is consumed overall (see, for example, Kalaitzakis and Papatheodorou 2026). Furthermore, strong inequalities are also closely related to the ecological limits of the previous section: as many studies show, the rich have a disproportionately larger ecological footprint than everyone else (Wilkinson and Pickett 2024).

Because of these various and interwoven developments, many observers also see in spatial income inequality the explanation for the rise of populist political parties and the accompanying sociopolitical destabilization. These in turn are linked to one of the fundamental questions of social coexistence: that of justice. To get to the bottom of the complexity of inequality, we devote the rest of the chapter to this topic, by looking in turn at different concepts of justice and different economic perspectives on the types and causes of inequality.

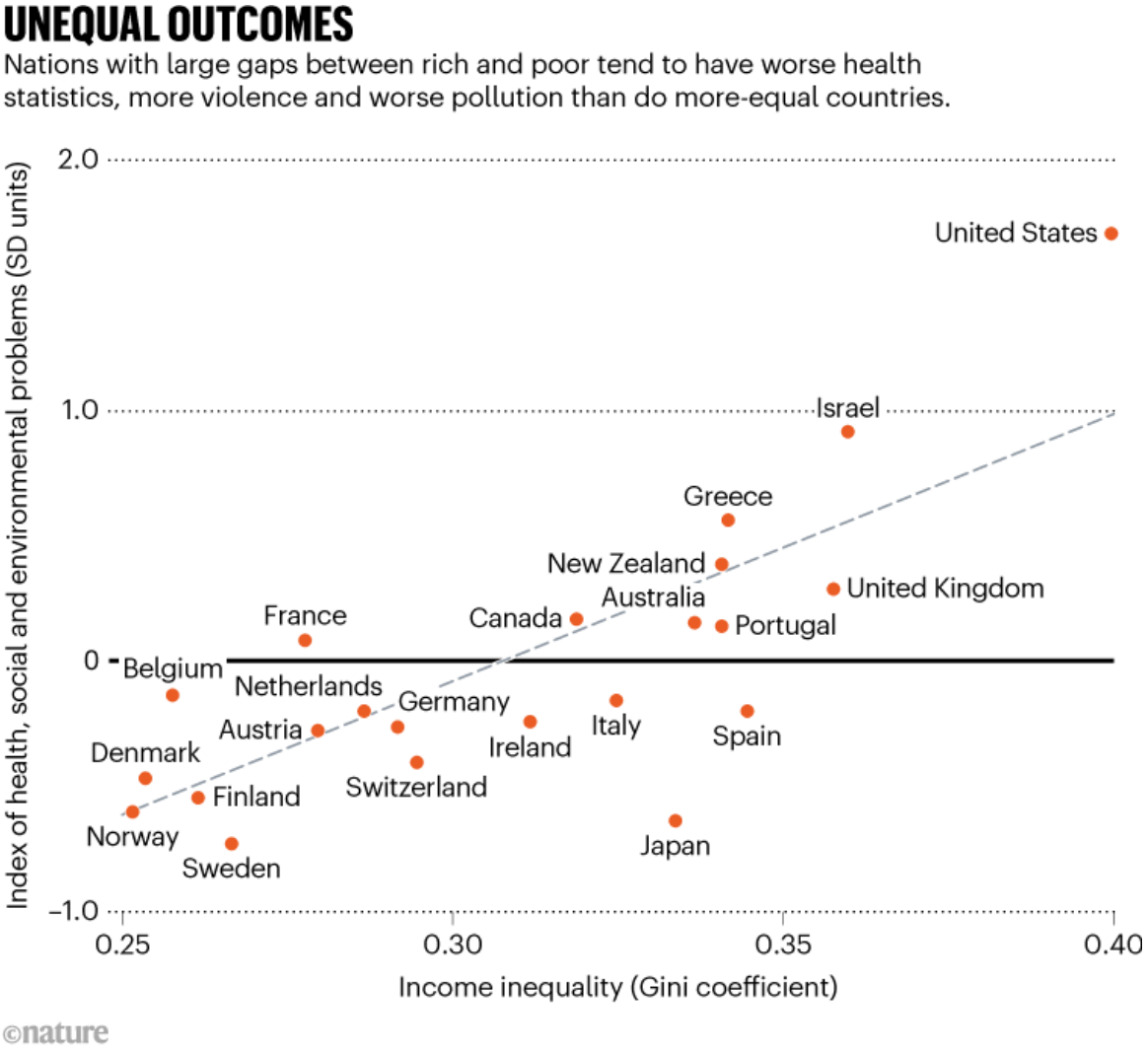


Figure 3.11: Income inequality and an index of health, social, and environmental problems. Source: Wilkinson and Pickett (2024).

3.2.2 Justice and inequality

Questions about justice and just distribution concern almost every social interaction in some way. While a "sense" of justice, much like the capacity for language, is part of the basic mental equipment of humans (although it can also be observed in non-human life), questions of justice have been in focus since the beginnings of Western philosophy. From Plato in ancient Greece to Martha Nussbaum in the 21st century, very different conceptions of what makes a just society have emerged. In what follows, we look at some approaches in more detail.

A good entry into the complex question of justice takes our intuitive sense of justice as its starting point. This is often based on a form of **reciprocity**, or mutuality. If I do you a big favour, we often find it fair or just if you return the favour at some point, even if possibly subject to our different

dispositions. A first differentiation is already hidden here: if we are on an equal footing or similarly disposed, one speaks of "balanced reciprocity", but if our positions differ fundamentally (for example rich/poor, king/subject), then justice requires "unbalanced reciprocity", for example: whoever has more gives more, but, depending on the case, also has more entitlement (Johnston 2011).

Although this reciprocity-based understanding of justice is still of great importance in social psychology today, it played hardly any role in the philosophy of justice for a long time – even though it actually stood at its very beginning. Plato formulates his definition of justice as a direct answer to such an intuitive understanding, which is put forward by Socrates' interlocutors (Plato's philosophy has been handed down to us largely through so-called "Socratic dialogues", that is, dialogues between Plato's teacher Socrates and other scholars). Plato's approach, later developed further by his student Aristotle, no longer defines justice through intuition but through a societal goal: what is just is what serves the good life, which is why one speaks here of a **teleological understanding of justice** (Greek *telos* = goal).

Although Plato and Aristotle do not clearly address many questions that are significant today, the step towards a goal-oriented (or consequentialist) concept of justice had major philosophical consequences. A more modern variant of teleological justice, **utilitarianism**, still represents an important philosophical current today, one that is widespread among many neoclassical economists in particular. At the core of utilitarianism is the search for the greatest possible overall societal benefit (*utility*), which is achieved by calculating and maximizing the total sum of positive and negative consequences. In utilitarian logic, it is therefore permissible, for example, to treat equals unequally, as long as the improvement (or utility gain) of one person is greater than the deterioration (or negative utility) of another. What counts is the total utility and thus also pure efficiency, or the maximum output that can be achieved with a given quantity of resources – regardless of its distribution and even if the poor become poorer and the already rich richer as a result. In this pure efficiency logic, however, the radicalism of the utilitarian position is also hidden.

A closely related but weakened variant of utilitarian logic is a definition of **justice as Pareto efficiency**. This maxim makes an important exception: while it retains the goal of maximizing overall societal utility, actions are accepted only if they make no one worse off. A state that is more efficient in itself thus counts as Pareto-superior only if no one is harmed at the same time. This restriction is also significant because, unlike utilitarianism, it also brings the well-being of the individual into focus: for us to define an action as just, we no longer need information only about the total sum of positive and negative utility but also about its distribution. This opens the door to theories of justice that are no longer derived from a societal goal but focus on the well-being of the individual and the ethics of an action as such (so-called deontological theories). As we saw in Chapter 2.5 (*Abstract Markets or Capitalist Production System*), this maxim forms the central reference point in neoclassical economics for judging whether a (market) outcome is efficient. Since this criterion can very rarely be met in reality, the Kaldor-Hicks criterion is also frequently applied, which states that a change is efficient for the economy as a whole if the winners could hypothetically compensate the losers for their losses and would still be better off afterwards. Whether the losers are then actually compensated for their losses often plays no central role. With this criterion, we are back at the utilitarian view.

Probably the most influential modern theory of justice was formulated in 1971 by the American philosopher **John Rawls**, drawing on his deontological predecessors such as Immanuel Kant. Rawls addressed the question of which rules the members of a society would agree on regarding the distribu-

tion of resources without knowing their own position, abilities, and conception of the good (Rawls 1999). Behind such a "veil of ignorance", Rawls argues, people would, on the one hand, agree on a system of equal basic liberties (such as respect for human dignity) and, on the other hand, accept inequalities only under two clear conditions:

> "first, they are to be attached to offices and positions open to all under conditions of fair equality of opportunity; and second, they are to be to the greatest benefit of the least advantaged members of society." (Rawls 1999)

The first condition is widely known – everyone should in principle get the same opportunities. The second condition, the difference principle, is less intuitive but also places well-founded limits on the equality of outcomes. Since behind a veil we would not know our abilities either (for example intelligence) – and thus whether we would really have equal opportunities in a world of pure equality of opportunity – we would accept inequalities only if they benefited the worst off the most. If we assume, for example, that inequality boosts economic growth, then this growth should benefit the poorest more than everyone else. A well-known phrase in the preamble to the Swiss Federal Constitution loosely takes up this thought: "that the strength of the people is measured by the well-being of its weakest members". Rawls's *A Theory of Justice* subsequently became a kind of core reference to which later theories of justice had to relate in order to distance themselves from it and develop it further (Rawls 1999). Important contributions to the debate criticized, on the one hand, the restriction of justice to questions of distribution. Nancy Fraser and others emphasized in particular the importance of fair processes, participation, and recognition as a dimension of justice that precedes and is independent of distribution (e.g. Fraser 1990).

On the other hand, the question gained in importance of what should actually be distributed justly – and, in subtle allusion to the ancient Greeks, what this has to do with the good life. Especially since the 1980s and 1990s, thinkers have engaged in various ways with the observation that human development and a life worth living can be measured only to a limited extent by material prosperity. Two of the most important voices in this debate are the economist and philosopher Amartya Sen and the philosopher Martha Nussbaum, whose work underlies the now widespread Human Development Index (HDI) (see Chapter 2.1.2, *Alternative indicators of prosperity*). Besides material prosperity, this index also captures life expectancy and the number of years of schooling, but it is based on Sen's and Nussbaum's concept of so-called "capabilities" (Nussbaum 2003; Sen 1999). Instead of a universal measure of development (such as material prosperity or resources), the capabilities approach puts people's own ideas of well-being at the centre, or rather each person's capabilities to realize these ideas. Since the resources required for this can differ greatly depending on the person and context, the focus on capabilities embodies a significant departure from concepts of justice and of the good life that rest exclusively on material prosperity. By moving non-material dimensions of well-being to the centre, such as the possibility of maintaining personal relationships or leading a long and satisfying life, the capabilities approach presents a slightly relativizing view of the dominance of economic realities and inequalities for human well-being.

As the Human Development Report published by the United Nations regularly shows, however, the HDI, too, is led by the materially richest countries. Regardless of which concept of prosperity and justice we use, it is therefore worth taking a closer look at the origins of economic inequality.

3.2.3 Types and causes of inequality

Inequalities in material prosperity, but also in the form of capabilities, have, on the one hand, grown historically, with global inequality in particular still being fundamentally shaped by colonial and postcolonial relations of domination. At the same time, these historically grown inequalities can also be observed "on a small scale": those who grow up with rich parents have a significantly higher chance of themselves being prosperous one day: access to education, hobbies, resources, networks, and so on explain a large part of intergenerational income inequality (Corak 2013). With regard to wealth inequality, the factor of inheritance also comes into play: those who start life with substantial assets usually not only keep them but can often count on equally substantial passive income (for example from renting out residential buildings) or on independent appreciation in value (for example if the residential buildings were built long ago in a location that is central today and could be sold for many times the price).

At the same time, every "historical" inequality is also to be understood as the result of current economic and political conditions. Whether wages, profits, inheritance, redistribution, or even expropriation are perceived as just ultimately depends on the prevailing ideas of justice and how these manifest themselves politically. Besides taxes and welfare-state questions, this concerns in particular various areas of economic policy and the underlying economic paradigms. But what do the questions about distribution and inequality refer to concretely? In economics, several important distinctions are made in the first place: between income and wealth, between primary and secondary distribution, and between national and global inequality:

- **Income and wealth**: The total product of an economy (as measured by GDP) can also be understood as the summed income of the factors of production, that is, wages for labour, ground rent or rent for land, and profits and interest on capital. The shares that these factors of production earn of total income, or how income is distributed within these groups, is called income distribution, and its unequal distribution is called income inequality. Distribution and inequality of wealth, in turn, refer to the already "distributed" ownership of accumulated property, understood as a result rather than a process, in the form of money or possession of land (such as a single-family house or agricultural land) and capital (such as tractors or company shares). Wealth stands in a twofold relationship to income: on the one hand, a person's wealth corresponds to the accumulation of their personal income, although inheritance, especially among very wealthy people, is usually at the origin of their riches. On the other hand, wealth opens up new sources of income when it is invested in capital or land. The phrase "letting your money work for you" refers to precisely this connection.
- **Primary and secondary distribution**: This concerns income exclusively: primary distribution refers to the direct distribution of total income to the factors of production (or to workers and owners of capital and land). Secondary distribution refers to the outcome of distribution after state intervention, in particular through redistribution via taxes, which is meant to correct an overly unequal primary distribution (for example by supporting the unemployed with state funds, financed by taxing everyone else). The division into primary and secondary distribution is intuitive but contested, since it hides the role of the state in primary distribution and presents the latter as a quasi natural market outcome. Instead of primary and secondary distribution, people therefore also speak of functional and personal distribution.

- **National and global inequality**: When inequality is talked about, it is often either about the unequal distribution of income or wealth within a country, but just as often about inequality between countries. This distinction is indispensable for a meaningful discussion and can lead to a lot of confusion. For example, we generally have an approximate understanding of poverty in countries of the Global South, but struggle to compare the life of a rich person there with that of a poor person living in Switzerland.

In what follows, we first turn to income inequality and in particular to primary distribution in order to explore the causes of inequality from an economic perspective, but we already do so with an eye to both national and international distribution.

3.2.4 Division of labour and comparative advantage

Besides the invisible hand, Adam Smith's pin factory is probably the best-known excerpt from his work. The first chapter deals with the division of labour in a pin factory and how it raises productivity to a great extent. Deriving from this, Adam Smith states more generally how productivity rises through the division of labour. Individuals, companies, or countries that specialize in an activity can perform it increasingly well. This creates an absolute advantage in production over others. Economic subjects increasingly specialize in those areas in which they are better than others. If a subject is superior to another in no area, it cannot take part in trade.

David Ricardo later criticizes Smith's theory of absolute advantages. If one economic subject were superior to another in all respects, then, according to Smith, it would also produce everything itself. Ricardo, by contrast, establishes the theory of comparative advantage. It states that subjects produce those goods for which they have the greatest *comparative advantage*, or the smallest relative disadvantage, over others. Here it still makes sense for subjects that are superior to others in no activity to participate in a shared economy. In doing so, they take on tasks for which others are better qualified but which those others do not prioritize because of a lack of resources.

Ricardo worked out his theory using the example of trade (cloth and wine) between England and Portugal and wanted to show in this way that free trade would be worthwhile for both sides. It was a plea against the then omnipresent import restrictions and tariffs between countries, which were based on mercantilist logic. This short video illustrates the basic idea behind Ricardo's theory of comparative cost advantages.

Ricardo is thus often drawn upon to illustrate that free trade is advantageous for all countries involved. This perspective, which is very widespread in neoclassical economics, should certainly be viewed critically, as many other schools of thought emphasize. For one thing, the notion of "free" trade quickly becomes relative in the context of European imperialism, which was emerging at the time when Ricardo developed his theory. In this period, the global market was shaped under the prevailing power relations primarily to the advantage of the Global North, was based on the exploitation of the Global South, and created unequal structures that in part persist to this day (see, for example, Anievas and Nişancıoğlu 2015; Davis 2017; Ghosh 2021; Hickel 2017; Patnaik and Patnaik 2021). In addition, advantageous participation in global trade requires a country to have competitive industries. Western Europe and the USA, too, built up their industries under the protection of the global market and with state industrial policy (Chang 2002). The forced further economic opening of many countries in the Global South from the end of the 1970s onwards by institutions such

as the International Monetary Fund, for example, made it harder to develop their own competitive manufacturing industries and pushed these countries back into the production and extraction of raw materials (see, e.g., Chang and Palma 2003). Dorninger et al. (2021) show how strongly this unequal exchange manifests and materializes in various aspects (for example land use and resource consumption) (see **?@fig-resourceuse**).

3.2.5 Power relations, labour share, and wealth inequality

The countries that benefited in this way from their industrial head start, while drawing on everyone else, also had to ensure at the national level that inequality stayed within (social) limits. In the middle of the 20th century, the American economist Simon Kuznets even put forward the hypothesis that in states with increasing economic growth, income inequality would first increase but would decrease again beyond a certain level. The so-called "Kuznets curve" arose in a time of flourishing social market economies in the 1950s and 1960s, when in the growing economies of the Global North **profits were broadly distributed to workers and the highest incomes were taxed strongly progressively**.

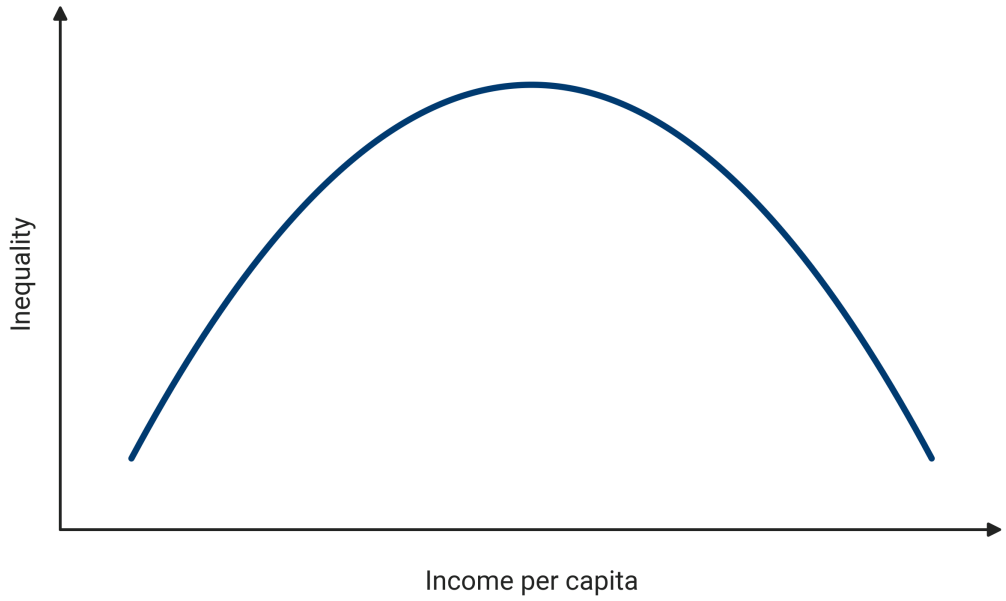


Figure created with the assistance of Claude Code (Anthropic) — transparency notice per EU AI Act Art. 50.

Figure 3.12: Kuznets curve. Source: Own illustration.

Excursus: The idea of the Kuznets curve persists stubbornly

Inequality in income and wealth – a question of belief?

In the following podcast (in German) by Swiss Radio and Television SRF, Graeme Maxton, economist and former Secretary General of the Club of Rome, and Samuel Rutz, economist and head of department at Avenir Suisse, discussed the extent and causes of inequality.

With the emergence of neoliberal reforms from the 1980s onwards, an increasing polarization of incomes became apparent strongly in the USA, but to a weaker degree also in Europe. Since then, the share of the richest 10% in total income has risen, while that of the bottom 50% has increasingly declined (see, for example, the World Inequality Database). Rather than a kind of law of nature, the causes of inequality are ultimately an interplay of existing economic conditions and political framework conditions. An important cause of rising income inequality is **the changing power relations between labour and capital in an opening world economy**. While a large part of financial capital can now travel around the world in fractions of a second at the click of a mouse, the mobility of employees has risen only comparatively little. Both institutional factors such as borders and social motives such as families or friendships slow down the factor of production labour. The compromise previously established between companies and employees organized in trade unions came under pressure in many places. As a consequence, the labour share, that is, the share of earned income in total economic output, is falling today in many countries, while the share of capital income is rising. While up to the 1970s clearly more than half of GDP was earned through labour, this share fell below the 50% mark in most countries in the 2000s. Besides economic globalization, this trend is also strongly shaped by technological development. In view of the rapid acceleration, in particular through the widespread application of artificial intelligence, no quick reversal of the trend is in sight (Dao et al. 2017). It is much more likely that income differences due to the division of labour will, over time, increasingly be overshadowed by differences between labour and capital income.

This in turn fuels the equally fast-growing **wealth inequality**. For however great the inequality of incomes is, it is overshadowed by the inequality of wealth – that is, the total of all goods a person is entitled to. In 2013, with his work *Capital in the Twenty-First Century*, the French economist Thomas Piketty showed how wealth inequality has historically grown again since the middle of the 20th century (Piketty and Goldhammer 2014). As the main driver behind this, he sees that capital incomes have grown faster than the real economy. The consequence is that rich people, who can invest in capital goods, gain access to wealth growth that remains closed to working people. Waves of political liberalization after the neoliberal turn of the 1980s also favour the increase in wealth inequality. Through low inheritance taxes or the privatization of state-owned companies, for example, wealth becomes concentrated among a few rich persons. The trend towards increased wealth inequality is reflected each year in even more extreme figures: as early as 2016, an Oxfam study (based on Credit Suisse data) noted that the richest one per cent held as much wealth as the rest of humanity combined – and that only 62 people owned as much as the poorer half of the world's population. In 2025, it took only 12 (Maitland et al. 2026).

3.2.6 National and global inequality

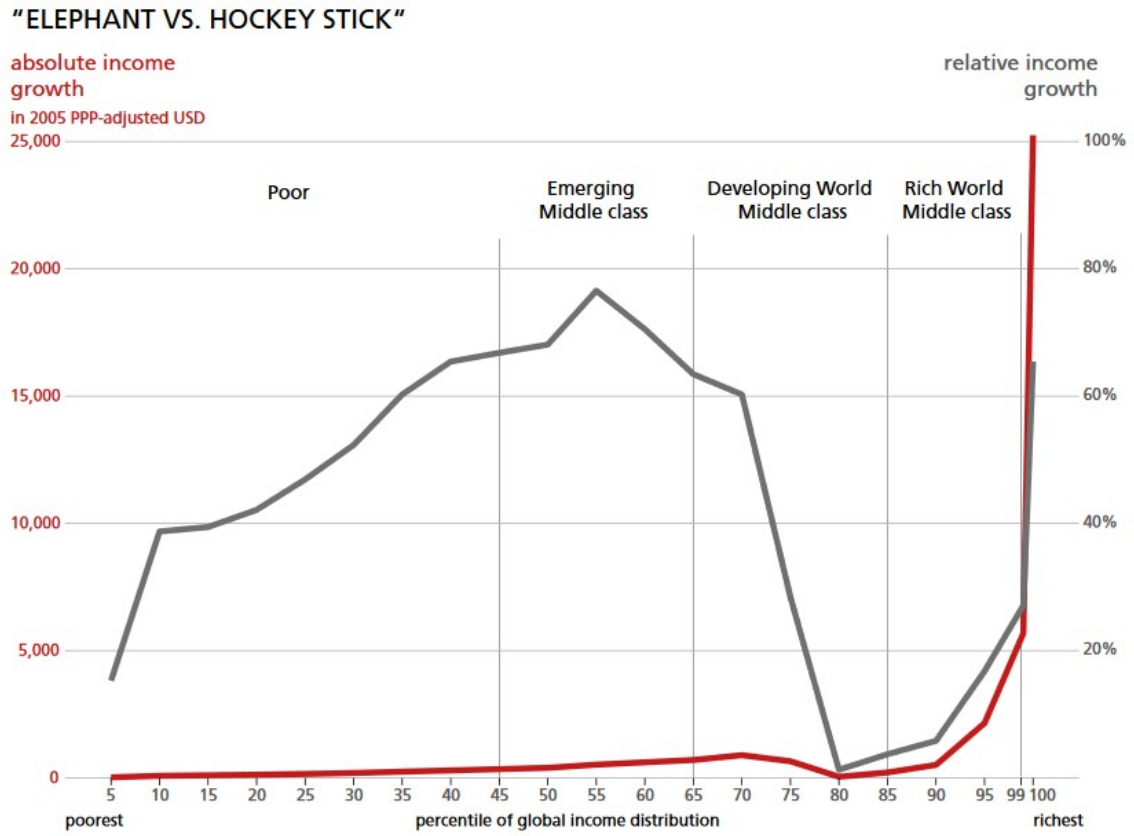


Figure 1: This chart offers a systemic perspective on income growth worldwide in US dollars (1988–2008), in which both absolute income growth ("hockey stick", red line) and *relative* income growth, i.e. percentage change in income ("elephant", black line), are displayed based on the *same* data source. (Data: Lackner & Milanovic 2013, 2016; Graph: C. Bader 2019)

Figure 3.13: Elephant or hockey stick? Source: Lannen et al. (2019).

At the global level, between 1988 and 2008, a picture emerges that is similar to that in high-income countries. The curve of **relative income growth (Figure 3.13 in black)** does show that incomes in the middle percentiles of global incomes have grown relatively strongly. This represents the rapid emergence of middle-income groups in countries such as China or Brazil. The lowest quantiles make clear how incomes in Africa and South Asia are growing only slowly. The trough between the richest 80 and 95 per cent depicts large parts of North America and Europe, where middle incomes have stagnated since the neoliberal reforms mentioned above. This depiction, known as "Milanovic's elephant", thus contains information on several phenomena at once. When looking at the **absolute income growth (Figure 3.13 in red)** – also referred to as the "hockey stick" – it becomes clear, however, how strongly the incomes of the richest people grew in this period (Green 2016). More than half of all income gains in this period went to the richest five per cent of the world's population. The poorer half of the world's population received only one tenth of the income.

While the globalization of the economy in recent decades (as well as the first globalization 150 years ago) is thus often portrayed in a positive light, and it is shown how free international trade leads to global prosperity, it also reveals new social limits. As the economist Branko Milanovic described more than 20 years ago, this **second face of globalization**, a face of exploitation and destruction (or colonialism), is overlooked (Milanovic 2003). Globalization is a phenomenon of enormous size and complexity and can accordingly show both its beautiful and its cruel face at the same time (depending on to whom the face reveals itself) (Milanovic 2003). Besides the fact that global capitalism has led to great growth in material prosperity in many places, an increase in inequality could also be observed, as explained above. Moreover, the economic system was based in many places on colonial and imperial violence. The structures still lead in many places to exploitation, dependencies, and unequal exchange (see, for example, Anievas and Nişancıoğlu 2015; Davis 2017; Ghosh 2021; Hickel 2017; Patnaik and Patnaik 2021).

Dorninger et al. (2021) show that high-income countries claim more resources, land, energy, and labour than they produce themselves. There is thus an **unequal exchange between the different groups of countries**. In addition, the globalized world economy and the corresponding institutions push many countries in the Global South into the production of raw materials (see, for example, Chang and Palma 2003). At the same time, many countries engage in more highly specialized industry and services, where more value can be captured. Global value chains are often controlled by companies in the Global North, which reinforces this unequal exchange and enables these companies to capture the bulk of the profit and shift the costs to the Global South (see, for example, Althouse et al. 2023; Durand and Milberg 2020; Ponte 2022). Such constellations increase the pressure on working conditions and favour exploitative relations in the Global South.

This is only a small and simplified excerpt from the complexity of the globalized economy. Many aspects are still insufficiently researched and in part contested. The point here is above all to show that, as Milanovic (2003) argues, globalization has at least two faces. Which one it shows has to be examined in each individual case.

Self-test Quiz

Test your understanding of this chapter with 9 short questions. The quiz is voluntary and ungraded, and you can repeat it as often as you like.

Question 3.1 (based on section 3.1.1)

Which two planetary boundaries are considered core boundaries, because they combine the processes of many other subsystems?

a) Ocean acidification and aerosol loading
b) Climate change and biosphere integrity
c) Land-use change and freshwater use
d) Stratospheric ozone and novel entities

Question 3.2 (based on section 3.1.2)

A government argues that extracting more oil is sustainable because the revenues are invested in schools and new technologies, so total capital is maintained. Which position does this reflect?

a) Strong sustainability
b) Weak sustainability
c) The precautionary principle
d) Sufficiency

Question 3.3 (based on section 3.1.2)

In ecological economics, the economy draws energy and raw materials from the ecosystem (source function) and releases waste and emissions back into it. What is this second function called? (One word.)

Answer: _____

Question 3.4 (based on section 3.1.3)

From a neoclassical perspective, why are CO_2 emissions too high?

a) Because firms are not aware of climate change
b) Because only private costs are reflected in the price, fossil-based goods are too cheap and the equilibrium quantity is too high
c) Because markets are generally unable to coordinate production
d) Because economic growth inevitably increases emissions

Question 3.5 (based on section 3.1.3)

GDP grows by 3% per year, while CO_2 emissions grow by 1% per year. What is this called?

a) Absolute decoupling
b) Relative decoupling
c) A rebound effect
d) No decoupling

Question 3.6 (based on section 3.1.3)

A household buys a fuel-efficient car and, because driving is now cheaper, drives longer distances. What is this an example of?

a) A direct rebound effect
b) An indirect rebound effect
c) Absolute decoupling
d) The sufficiency strategy

Question 3.7 (based on section 3.2.2)

A policy makes some people better off by CHF 100 in total and others worse off by CHF 40 in total. The losers are not compensated. How do the two efficiency criteria judge this policy?

a) It is a Pareto improvement and meets the Kaldor-Hicks criterion
b) It is not a Pareto improvement, but it meets the Kaldor-Hicks criterion

c) It is a Pareto improvement, but it does not meet the Kaldor-Hicks criterion
d) It meets neither criterion

Question 3.8 (based on section 3.2.2)

According to John Rawls's difference principle, when are social and economic inequalities acceptable?

a) When they result from free market exchange
b) When they maximize total utility in society
c) When they are to the greatest benefit of the least advantaged, and positions are open to all under fair equality of opportunity
d) Never, because justice requires equal outcomes

Question 3.9 (based on section 3.2.5)

According to the script, what is an important reason why the labour share of income has been falling in many countries?

a) Workers have become less productive
b) Power relations have shifted in favour of capital, because capital is highly mobile in an open world economy while labour is not
c) Wages have risen faster than profits since the 1980s
d) Most people now live from capital income

Answers and explanations

3.1 b). Climate change and biosphere integrity act at a supraregional level and integrate many other subsystems. Reaching tipping points in these two systems can push the entire Earth system into a new state. According to the 2025 Planetary Health Check, seven of the nine boundaries have been crossed.

3.2 b). Weak sustainability assumes that natural capital can in principle be substituted by manufactured or human capital, as long as total capital and total utility are maintained. This requires putting a monetary value on everything. Strong sustainability regards natural capital as only partly substitutable and relies on prevention rather than compensation.

3.3 "sink" / "sink function". The sink function. The economy is a subsystem of a finite global ecosystem and depends on both functions. Only solar energy enters the system from outside, while waste heat leaves it.

3.4 b). Climate change is a negative externality and therefore a market failure. The social marginal costs lie above the private ones, so the market equilibrium (A) produces more than the social optimum (B), causing a welfare loss. The policy answer is to internalize the costs, for example with a CO_2 tax or emissions trading. Option D reflects the ecological economics view.

3.5 b). Relative decoupling means emissions or resource use grow more slowly than GDP but still grow. Absolute decoupling would require emissions to fall while GDP grows. Absolute decoupling

of CO_2 can be observed in some countries of the Global North, but not at the global level, and not for resource consumption.

3.6 a). A direct rebound effect occurs when efficiency gains make the same service cheaper and people use more of it. If the household instead spent the savings on a holiday flight, that would be an indirect rebound effect. Rebound effects are why ecological economics sees efficiency as necessary but not sufficient.

3.7 b). A Pareto improvement requires that nobody is made worse off, which is not the case here. The Kaldor-Hicks criterion only requires that the winners could hypothetically compensate the losers and still be better off (100 > 40), whether or not compensation actually happens. That is why the script says Kaldor-Hicks brings us back to the utilitarian view.

3.8 c). Behind a "veil of ignorance", people who do not know their own position or abilities would accept inequalities only under these two conditions. Option B is utilitarianism, which accepts inequality whenever total utility rises, regardless of distribution.

3.9 b). Financial capital can move around the world in seconds, while borders, families and friendships keep workers in place. This weakened the post-war compromise between firms and trade unions. Technological change, including artificial intelligence, reinforces the trend. Until the 1970s, more than half of GDP was earned through labour; in the 2000s the share fell below 50% in most countries.

4 Strategies for a Sustainable Economy: Efficiency, Consistency, Sufficiency

> Guiding question
>
> In Chapter 3, we saw why our economy fails to achieve its goal – prosperity for all, within ecological limits. What fundamental levers are there to change this?

In view of the ecological limits of our planet, most highly industrialized countries share a broad scientific and political consensus that ecological limits (planetary boundaries) should be respected in the long term. The 1.5-degree and 2-degree targets, a central climate policy goal, have also been enshrined internationally in the Paris Agreement, which Switzerland has signed. With regard to economic and social limits, too, the global community has committed itself through the 2030 Agenda, and Switzerland specifically through the goals of the Sustainable Development Strategy 2030. These goals require a fundamental transformation of how we do business within a few decades.

This raises the decisive question: *how* exactly should this transformation succeed? The sustainability debate discusses three fundamental strategies: **efficiency**, **consistency**, and **sufficiency**. Practically all approaches to sustainable development build on these three strategies, but they differ in how they weigh their interplay and prioritization. Ultimately, achieving the sustainability goals probably requires all three working together:

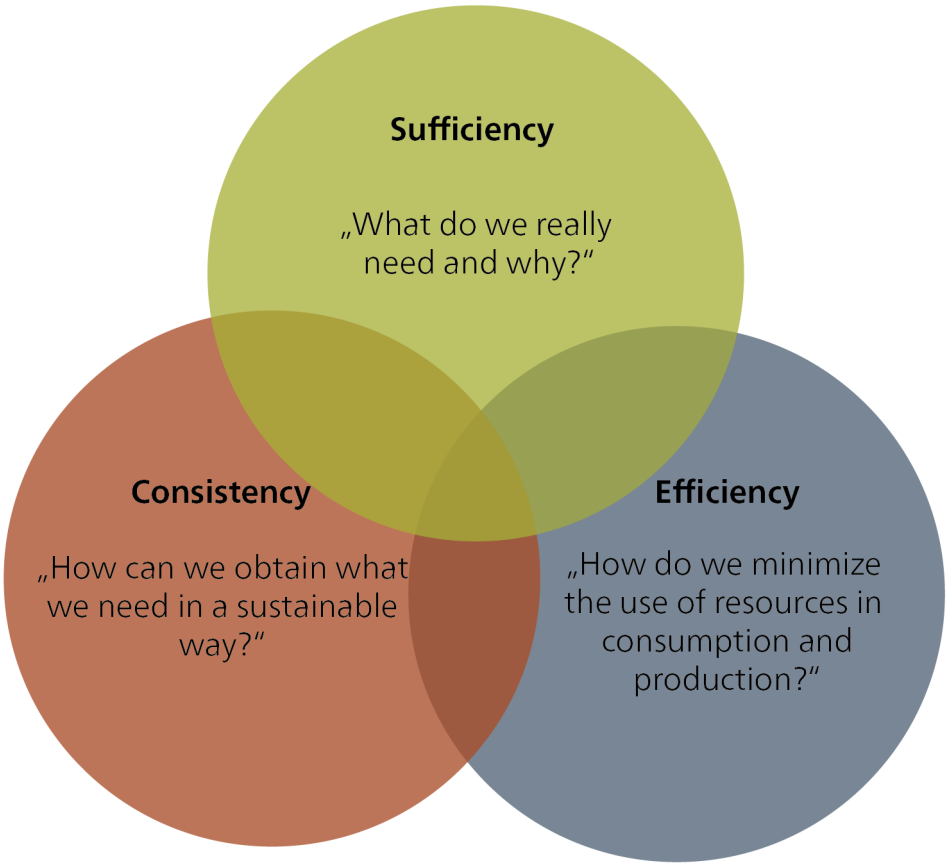


Figure 4.1: The three sustainability strategies. Source: Own illustration.

4.1 Efficiency Strategy

The efficiency strategy aims to use fewer resources (raw materials and energy) than before when manufacturing products or providing services, without reducing the quantity produced. This includes minimizing material consumption (material intensity), energy consumption (energy intensity), and the emission of harmful substances such as CO_2. This concept is often called **eco-efficiency**. Business and society regard it as promising because it can reduce costs, resource consumption, and environmental impacts. The efficiency strategy starts on the production side, where change is sought primarily through technological progress. Proponents of this approach believe it is possible to double prosperity while halving the use of nature (Weizsäcker et al. 1997). Critics of the efficiency strategy are less optimistic, however, and warn against overestimating its effects on sustainable economic activity. They point to so-called *rebound effects* (see Chapter 3.1.3, *Climate crisis – market failure or systemic problem*), which can cause the gains from efficiency improvements to be smaller or even cancelled out (Paech 2012; Santarius 2014).

Examples of the efficiency strategy

- **Energy-efficient lighting**: Replacing conventional incandescent bulbs with energy-efficient LED lamps reduces electricity consumption and helps to lower energy demand.
- **Fuel-efficient vehicles**: Developing hybrid or electric vehicles with improved fuel consumption reduces fuel demand and road-traffic emissions.
- **Efficient building technology**: Installing smart heating, ventilation, and air-conditioning systems in buildings helps to reduce the energy used for heating and cooling.
- **Efficient water use**: Using water-saving fittings and systems in households and industrial companies reduces water consumption and minimizes waste.

4.2 Consistency Strategy

The efficiency strategy is quantity-oriented – less resource use for more output. By contrast, the consistency strategy seeks compatibility between nature and technology, aiming to use resources repeatedly rather than consuming them only once. This implies replacing materials, products, and technologies that often rely on fossil resources with ones that match natural material cycles and operate in harmony with natural processes (Pufé 2017). This is often called **eco-effectiveness** and follows the "cradle to cradle" principle, in which products go from "cradle to cradle" instead of from "cradle to grave". The idea behind it is that in intelligent systems there is no waste, only products. This can happen in two ways: materials can either be biodegradable, as with shampoo free of synthetic ingredients, or they can be designed as "technical nutrients" that remain in the technical cycle. This means that a product at the end of its life does not end up in the bin but passes into a next cycle of use, for example by being upcycled. A computer casing, for instance, could be reused again and again or converted into a shelving system. The consistency strategy, too, starts on the production side. It is expected to offer great problem-solving potential, a wider reach, and more profound change than the efficiency strategy.

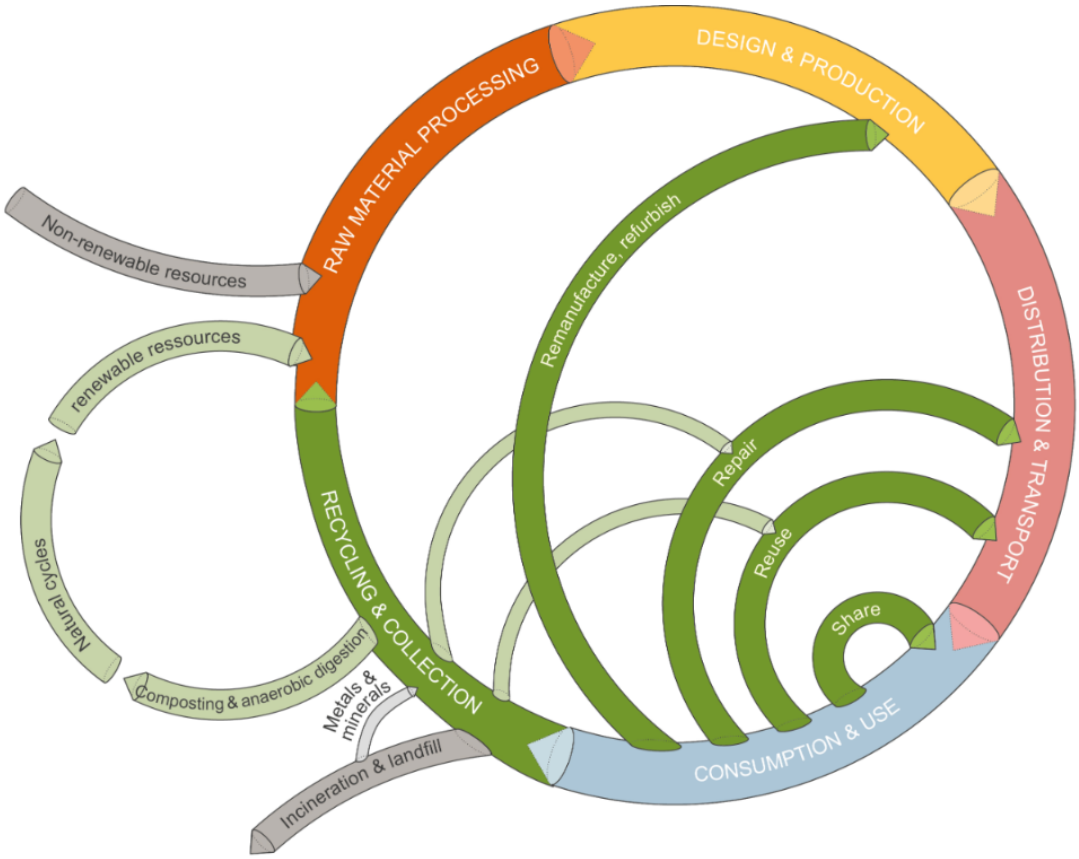


Figure 4.2: Schematic illustration of the circular economy. Source: BAFU (2019).

Nevertheless, the economy's material cycles cannot be achieved without losses of mass and energy, so absolute consistency remains an unattainable ideal. Even products that are 100% biodegradable consume energy in their production. Consistency is nonetheless understood as an impetus for industry to strive for this ideal and to reduce both resource consumption and emissions as far as possible.

Examples of the consistency strategy

- **Renewable energy systems**: The transition from fossil fuels to renewable energy sources such as solar, wind, and hydropower ensures that energy is produced in harmony with the natural rhythms and cycles of the environment.
- **Electronics designed for circularity**: Electronic devices are designed so that they can be easily repaired, upgraded, and recycled, in order to minimize resource consumption and environmental impact.
- **Sustainable construction**: Buildings are constructed using sustainable materials that have low environmental impacts and can be reused or recycled at the end of their service life.

4.3 Sufficiency Strategy

Sufficiency questions the level of need for a good life and aims to reduce resource and energy consumption through lower demand for resource-intensive goods and services. It is often also translated as moderation or appropriateness. Sufficiency takes a critical view of the new needs created by technology and advertising in relation to limited natural resources. It fosters an understanding that we need not "chase after" every newly created need, and that needs can be satisfied without consumption. Unlike efficiency and consistency, sufficiency focuses on the consumption side, though not exclusively. It can be practised to different degrees and at different levels, from smaller changes in behaviour (sharing rather than buying) to substantial changes in how we live (giving up air travel). Although it begins at the individual level, sufficiency can be applied at various levels, including companies (sufficiency-oriented product design) and governments (sufficiency policy). Sufficiency thus asks what the right measure is: How much do we need for a good life? And what do we not need? The sufficiency strategy is the subject of a passionate and intensive debate (Sedlacek 2012).

Critics consider the sufficiency strategy limited in its savings potential and its sociocultural resonance. They doubt that it can find broad acceptance. By contrast, proponents argue that the sufficiency strategy occupies an essential place in sustainability policy, especially where the efficiency and consistency strategies reach their limits.

Examples of the sufficiency strategy

- **Sharing and communal use**: Platforms and initiatives for sharing objects, tools, or vehicles make it possible to use resources more efficiently and to reduce the number of products manufactured.
- **Plant-based diets**: Switching to a mainly plant-based diet reduces the demand for resources such as water and land compared with meat production.
- **Working-time reduction**: Reducing working time can lead to lower resource consumption, since less energy and fewer materials are needed to produce goods and services.
- **Local consumption**: Supporting local producers and markets helps to reduce transport distances and emissions.

Further reading

To explore the sufficiency strategy in more depth, read the Denknetz article "The New Debate on Sufficiency" (in German).

Self-test Quiz

Test your understanding of this chapter with 7 short questions. The quiz is voluntary and ungraded, and you can repeat it as often as you like.

Question 4.1 (based on section 4.1–4.3)

Replacing incandescent bulbs with LED lamps is an example of which strategy?

a) Efficiency
b) Consistency
c) Sufficiency

Question 4.2 (based on section 4.1–4.3)

Switching from fossil fuels to solar, wind and hydropower is an example of which strategy?

a) Efficiency
b) Consistency
c) Sufficiency

Question 4.3 (based on section 4.1–4.3)

Reducing working time in order to lower production and resource use is an example of which strategy?

a) Efficiency
b) Consistency
c) Sufficiency

Question 4.4 (based on section 4.1–4.3)

Which strategy focuses mainly on the consumption side rather than the production side?

a) Efficiency
b) Consistency
c) Sufficiency

Question 4.5 (based on section 4.1)

What is the main criticism of relying on the efficiency strategy alone?

a) It is too expensive for companies
b) Rebound effects can reduce or even cancel out the savings from efficiency gains
c) It requires people to give up their standard of living
d) It only works for renewable energy

Question 4.6 (based on section 4.2)

With today's best technology, material cycles can be fully closed without any losses of mass or energy.

a) True
b) False

Question 4.7 (based on section 4 (Introduction))

What does the script conclude about the three strategies?

a) Efficiency alone will be sufficient if technology advances fast enough
b) Consistency should replace the other two strategies

c) Achieving the sustainability goals probably requires all three strategies working together
d) Sufficiency is the only strategy that really works

Answers and explanations

4.1 a). Efficiency means using fewer resources and less energy for the same output. LEDs provide the same light with less electricity. The strategy starts on the production side and relies mainly on technological progress.

4.2 b). Consistency (eco-effectiveness) seeks compatibility between technology and natural cycles rather than simply using less. Renewable energy works with natural rhythms instead of consuming fossil stocks. Electronics designed for repair and recycling ("cradle to cradle") are another example.

4.3 c). Sufficiency asks how much we need for a good life and aims to reduce demand for resource-intensive goods and services. It can be practised by individuals (sharing instead of buying), companies (sufficiency-oriented product design) and governments (sufficiency policy, such as working-time reduction).

4.4 c). Efficiency and consistency both start on the production side. Sufficiency focuses on consumption and needs, though not exclusively.

4.5 b). Efficiency gains often make goods and services cheaper, which can increase demand (see Chapter 3.1.3). Critics therefore warn against overestimating the effect of efficiency on absolute resource use.

4.6 False. False. Material cycles always involve losses of mass and energy, and even fully biodegradable products need energy to produce. Absolute consistency is an unattainable ideal, but it still serves as a target for reducing resource use and emissions as far as possible.

4.7 c). Practically all approaches to sustainable development build on the three strategies, but they weigh and prioritize them differently. The script's conclusion is that the goals will probably only be achieved if all three work together.

5 Role of the State and Policy Paradigm for Sustainability

Guiding question

How can a society ensure that its economy actually achieves the goal we took as our starting point in Chapter 2 – prosperity for all, within ecological limits?

We have covered a lot of ground in this script. In Chapter 1, we saw that there is no single economics, but rather different schools of thought, each with its own basic assumptions. In Chapter 2, we examined prosperity as the actual goal of economic activity. We asked what is used to provide this prosperity (factors of production, goods) and how this provision is coordinated. In Chapter 3, we saw that today's economy misses this goal in several respects, and in Chapter 4, we encountered three strategies for addressing these problems: efficiency, consistency, and sufficiency. This closes the circle with the original question from Chapter 2: How must an economy be coordinated so that prosperity for all, within ecological limits, is actually achieved?

This brings us to the question of implementation: Who actually ensures that we get there? The market alone, or do we need the state? This question cannot be answered independently of what we have already seen. Depending on one's understanding of the economy, markets, and capitalism, the answer on the role of the state differs as well – from an intervention limited to market failure to a state that actively shapes the economy or even sets limits.

These different ideas about the state and the market do not simply stand side by side in isolation. In practice, they condense into larger, coherent **economic policy paradigms** – that is, overall pictures of how the state, the market, and society should interact in order to achieve the shared goal. In this course, following Novy et al. (2023), we distinguish three basic economic policy paradigms that can be observed today as the result of the influence of different schools of thought: **market liberalism**, **welfare capitalism**, and **post-growth**. Each of them can be traced back to the styles of thought that accompany us throughout the script.

5.1 Why "Market vs. State" Is Historically Not a Given



Figure 5.1: SRF report, 15 March 2023. Translation: "Housing is becoming scarcer. State promotion of housing construction or more free market? There are many recipes against the looming housing shortage. One thing seems clear: something must be done soon." Source: https://www.srf.ch/news/schweiz/wohnungen-werden-knapper-staatliche-wohnbaufoerderung-oder-mehr-freier-markt.

That we can speak at all of the state "intervening in the market" is not a universal historical truth. For people who lived before the emergence of modern capitalism, this formulation would simply have made no sense. In this chapter, we explore the specific role of the state in capitalism and how the actions of state institutions in relation to the economy can be explained.

As we saw in the introductory chapter, it was only with the transition from feudalism to industrial capitalism that the economy came to be thought of as a supposedly separate sphere, detached from political power. Ellen Meiksins Wood (2017) shows that under feudalism, economic and political rule were inseparably linked: the nobility appropriated part of the serfs' production through direct political coercion. Wage labour was a marginal phenomenon at the time, and production was shaped by craft workshops, agriculture, and subsistence production. In the transition to industrial capitalism and emerging liberalism, the state came to be understood as the sphere of power, of its distribution and limitation, and increasingly of democratic participation. In contrast stood civil society, which was understood as a part of society separate from the state, protected by rights, and characterized by the individual, rational pursuit of economic interests. The economy was thus conceived, in particular through the right to private property, as a separate space protected from the state and understood as largely free of power relations.

When we talk today about the right form of the state, we usually mean the liberal parliamentary democracy, in which elections take place but which also protects private property from state interference. The economy is thus not an area in which democratic procedures have a place. When we go to the supermarket, we do not face goods that we have agreed on as a society, but goods that the supermarket chain considers profitable. When we go to work, we do not expect to have a say in production goals and the distribution of profits, but accept our employer's decisions in return for a wage. Democracy thus takes place in the political sphere through elections, but in everyday life, which is shaped above all by wage labour and the purchase of goods, the possibility of exerting democratic influence on our living conditions is limited (Anderson 2019).

That we can meaningfully speak of "state interventions in the market" today is therefore itself the product of a specific historical and political constellation, and hence not a neutral starting point. This insight is central to the following sections: if the separation between market and state is historically made, then the question of *how* the state and the market should relate to each other is not a purely technical one but always also a political one.

5.2 Market-Liberal Paradigm: The State as Night-Watchman

In the most widely used economics textbook (Mankiw and Taylor 2020), the role of the state is defined through four concepts: property rights, market power, market failure, and externalities. The market is considered fundamentally efficient, and deviations require justification. This gives rise to a certain circular reasoning, however: if the market leads to poor outcomes, the market itself is never at fault, but rather incomplete property rights or incomplete markets. Accordingly, new markets must be created instead of questioning the market as an organizing principle. This very circular reasoning is the starting point for Samuel Bowles's critique (see below).

David Harvey summarizes the prevailing basic idea of the role of the state from a neoclassical perspective as follows:

> "State interventions in markets (once they have been created) must be kept to an absolute minimum, because according to the theory the state cannot possibly have enough information to anticipate market signals (prices)" (translated from Harvey 2005, 3)

Today, this understanding is often referred to by the term *night-watchman state*. In contrast, critics such as Michel Foucault and Quinn Slobodian point out that enforcing, maintaining, and managing markets by no means results in a weak state but requires a strong one (Slobodian 2018).

We call the role of the state described here on the basis of a neoclassical understanding – limited to guaranteeing private ownership of the means of production (property rights), preventing monopolies (market power), and limited interventions to correct market outcomes (market failure and externalities) – the **market-liberal paradigm**.

5.2.1 Critique of the market-liberal paradigm: markets are not neutral – they constitute

An important critique of the market-liberal paradigm and the role it assigns to the state arises from the observation that its concept of market failure is often too narrow. The economist Samuel Bowles, for instance, argues that markets not only allocate resources and distribute income but also **shape culture, preferences, and power structures**. They are thus "as much political and cultural institutions as economic ones" (Herzog et al. 2014, 471). Four building blocks support this argument:

- **Endogenous vs. exogenous participants.** The abstract textbook market, as we discussed it in Chapter 2.5, assumes exogenous, unchanging preferences. Once participants are *endogenous* (the market itself affects their abilities, values, and wishes), allocation in markets no longer works as efficiently as described in the model, because the model relies on precisely this exogeneity assumption (Herzog et al. 2014, 481).

- **Contested markets.** Where claims are enforced endogenously (by the participants themselves rather than by third parties), a power relationship arises between the side facing greater scarcity and the other side, "even if the exchange takes place under full information, without coercion, and under fully competitive conditions" (Herzog et al. 2014, 478).
- **Incomplete contracts as the rule, not the exception.** Because contracts can never cover all contingencies, political or social mechanisms are needed to fill the gaps. This is precisely why markets are political.
- **Constitutive exchange processes.** "When friends give each other gifts, gifts make friends" (Herzog et al. 2014, 475) – how work is organized and distributed shapes the development of our abilities and norms. This means that markets are constitutive.

If we follow Bowles's arguments, market failure is not a rare exception but is structurally present almost everywhere outside the abstract model of the (market-liberal) textbook.

5.3 Welfare-Capitalist Paradigm: The State as Stabilizer and Co-Shaper

Another widely recognized critique of the market-liberal paradigm comes from the economist John Maynard Keynes, who formulated it in the aftermath of the Great Depression (from 1929). A central insight of Keynes is that market economies generate uncertainty and instability out of their own logics. Markets, and in particular aggregate demand, do not stabilize themselves but are always prone to crises, since long-term, rational decisions are hardly possible in an environment marked by strong uncertainty. In times of crisis, uncertainty about the future can lead to underinvestment and reduced consumption, and thus to a suboptimal use of societal resources. According to Keynesian theory, the state therefore has the role of stabilizing aggregate demand through countercyclical action (that is, if necessary, also by increasing its spending in times of crisis through debt financing). In principle, the state should also stabilize societal investment, for example through public investment programmes.

> "The important thing for government is not to do things which individuals are doing already, and to do them a little better or a little better; but to do those things which at present are not done at all." Keynes (1978, 291) John Maynard Keynes (1926) in "The End of Laissez-Faire" (**keynesEndLaissezFaireEconomic2020?**)

The role of the state expands compared with that in neoclassical economics. Based on Keynes's ideas, the state has the task of securing and protecting material prosperity, for example by building a welfare state. In this way, competitive economic measures are to be combined with social justice and, in green variants, also with environmental protection. We describe this understanding of the role of the state as the **welfare-capitalist paradigm**.

5.3.1 Critique of the welfare-capitalist paradigm: from "correcting" to "shaping"

Taking up Bowles's observation, the economist Mariana Mazzucato argues that the corrective function of the state, as Keynes justified it, will ultimately not be sufficient: if markets have a constitutive

effect, correction is not enough – the state must proactively help to shape markets ("market shaping, not fixing"). Mazzucato (2022, 2018) turns this into a government programme: market failure should be treated not as an exception but as the rule. Her work rests on the historical observation that many groundbreaking technologies were, at least initially, made possible only by (often massive) state investment. The most famous example is the internet, whose precursor, Arpanet, was developed as a military technology. She also points to countless civilian applications of innovations that were originally developed with large amounts of public money for NASA's space programmes. In other words, had states not made the first move and made the investments in core technologies that were far too expensive and risky for private funders, many later innovations building on them could not have been developed either. Such public upfront financing could accordingly be understood as "anticipatory" correction of market failure, but it would be more accurate to speak instead of active shaping and co-design.

Mazzucato and Bowles thus both ascribe a much more active and present role to the state than a neoclassical (or welfare-capitalist) perspective does. Bornemann et al. (2018) describe three different roles that a state can take on in societal transformation: (1) **regulator**, with a focus on limiting and preserving; (2) innovator and exnovator, with a focus on **shaping** markets; and (3) **enabler**, for example of experiments and social innovations for more sustainable patterns of production and consumption.

The different state interventions during the COVID-19 pandemic illustrate the different ideas about the role and function of a state. While in Switzerland COVID loans were granted almost entirely without conditions (for example, Swiss International Air Lines was granted a loan of billions of francs without conditions such as a ban on dividend payments in the years following the pandemic), other countries attached conditions to COVID loans: Denmark did not grant rescue loans to companies in tax havens, and France tied its loans to Air France to climate conditions (for example, fewer domestic flights, although this was not legally binding).

5.4 Post-Growth Paradigm: The State as Guardian of Limits – or Part of the Problem?

The main goal of the post-growth approach is a fulfilling life in a sustainable society that relates to nature in a harmonious way by overcoming the compulsion for constant economic growth in the capitalist system. It aims at an economy that remains in a stable state (*steady-state economy*) or can function independently of economic growth and that ensures people's basic provision, in particular through the principle of sufficiency (the right amount of resource use, not the maximum).

Ecological economics understands the economy as an embedded subsystem of the Earth rather than as an isolated circuit between households and firms. This also shifts the question of the role of the state: the primary concern is not to correct individual market failures but to keep the entire economic throughput within the carrying capacity of the planet (Rockström et al. 2009). Herman Daly (2017), a founding figure of ecological economics, derives from this an explicitly state-centred institutional design for a "steady-state economy". It involves three instruments that require a strong, shaping state, because no market can give them to itself:

- Cap-auction-trade systems: The state sets a physical upper limit for resource extraction and pollutant emissions (the "cap"), auctions usage rights, and only within this limit leaves the allocation to the market.
- Limits on the distribution of income and wealth (minimum and maximum incomes), so that capped resource use does not simply lead to sharper distributional competition over a fixed "slice of the cake".
- Population policy instruments to stabilize resource demand.

The market remains desirable as an allocation mechanism within the limits, but only the state can legitimately set and enforce the limit itself. This logic coincides with the functions of a "shaping state" discussed above, and also with Gabor and Braun's ((Schmelzer and Vetter 2019)) "big green state", who distinguish four models of state-led ecological transformation. These models range between two poles: (1) **carbon shock therapy**: forcing the transformation through a very high CO_2 price commensurate with the climate crisis, an example of fully market-based steering; and (2) **big green state**: a massively interventionist state with its own stakes in the economy that engages in ecological planning and forces private investors to act ecologically.

5.4.1 Critique of the post-growth paradigm: the state as ideal collective capitalist

However far-reaching the toolkit of ecological economics may be, it shares with the welfare-capitalist paradigm a basic assumption that, from the perspective of Marxist political economy (MPE), is the real problem: that the state is in principle a neutral body acting in the common good, which only needs to be equipped with the right competences and the right mandate. MPE, by contrast, understands the state decidedly not as neutral but as an instrument of the respective ruling class: "The modern state, whatever its form, is [...] the ideal collective capitalist" (Engels 1975, p. 260). The state does not act in the interest of individual fractions of capital but in the interest of maintaining capitalism as a whole, which may well include social policy or co-determination, as long as private ownership of the means of production is not called into question.

For the role of the state in post-growth, this means, among other things, that even a "big green state" remains structurally tied to the reproduction of the capital relation. A cap-auction-trade system, for example, auctions usage rights to the same actors who previously profited from overuse, and the proceeds flow back to the same state, which remains structurally bound to this very capitalist production through growth, employment, and tax revenue. The state can administer limits – but, according to MPE, it can hardly dissolve growth dependencies from within, because it is itself part of them (growth imperative via the tax system and social insurance, legitimation via employment).

MPE therefore looks more consistently for the alternative outside the market and the (capitalist) state: in commons, cooperatives, and collective self-management.

5.5 Beyond Market and State

With the commons, cooperatives, and forms of collective self-management that MPE puts forward as an alternative, a path beyond the alternative of "market or state" is already hinted at. At the same time, a plural view that places the different perspectives presented side by side makes clear that what the state does and may do is the result of political and societal negotiation processes. If – as we saw in Chapter 5.1 (*Why "Market vs. State" Is Historically Not a Given*) – even the separation between market and state is itself historically produced and by no means a given, then new questions arise all the more urgently: What does democratic and common-good-oriented control of economic processes look like in concrete terms? Who decides, and in what way, what is produced, how much, and how? How can market concentration and abuse be prevented and a long-term orientation towards the common good be ensured?

In this chapter, we have become acquainted with three economic policy paradigms – market liberalism, welfare capitalism, and post-growth – that interpret the role of the state differently. They draw on the scientific concepts of the different styles of thought but in turn also shape the scientific debate itself – after all, as discussed at the outset, economics too is not free of normative influences.

In the in-depth block, we orient ourselves towards these paradigms and the styles of thought underlying them, with a focus on the post-growth paradigm and, to a lesser extent, on green growth approaches within the welfare-capitalist paradigm. The following overview shows how the styles of thought feed into the three paradigms. This assignment is not set in stone, however: it can vary depending on one's standpoint, and some schools of thought cannot be clearly assigned to one of the three paradigms.

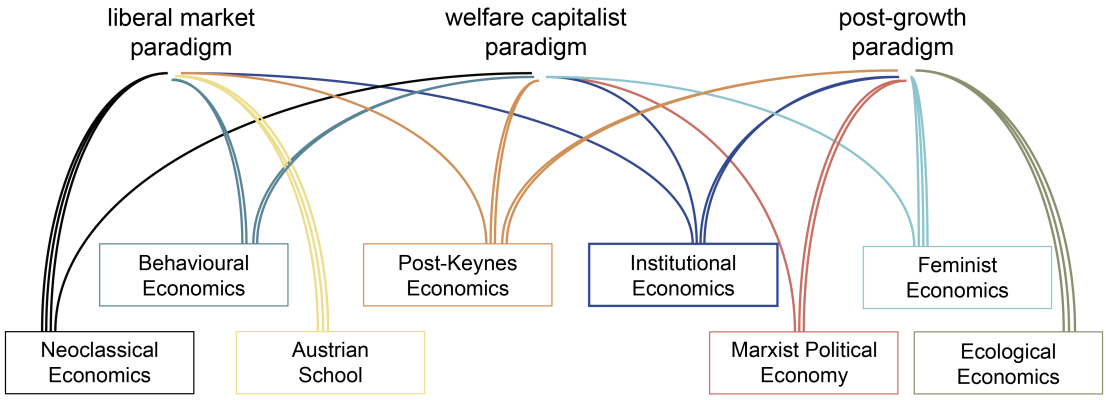


Figure 5.2: Economic policy paradigms and schools of thought. Source: Own illustration based on the illustrations of Exploring Economics.

Self-test Quiz

Test your understanding of this chapter with 8 short questions. The quiz is voluntary and ungraded, and you can repeat it as often as you like.

Question 5.1 (based on section 5.1)

Why does the script say that speaking of "state intervention in the market" is not a historical given?

a) Because markets did not exist before the 19th century
b) Because under feudalism economic and political rule were inseparably linked; the economy came to be seen as a separate sphere only with industrial capitalism and liberalism
c) Because states have always controlled all economic activity
d) Because the term was invented by Keynes in the 1930s

Question 5.2 (based on section 5.2)

In the market-liberal paradigm, which set of tasks is assigned to the state?

a) Stabilizing aggregate demand and building a welfare state
b) Setting physical limits on resource use and limiting income inequality
c) Guaranteeing property rights, preventing market power, and correcting market failure and externalities
d) Actively shaping markets through mission-oriented investment

Question 5.3 (based on section 5.2)

Michel Foucault and Quinn Slobodian criticize the idea of the "night-watchman state". What is their point?

a) The state should own all means of production
b) Enforcing, maintaining and managing markets requires a strong state, not a weak one
c) Markets can function entirely without a state
d) The night-watchman state is the most efficient form of government

Question 5.4 (based on section 5.2.1)

What is the core of Samuel Bowles's critique of the market-liberal paradigm?

a) Markets are inefficient because firms lack information about prices
b) Markets not only allocate resources but also shape culture, preferences and power, so market failure is structural rather than an exception
c) Markets work perfectly as long as contracts are complete
d) Markets should be replaced by central planning

Question 5.5 (based on section 5.3)

According to Keynes, what should the state do in an economic crisis?

a) Cut spending to balance the budget
b) Stay out, because markets return to equilibrium on their own
c) Stabilize aggregate demand countercyclically, if necessary by increasing debt-financed spending
d) Nationalize all banks

Question 5.6 (based on section 5.3.1)

What does Mariana Mazzucato mean by "market shaping, not fixing"?

a) The state should only correct market failures once they occur
b) The state should actively co-shape markets, for example through risky early investment in core technologies such as the forerunner of the internet
c) The state should withdraw from markets entirely
d) Markets should be fixed in size so they cannot grow

Question 5.7 (based on section 5.4)

How does a cap-auction-trade system, as proposed by Herman Daly, work?

a) The market sets the cap, and the state auctions the remaining resources
b) The state sets a physical upper limit for resource use and emissions, auctions the usage rights, and leaves allocation to the market only within this limit
c) Firms trade emission rights freely without any upper limit
d) The state distributes resources directly to households without a market

Question 5.8 (based on section 5.4.1)

Why does Marxist political economy doubt that even a "big green state" can overcome the growth imperative?

a) Because the state lacks the technical knowledge to plan the economy
b) Because the state is not neutral: as the "ideal collective capitalist" it depends on growth via tax revenue, employment and social insurance
c) Because green technologies are too expensive
d) Because post-growth policies are unpopular with voters

Answers and explanations

5.1 b). Ellen Meiksins Wood shows that under feudalism the nobility appropriated production through direct political coercion. Only with industrial capitalism was the economy conceived as a sphere protected from the state, especially through private property. If the separation is historically made, how state and market should relate is always also a political question.

5.2 c). These four concepts define the role of the state in the most widely used economics textbook (Mankiw). The market is considered efficient, and deviations require justification. Option A is the welfare-capitalist paradigm, option B the post-growth paradigm, and option D Mazzucato's shaping state.

5.3 b). Even the market-liberal paradigm depends on a state that creates and protects markets, property rights and competition. So the opposition between "free market" and "strong state" is misleading.

5.4 b). Bowles argues that markets are "as much political and cultural institutions as economic ones". Participants are endogenous (markets change their preferences and abilities), many markets

are contested (power relations arise), contracts are always incomplete, and exchange is constitutive ("gifts make friends").

5.5 c). Keynes argued that market economies generate uncertainty and instability from their own logic. In crises, uncertainty leads to underinvestment and lower consumption, so the state should stabilize demand and investment. This idea underlies the welfare-capitalist paradigm.

5.6 b). Many groundbreaking technologies (Arpanet, innovations from NASA's space programmes) were first made possible by massive public investment that was too expensive and risky for private funders. Mazzucato therefore treats market failure as the rule, not the exception, and calls for a proactive, shaping state.

5.7 b). In Daly's steady-state economy, the market remains useful as an allocation mechanism, but only the state can legitimately set and enforce the limit itself. Daly combines this with limits on income and wealth inequality and with population policy.

5.8 b). MPE sees the state as acting in the interest of maintaining capitalism as a whole. A cap-auction-trade system, for example, auctions usage rights to the same actors who profited from overuse. MPE therefore looks for alternatives outside market and capitalist state: commons, co-operatives and collective self-management.

6 Sustainable Economics: A Synthesis

Guiding question

How can we build a coherent answer to our initial question from everything we have learned so far – how can we, as a society, achieve prosperity within ecological limits?

At the end of Chapter 1, we announced that this script follows the **sustainable economics** approach – and deliberately left open what that means in concrete terms. We noted at the time that sustainable economics does not start with a fixed answer ("more green growth" or "shrinkage") but with the **goals** of economic activity itself. Ever since, we have pursued precisely this question – what the goal of economic activity actually is – throughout the script.

In this final chapter, we bring these threads together. Sustainable economics is not a further, fourth economic policy paradigm alongside market liberalism, welfare capitalism, and post-growth. Rather, it is a framework that derives what is needed from the goals themselves, and it therefore cuts across the economic policy paradigms discussed so far. We now make this approach concrete along three goals – global ecological justice, a good life for all, and growth independence – and six core elements. Together, they allow us to give a coherent answer to the initial question of this script: how must an economy be designed so that it actually achieves its real purpose?

Drawing on Schmelzer and Vetter (2019) and Rogall (2012), sustainable economics can be captured through three goals:

- **Global ecological justice**: This includes environmental and climate justice, as well as debates on post-extractivism, that is, the critique of resource-intensive logics of extraction and export (Acosta 2017), and post-development approaches that question the Western development model as Eurocentric and normative (Klapeer et al. 2023).
- **A good life for all**: not growth as an end in itself, but an economy that meets real needs.
- **Growth independence**: The ability of an economy to fulfil its core functions (employment, social security, public finances) without structurally depending on continuous GDP growth.

These three goals are deliberately not built on a logic of growth or shrinkage. They leave open whether and where growth takes place, as long as the three goals are achieved. This is precisely where their bridging function lies: sustainable economics does not ask "growth: yes or no?" but *"what is needed to make ecological justice and a good life for all possible, regardless of whether the economy grows, stagnates, or shrinks in the process?"*

6.1 Six Core Elements

Following Rogall (2012) and Novy et al. (2020; 2023), the approach can be made concrete in six core elements:

1. **Strong rather than weak sustainability**: Natural capital can be substituted by manufactured or human capital only to a limited extent.
2. **Pluralist approach**: Sustainable economics does not draw on a single economic theory. Instead, it integrates ecological, institutional, feminist, and post-Keynesian perspectives.
3. **Sustainability paradigm rather than growth paradigm**: This is the real controversy in the growth debate: How can the growth paradigm be replaced by a sustainability paradigm? The answer is "selective growth": sectors that contribute to sustainability goals (such as renewable energy, care, and education) can grow, while others (such as fossil fuels and armaments) must shrink. Growth thus turns from a goal into a dependent variable.
4. **Ethical principles**: Precaution rather than remediation, and intra- and intergenerational justice rather than pure Pareto efficiency. The concern is not only efficiency goals but explicitly also questions of distribution. Responsibility is understood in two ways here: as consumer sovereignty (responsibility for oneself) and as responsibility for the environment and the more-than-human world.
5. **Changing the framework conditions**: political and legal instruments such as standard-price approaches or cap-and-trade, merit goods, and a sustainable economic policy, for example financing work rather than unemployment, reducing working time, and promoting the regional economy.
6. **Inter- and transdisciplinary collaboration**: The complexity of the challenges requires collaboration across disciplinary boundaries, between science, politics, and society.

> Further reading
>
> If you are interested, you can watch the statement by Paolo Gentiloni (European Commissioner for Economy) at the Beyond Growth conference to illustrate the green growth position. The post-growth position is well illustrated in a talk by Timothée Parrique at the same conference.

7 Approaches to Sustainable Economics

Guiding question

What concrete approaches are there for achieving the goals of sustainable economics, and how do they differ?

As outlined in Chapters 5 and 6, the debates around sustainable economics are shaped by different economic policy paradigms, which draw on various schools of thought within pluralist economics and differ in their understanding of sustainability. Most approaches within sustainable economics can be assigned to the post-growth paradigm. Approaches such as green growth and the circular economy can be derived from the welfare-capitalist paradigm. The market-liberal paradigm lies largely outside the debate on sustainable economics, although certain aspects of it are discussed under green growth, such as the internalization of external effects.

In what follows, we engage more deeply with specific approaches. Since the debates within sustainable economics are very diverse, no definitive or fixed classification of the approaches is possible. Nevertheless, we distinguish between several approaches here. While there are many overlaps among them, in our view they can still be told apart by their distinct emphases. This classification is intended as an orientation. For the approaches within the post-growth paradigm, we follow Schmelzer and Vetter (2023).

Learning outcomes

By the end of this chapter, students…

- are able to distinguish between the main approaches of sustainable economics.
- know the key characteristics, solutions, and concepts of one approach and are able to explain it using examples.

7.1 Approaches of Sustainable Economics

Exercise – Approaches of sustainable economics

Below you will find a list of different approaches within sustainable economics and corresponding material for each approach. Please choose two of the following approaches and work through the material for each of them.
Use the following guiding questions to work through each of the two approaches you have chosen:

- What goals and visions are pursued in this approach?
- How can the degree of sustainability of this approach be described (weak or strong sustainability)?
- What causes of the overuse of natural resources does this approach identify?
- What are the strengths, weaknesses, and blind spots of this approach?

In the online version, the material for each approach is embedded directly (videos and texts). Here you find the links. Texts on ILIAS are only accessible with a UniBE login.

7.1.1 Welfare-capitalist paradigm

7.1.1.1 Green growth

The report by Petschow et al. (2020) serves as an introduction and provides an overview of green growth. A video excerpt from a contribution by Michael Jacobs to the panel "Focus Panel 1 – Which Prosperous Future? Confronting Narratives of Growth" at the Beyond Growth Conference 2023 adds a further perspective. If you are interested, you can watch the full panel discussion.

- Text (ILIAS): https://ilias.unibe.ch/go/file/3512925/download
- Video excerpt: https://tobira.unibe.ch/!v/B96EQrQRZHm
- Full panel discussion (optional): https://www.youtube.com/watch?v=KqtlQcU3PaQ

7.1.1.2 Circular economy

A short video serves as a brief introduction. The text by Lazarevic and Brandão (2020) provides a more in-depth introduction to the topic. If you are interested, you can read the text by Bocken et al. (2022) as an additional resource.

- Video: https://www.youtube.com/watch?v=NBEvJwTxs4w
- Lazarevic and Brandão (2020) (ILIAS): https://ilias.unibe.ch/go/file/3512927/download
- Bocken et al. (2022), optional (ILIAS): https://ilias.unibe.ch/go/file/3512928/download

7.1.2 Post-growth paradigm

7.1.2.1 Growth-independent institutions

The paper by Corlet Walker et al. (2024) presents this approach using the example of the UK welfare system. A video discusses further how the UK welfare system could be made more independent of growth.

- Corlet Walker et al. (2024) (ILIAS): https://ilias.unibe.ch/go/file/3512933/download
- Video: https://www.youtube.com/watch?v=gK4iuguEDgQ

7.1.2.2 Sufficient lifestyles

A video by Niko Paech, a prominent proponent of this approach, provides an introduction to its main ideas. The text by Juliet Schor (2008), a pioneer of research on consumerism in the US, discusses this approach further. The text is relatively old but provides a good overview of the central issues. Especially with regard to working-time reduction, there has been a lot of new research since then, which you are encouraged to explore if you are interested. A second video complements the text.

- Video by Niko Paech: https://www.youtube.com/watch?v=rS3ldLZ_kYE
- Schor (2008) (ILIAS): https://ilias.unibe.ch/go/file/3512937/download
- Second video: https://www.youtube.com/watch?v=H3eSFIu3DoE

7.1.2.3 Collective action

The paper by David Bollier (2015) conveys the core idea of such approaches and gives a good overview. A video on commoning the city discusses more concrete ideas of what this approach might look like.

- Bollier (2015) (ILIAS): https://ilias.unibe.ch/go/file/3512934/download
- Video on commoning the city: https://www.youtube.com/watch?v=qCmJcPkpI98
- Further videos: https://www.youtube.com/watch?v=4MKu_jB-UkM and https://www.youtube.com/watch?v=gydkOrGrs8Q

7.1.2.4 Overcoming the imperial mode of living

The text by the I.L.A. Kollektiv provides a concise introduction to the idea of this approach. A video taken from a talk by Ulrich Brand, one of the main proponents of this approach, discusses it further.

- I.L.A. Kollektiv (ILIAS): https://ilias.unibe.ch/go/file/3512940/download
- Video by Ulrich Brand: https://www.youtube.com/watch?v=4UiT5JoHPHc

References

Acosta, Alberto. 2017. 'Post-Extractivism: From Discourse to Practice—Reflections for Action'. *International Development Policy | Revue Internationale de Politique de Développement*, no. 9: 77–101. https://doi.org/10/ggh26b.

Althouse, Jeffrey, Louison Cahen-Fourot, Bruno Carballa-Smichowski, Cédric Durand, and Steven Knauss. 2023. 'Ecologically Unequal Exchange and Uneven Development Patterns Along Global Value Chains'. *World Development* 170: 106308. https://doi.org/10.1016/j.worlddev.2023.106308.

Anievas, Alexander, and Kerem Nişancıoğlu. 2015. *How the West came to rule: the geopolitical origins of capitalism*. Pluto Press.

Atkinson, Anthony B. 2009. 'Economics as a Moral Science'. *Economica* 76: 791–804. https://doi.org/10.1111/j.1468-0335.2009.00788.x.

BAFU. 2019. 'Kreislaufwirtschaft'. In *Bundesamt Für Umwelt (BAFU)*. https://www.bafu.admin.ch/de/kreislaufwirtschaft.

Bäuerle, Lukas, Anna Katharina Keil, Rouven Reinke, and Stella Wasenitz, eds. 2026. *Grundfragen Der Ökonomie*. 1. Auflage. Schäffer-Poeschel Verlag.

Baumgartner, A. Doris, Beat Fux, and Peter Streckeisen. 2019. 'Von der unsichtbaren Prekarität zur Beschäftigung ohne Qualität. Politische Programmatik und das Streben nach kognitiver Hegemonie'. In *Sozialstaat unter Zugzwang?: Zwischen Reform und radikaler Neuorientierung*. Springer Fachmedien Wiesbaden. http://link.springer.com/10.1007/978-3-658-22444-8.

Beckert, Sven, and Helmut Dierlamm. 2026. *Kapitalismus: Geschichte einer Weltrevolution*. 3. Auflage, deutsche Erstausgabe. Rowohlt.

Benelli, Natalie, and Peter Streckeisen. 2014. 'Weniger Wert? Temporär Beschäftigte in Der Schweiz'.

Bernet, Brigitta. 2021. 'Arbeit Anders Denken'. In *Sozialalmanach 2021: Armut Grenzt Aus*. Caritas.

BFS. 2026. *Landesindex Der Konsumentenpreise (LIK)*. Bundesamt für Statistik (BFS). www.LIK.bfs.admin.ch.

Binswanger, Mathias. 2019. *Der Wachstumszwang: warum die Volkswirtschaft immer weiterwachsen muss, selbst wenn wir genug haben*. Wiley-VCH Verlag GmbH & Co. KGaA.

Bontrup, Heinz-J., and Ralf-M. Marquardt. 2021. *Volkswirtschaftslehre Aus Orthodoxer Und Heterodoxer Sicht*. De Gruyter Oldenbourg.

Ceasar, L., B. Sakschweski, L. S. Andersen, et al. 2024. *Planetary Health Check Report 2024*. Potsdam Institute for Climate Impact Research.

Chang, Ha-Joon. 2002. *Kicking Away the Ladder: Development Strategy in Historical Perspective*. 1st ed. Anthem Press.

Chang, Ha-Joon, and Ha-Joon Chang. 2014. 'Life, the Universe and Everything. What Is Economics?' In *Economics: The User's Guide*. Bloomsbury Press.

Chang, Ha-Joon, and Jose Gabriel Palma. 2003. 'The Latin American Economies During the Second Half of the Twentieth Century - from the Age Og 'ISI' to the Age of 'the End of History". In *Rethinking Development Economics*. Anthem Press.

Coase, Ronald H. 1960. 'The Problem of Social Cost'. *The Journal of Law & Economics* 3: 1–44.

Corak, Miles. 2013. 'Income Inequality, Equality of Opportunity, and Intergenerational Mobility'. *Journal of Economic Perspectives* 27 (3): 79–102. https://doi.org/10.1257/jep.27.3.79.

D'Alisa, Giacoma, Giorgos Kallis, and Federico Demaria. 2016. 'Nachwort: Von Der Austerität Zur Dépense'. In *Degrowth: Handbuch Für Eine Neue Ära*, edited by Giacoma D'Alisa, Federico Demaria, and Giorgios Kallis. Oekom Verlag.

Daly, Herman. 2017. 'A New Economics for Our Full World'. *Victor, P., Dolter, B., Handbook of Growth and Sustainability. Cheltenham: Edward Elgar*, 85–106.

Dao, Mai, Mitali Das, Zsoka Koczan, and Weicheng Lian. 2017. 'Why Is Labor Receiving a Smaller Share of Global Income? Theory and Empirical Evidence'. *IMF Working Papers* 2017 (169). https://doi.org/10.5089/9781484311042.001.

Davis, Mike. 2017. *Late Victorian holocausts: El Niño famines and the making of the Third World*. Paperback edition. Verso.

De Keyzer, Maïka. 2018. *Inclusive Commons and the Sustainability of Peasant Communities in the Medieval Low Countries*. Routledge.

De Moor, Tine. 2015. *The Dilemma of the Commoners: Understanding the Use of Common-Pool Resources in Long-Term Perspective*. Cambridge University Press. https://www.cambridge.org/core/books/dilemma-of-the-commoners/BA497A670AD2F573178D89947EC38F7D.

DeMartino, George F., Deirdre McCloskey, John B. Davis, and John B. Davis. 2016. 'Economists' Odd Stand on the Positive–Normative Distinction: A Behavioral Economics View'. In *The Oxford Handbook of Professional Economic Ethics*. Oxford University Press. https://academic.oup.com/edited-volume/28011/chapter/211790248.

Dorninger, Christian, Alf Hornborg, David J. Abson, et al. 2021. 'Global Patterns of Ecologically Unequal Exchange: Implications for Sustainability in the 21st Century'. *Ecological Economics* 179: 106824. https://doi.org/10.1016/j.ecolecon.2020.106824.

Durand, Cédric, and Wiliiam Milberg. 2020. 'Intellectual Monopoly in Global Value Chains'. *Review of International Political Economy* 27 (2): 404–29. https://doi.org/10.1080/09692290.2019.1660703.

EEA. 2015. *The European Environment — State and Outlook 2015: Synthesis Report*. European Environment Agency (EEA). https://doi.org/10.2800/944899.

Elson, Diane. 2000. 'The Progress of Women: Empowerment and Economics'. In *Progress of the World's Women 2000: UNIFEM Biennial Report*, edited by Karen Judd. United Nations Development Fund for Women (UNIFEM).

Engelkamp, Paul, and Friedrich L. Sell. 2013. *Einführung in die Volkswirtschaftslehre*. 6., überarbeitete und erweiterte Aufl age. Springer Berlin Heidelberg. https://link.springer.com/10.1007/978-3-642-36522-5.

Fama, Eugene F. 1970. 'Efficient Capital Markets: A Review of Theory and Empirical Work'. *The Journal of Finance* 25 (2): 383. https://doi.org/10.2307/2325486.

Fisher, Irving. 1922. *The Purchasing Power of Money, Its Determination and Relation to Credit, Interest and Crises*. Macmillan.

Fraser, Nancy. 1990. 'Rethinking the Public Sphere: A Contribution to the Critique of Actually Existing Democracy'. *Social Text*, no. 25/26: 56–80. https://doi.org/10.2307/466240.

Fraser, Nancy. 2022. *Cannibal Capitalism: How Our System Is Devouring Democracy, Care, and the Planet - and What We Can Do about It*. Verso.

Friedli, Vera, Renato Müller Vasquez Callo, and Rahel Balmer-Zahnd. 2019. *Betriebswirtschaftslehre: Zusammenhänge Verstehen*. 4. Auflage. Hep Verlag AG.

Friedman, Milton. 1953. *The Methodology of Positive Economics*. University of Chicago. https://www.marxists.org/reference/subject/philosophy/works/us/friedman.htm.

Ghosh, Amitav. 2021. *The Nutmeg's Curse: Parables for a Planet in Crisis*. University of Chicago Press.

Green, Duncan. 2016. 'What's Happening on Global Inequality? Putting the "Elephant Graph" to Sleep with a "Hockey Stick"'. In *From Poverty to Power*. https://frompoverty.oxfam.org.uk/whats-happening-on-global-inequality-putting-the-elephant-graph-to-sleep-with-a-hockey-stick/.

Halbeisen, Patrick, Margrit Müller, Béatrice Veyrassat, and Bernard Degen. 2012. 'Arbeit Und Kapital'. In *Wirtschaftsgeschichte Der Schweiz Im 20. Jahrhundert*. Schwabe Verlag.

Hardin, Garrett. 1968. 'The Tragedy of the Commons'. *Science* 162 (3859): 1243–48. https://doi.org/10.1126/science.162.3859.1243.

Harvey, David. 2005. *A Brief History of Neoliberalism*. Oxford Scholarship Online. Oxford University Press. https://doi.org/10.1093/oso/9780199283262.001.0001.

Herzog, Lisa, Samuel Bowles, and Axel Honneth. 2014. 'Welche Güter sollen in Märkten gehandlet werden? Was Märkte können - und was nicht'. In *Der Wert des Marktes: ein ökonomisch-philosophischer Diskurs vom 18. Jahrhundert bis zur Gegenwart*. Suhrkamp.

Hickel, Jason. 2017. *The Divide - a Brief Guide to Global Inequality and Its Solutions*. William Heinemann.

Hill, Lisa. 2012. 'Adam Smith on 'Thumos' and Irrational Economic "Man"'. *The European Journal of the History of Economic Thought* 19 (1): 1–22. https://doi.org/10.1080/09672561003632550.

Hodgson, Geoffrey M. 2014. 'What Is Capital? Economists and Sociologists Have Changed Its Meaning: Should It Be Changed Back?' *Cambridge Journal of Economics* 38 (5): 1063–86. https://doi.org/10.1093/cje/beu013.

Huber, Matt. 2013. 'Fueling Capitalism: Oil, the Regulation Approach, and the Ecology of Capital'. *Economic Geography* 89 (2): 171–94. https://doi.org/10.1111/ecge.12006.

Jackson, Tim. 2009. *Prosperity Without Growth? The Transition to a Sustainable Economy*. Sustainable Development Commission.

Jarrige, François, Thomas Le Roux, Janice Egan, and Michael Egan. 2020. *The Contamination of the Earth: A History of Pollutions in the Industrial Age*. The MIT Press.

Jevons, William Stanley. 1879. *The Theory of Political Economy*. 2nd edn. Macmillan.

Johnston, David. 2011. *A Brief History of Justice*. Brief Histories of Philosophy. Wiley-Blackwell. https://doi.org/10.1002/9781444397550.

Kalaitzakis, Alexandros, and Christos Papatheodorou. 2026. 'Functional and Personal Income Distribution and Their Effect on Demand: An Empirical Assessment on the EU-15 Countries'. *Journal of Post Keynesian Economics* 49 (3): 560–604. https://doi.org/10.1080/01603477.2026.2626368.

Kalecki, Michal. 1943. 'The Political Aspects of Full Employment'. *The Political Quarterly* 14 (4): 322–30. https://doi.org/10.1111/j.1467-923X.1943.tb01016.x.

Kallis, Giorgos, and Jalel Sager. 2017. 'Oil and the Economy: A Systematic Review of the Literature for Ecological Economists'. *Ecological Economics* 131: 561–71. https://doi.org/10.1016/j.ecolecon.2016.08.011.

Kapp, K. William. 1974. *Environmental Policies and Development Planning in Contemporary China and Other Essays*. De Gruyter. https://www.degruyter.com/document/doi/10.1515/9783112317273/html.

Kapp, William. 1950. *The Social Costs of Private Enterprise*. Harvard University Press.

Keen, Steve. 2010. *Debunking economics: the naked emperor of the social sciences*. 5. impression. Zed Books.

Keynes, John Maynard. 1978. 'THE END OF LAISSEZ-FAIRE (1926)'. In *The Collected Writings of John Maynard Keynes*, edited by Donald Moggridge and Elizabeth Johnson, Volume 9: Essays in Persuasion. Royal Economic Society. https://doi.org/10.1017/UPO9781139524162.027.

Klapeer, Christine M., Karin Fischer, Gerhard Hauck, and Manuela Boatcă. 2023. 'Post-Development'. In *Handbuch Entwicklungsforschung*. Springer Fachmedien. https://doi.org/10.1007/978-3-658-32946-4_12.

Kubiszewski, Ida, Robert Costanza, Carol Franco, et al. 2013. 'Beyond GDP: Measuring and Achieving Global Genuine Progress'. *Ecological Economics* 93: 57–68. https://doi.org/10.1016/j.ecolecon.2013.04.019.

Lannen, Anu, Sabin Bieri, and Christoph Bader. 2019. *Inequality: What?s in a Word?* Centre for Development; Environment (CDE), University of Bern. https://boris.unibe.ch/130698/.

Lawrence, Michael, Thomas Homer-Dixon, Scott Janzwood, Johan Rockström, Ortwin Renn, and Jonathan F. Donges. 2024. 'Global Polycrisis: The Causal Mechanisms of Crisis Entanglement'. *Global Sustainability* 7: e6. https://doi.org/10.1017/sus.2024.1.

Lepenies, Philipp. 2013. *Die Macht Der Einen Zahl: Eine Politische Geschichte Des Bruttoinlandsprodukts*. Erste Auflage, Originalausgabe. Suhrkamp.

Lessenich, Stephan. 2018. *Neben Uns Die Sintflut: Wie Wir Auf Kosten Anderer Leben*.

Lessenich, Stephan. 2020. *Neben uns die Sintflut: die Externalisierungsgesellschaft und ihr Preis*. 6. Auflage. Hanser Berlin.

Magdoff, Fred, and Harry Magdoff. 2004. 'Disposable Workers: Today's Reserve Army of Labor'. *Monthly Review* 55 (11): 18. https://doi.org/10.14452/MR-055-11-2004-04_2.

Maitland, Alex, Anjela Taneja, Anthony Kamande, et al. 2026. *Resisting the Rule of the Rich*. Oxfam International. https://doi.org/10.21201/2025.000113.

Malm, Andreas. 2016. *Fossil capital: the rise of steam-power and the roots of global warming*. Verso.

Mankiw, Nicholas Gregory, and Mark P. Taylor. 2020. *Economics*. Fifth edition. Cengage.

Marshall, Alfred. 1919. *Industry and Trade: A Study of Industrial Technique and Business Organization; and of Their Influences on the Conditions of Various Classes and Nations*. Macmillan.

Marshall, Alfred. 2013. *Principles of Economics*. 8th edn. Palgrave Macmillan UK.

Maslow, Abraham H. 1966. *The Psychology of Science: A Reconnaissance*. Harper & Row.

Mattei, Ugo. 2014. 'Eine Kurze Phänomenologie Der Commons'. In *Commons*, edited by Silke Helfrich and Heinrich-Böll-Stiftung. Transcript Verlag. https://doi.org/10.14361/9783839428351-010.

Mazzucato, Mariana. 2018. *The Value of Everything: Making and Taking in the Global Economy*. Hachette UK.

Mazzucato, Mariana. 2022. *Mission economy: a moonshot guide to changing capitalism*. Penguin Books.

Milanovic, Branko. 2003. 'The Two Faces of Globalization: Against Globalization as We Know It'. *World Development* 31 (4): 667–83. https://doi.org/10.1016/S0305-750X(03)00002-0.

Minsky, Hyman. 1992. *The Financial Instability Hypothesis*. No. 74. The Jerome Levy Economics Institute of Bard College.

Mirowski, Philip. 1989. *More Heat Than Light: Economics as Social Physics, Physics as Nature's Economics*. Cambridge University Press. https://www.cambridge.org/core/books/more-heat-than-light/4CD2ADE8D5DE8665E43E2922D7E360B3.

Morgan, Mary S. 2006. 'Economic Man as Model Man: Ideal Types, Idealization and Caricatures'. *Journal of the History of Economic Thought* 28 (1): 1–27. https://doi.org/10.1080/10427710500509763.

Morgan, Mary S. 2012. *The World in the Model: How Economists Work and Think*. Cambridge University Press. https://www.cambridge.org/core/books/world-in-the-model/6FD82E4D498F94CBE5F56078FD007729.

Nordhaus, William. 2019. 'Climate Change: The Ultimate Challenge for Economics'. *American Economic Review* 109 (6): 1991–2014. https://doi.org/10.1257/aer.109.6.1991.

Novy, Andreas, Richard Bärnthaler, and Veronika Heimerl. 2020. *Zukunftfähiges Wirtscchaften*. Beltz Juventa.

Novy, Andreas, Richard Bärnthaler, and Magdalena Prieler. 2023. *Zukunftsfähiges Wirtschaften: Herausforderungen der sozialökologischen Transformation*. 2. Auflage. Juventa Verlag.

Nubbemeyer, Elmar. 2010. 'A Reconsideration of Full-Cost Pricing - Methodological Aspects of Marginalism and Theoretical Explanations of Pricing Behaviour'. PhD thesis, Ludwig-Maximilians-Universität.

Nussbaum, Martha. 2003. 'Capabilites as Fundamental Entitlements: Sen and Social Justice'. *Feminist Economics* 9 (2/3): 33–59.

OECD. 2011. *Towards Green Growth*. OECD Publishing. https://doi.org/10.1787/9789264111318-en.

Ostrom, Elinor. 2010. 'Beyond Markets and States: Polycentric Governance of Complex Economic Systems'. *American Economic Review* 100 (3): 1–33.

Ott, Konrad, and Ralf Döring. 2004. *Theorie Und Praxis Starker Nachhaltigkeit*. Metropolis.

Paech, Niko. 2012. *Befreiung Vom Überfluss. Auf Dem Weg in Die Postwachstumsökonomie*. München.

Pareto, Vilfredo, and Wilhelm Franz Meyer. 1902. 'Anwendungen Der Mathematik Auf Nationalokonomie'. In *Encyklopädie Der Mathematischen Wissenschaften Mit Einschluss Ihrer Anwendungen*. B.G. Teubner.

Patnaik, Utsa, and Prabhat Patnaik. 2021. *Capital and Imperialism: Theory, History, and the Present*. Monthly Review Press.

Petschow, Ulrich, Steffen Lange, and David Hofmann. 2020. *Social Well-Being Within Planetary Boundaries: The Precautionary Post-Growth Approach*. Umweltbundesamt. https://www.umweltbundesamt.de/en/publikationen/social-well-being-within-planetary-boundaries-the.

Petschow, Ulrich, Eugen Pissarskoi, Nils aus dem Moore, Hermann Ott, and Umweltbundesamt. 2018. *Gesellschaftliches Wohlergehen innerhalb planetarer Grenzen: Der Ansatz einer vorsorgeorientierten Postwachstumsposition*. No. 89. Umweltbundesamt.

Piketty, Thomas, and Arthur Goldhammer. 2014. *Capital in the twenty-first century*. The Belknap Press of Harvard University Press.

Pirgmaier, Elke. 2017. 'The Neoclassical Trojan Horse of Steady-State Economics'. *Ecological Economics* 133: 52–61. https://doi.org/10.1016/j.ecolecon.2016.11.010.

Polanyi, Karl. 2017. *The Great Transformation*. 13. Auflage. Suhrkamp Taschenbuch Verlag.

Polanyi, Karl, Karl Polanyi, Conrad M. Arensberg, and Harry W. Pearson. 1957. 'The Economy as Instituted Process'. In *Trade and Market in the Early empires*. The Free Press & The Falcon's Wing Press.

Ponte, Stefano. 2022. 'The Hidden Costs of Environmental Upgrading in Global Value Chains'. *Review of International Political Economy* 29 (3): 818–43. https://doi.org/10.1080/09692290.2020.1816199.

Pufé, Iris. 2017. *Nachhaltigkeit*. 3., überarbeitete und erweiterte Auflage. UVK Verlagsgesellschaft mbH.

Putnam, Hilary. 2002. *The Collapse of the Fact-Value Dichotomy and Other Essays*. Harvard University Press.

Rawls, John. 1999. *A Theory of Justice: Revised Edition*. Harvard University Press. https://doi.org/10.2307/j.ctvkjb25m.

Raworth, Kate. 2017. 'A Doughnut for the Anthropocene: Humanity's Compass in the 21st Century'. *The Lancet Planetary Health* 1 (2): e48–49. https://doi.org/10.1016/S2542-5196(17)30028-1.

Reiss, Julian. 2000. 'Mathematics in Economics: Schmoller, Menger and Jevons'. *Journal of Economic Studies* 27 (4/5): 477–91. https://doi.org/10.1108/01443580010342393.

Robbins, Lionel. 1932. *An Essay on the Nature and Significance of Economic Science*. Macmillan & Co. Limited.

Robson, John M., and John Stuart Mill. 1996. *Essays on Economics and Society*. University of Toronto Press.

Rockström, Johan, Will Steffen, Kevin Noone, et al. 2009. 'A Safe Operating Space for Humanity'. *Nature* 461 (7263): 472–75. https://doi.org/10.1038/461472a.

Rogall, Holger. 2012. *Nachhaltige Ökonomie. Ökonomische Theorie Und Praxis Einer Nachhaltigen Entwicklung*. 2., überarbeitete und stark erweiterte Auflage. Metropolis.

Rogall, Holger. 2013. *Volkswirtschaftslehre Für Sozialwissenschaftler: Einführung in Eine Zukunftsfähige Wirtschaftslehre*. 2. Aufl. Lehrbuch. Springer VS. https://doi.org/10.1007/978-3-658-01980-8.

Rosa, Hartmut. 2003. 'Social Acceleration: Ethical and Political Consequences of a Desynchronized High–Speed Society'. *Constellations* 10 (1): 3–33. https://doi.org/10.1111/1467-8675.00309.

Rosa, Hartmut, and Robin Celikates. 2024. *Beschleunigung und Entfremdung: Entwurf einer kritischen Theorie spätmoderner Zeitlichkeit*. 10. Auflage. Suhrkamp.

Saave, Anna, and Friedrich-Schiller-Universität Jena. 2022. *Einverleiben und Externalisieren: zur Innen-Außen-Beziehung der kapitalistischen Produktionsweise*. Transcript.

Sakschewski, Boris, Levke Caesar, Lauren Andersen, et al. 2025. *Planetary Health Check 2025: A Scientific Assessment of the State of the Planet*. Potsdam Institute for Climate Impact Research (PIK). https://publications.pik-potsdam.de/pubman/item/item_32589.

Santarius, Tilman. 2014. *Der Rebound-Effekt: Über Die Unerwünschten Folgen Der Erwünschten Energieeffizienz*. Wuppertal Institut für Klima, Umwelt und Energie.

Schaupp, Simon. 2024. *Stoffwechselpolitik: Arbeit, Natur und die Zukunft des Planeten*. Erste Auflage, Sonderdruck, Originalausgabe. Suhrkamp.

Scherschel, Karin, Peter Streckeisen, Manfred Krenn, and Peter Streckeisen. 2012. 'Steigende Erwerbslosigkeit Und Prekarität in Der Schweiz: Das Ende Eines "Sonderfalls"'. In *Neue Prekarität: Die Folgen Aktivierender Arbeitsmarktpolitik: Europäische Länder Im Vergleich*. Campus.

Schmelzer, Matthias, and Andrea Vetter. 2019. *Degrowth/Postwachstum*. Junius.

Schmelzer, Matthias, and Andrea Vetter. 2023. *Degrowth/Postwachstum zur Einführung*. 3., unveränderte Auflage. Junius.

Sedlacek, Tomas. 2012. *Die Ökonomie von Gut Und Böse*. Hanser.

Seidl, Irmi, and Angelika Zahrnt, eds. 2010. *Postwachstumsgesellschaft. Konzepte Für Die Zukunft*. Ökologie Und Wirtschaftsforschung. Marbug.

Sen, Amartya. 1999. *Commodities and Capabilities*. Oxford University Press.

Sheldon, George. 2007. *Migration, Integration und Wachstum: Die Performance und wirtschaftliche Auswirkung der Ausländer in der Schweiz*. 01/07. Center of Business; Economics (WWZ) / Universität Basel.

Slobodian, Quinn. 2018. *Globalists: The End of Empire and the Birth of Neoliberalism*. Harvard University Press,. https://slsp-ube.primo.exlibrisgroup.com/view/action/uresolver.do?operation=resolveService&package_service_id=28697326480005511&institutionId=5511&customerId=5500&VE=true.

Smith, Adam, and Horst Claus Recktenwald. 2013. *Der Wohlstand der Nationen: eine Untersuchung seiner Natur und seiner Ursachen [1789]*. 13. Auflage. Deutscher Taschenbuch Verlag.

Steffen, Will, Wendy Broadgate, Lisa Deutsch, Owen Gaffney, and Cornelia Ludwig. 2015. 'The Trajectory of the Anthropocene: The Great Acceleration'. *The Anthropocene Review* 2 (1): 81–98. https://doi.org/10.1177/2053019614564785.

Vazquez, Andres. 1995. 'Marshall and the Mathematization of Economics'. *Journal of the History of Economic Thought* 17 (2): 247–65. https://doi.org/10.1017/S1053837200002625.

Weizsäcker, Ernst Ulrich von, Amory Lovins, and L. Hunter Lovins. 1997. *Faktor Vier - doppelter Wohlstand - halbierter Naturverbrauch. Der neue Bericht an den Club of Rome.* 10. Aufl. Droemer Knaur.

Wilkinson, Richard G., and Kate Pickett. 2011. *The Spirit Level: Why Greater Equality Makes Societies Stronger*. Bloomsbury.

Wilkinson, Richard G., and Kate E. Pickett. 2024. 'Why the World Cannot Afford the Rich'. *Nature* 627 (8003): 268–70. https://doi.org/10.1038/d41586-024-00723-3.

Wilson, David, and William Dixon. 2014. *A History of Homo Economicus - the Nature of the Moral in Economic Theory*. Paperback Edition. Routledge.

Wood, Ellen. 2015. *Der Ursprung Des Kapitalismus - Eine Spurensuche*. 1. Auflage. LAIKA-Verlag.

Wood, Ellen Meiksins, and Verso (Firm : London, England). 2017. *The Origin of Capitalism: A Longer View*. New, revised and expanded edition. Verso.

World Bank. 2012. *Inclusive Green Growth: The Pathway to Sustainable Development*. https://hdl.handle.net/10986/6058.